

FEDERAL REPUBLIC OF SOMALIA

Somali Livestock Sector Development Strategy 2020 - 2030



MINISTRY OF LIVESTOCK
FORESTRY AND RANGE





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Sector Strategy Paper

SOMALIA LIVESTOCK SECTOR DEVELOPMENT STRATEGY

MINISTRY OF LIVESTOCK, FORESTRY AND RANGE

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ABBREVIATIONS

CAHW	Community-based Animal Health Worker
CBPP	Contagious Bovine Pleuropneumonia
CCPP	Contagious Caprine Pleuropneumonia
CGIAR	Consultative Group on International Agricultural Research
CLT	Critical Livestock Threshold
COMTRADE	Common Format for Transient Data Exchange
CSO	Community Service Organization
DFID	U.K. Department for International Development
DWLM	Department of Water and Land Management
EU	European Union
FAN Meat	Farm Assured Namibian Meat
FAO	Food and Agriculture Organization
FAOSTAT	Food and Agriculture Organization Statistical Database
FEWS	Famine Early Warning System
FGS	Federal Government of Somalia
FMD	Foot and Mouth Disease
FMS	Federal Member State
FSNAU	Food Security and Nutrition Analysis Unit
FSNWG	Food Security and Nutrition Working Group
GDP	Gross Domestic Product
GEEL	Growth, Enterprise, Employment and Livelihoods
HACCP	Hazard Analysis and Critical Control Points
HoA-REC&N	Horn of Africa Regional Environment Centre and Network
ICPALD	IGAD Center for Pastoral Areas and Livestock Development
ICT	Information and Communication Technology
IDP	Internally Displaced Person
IGAD	Intergovernmental Authority on Development
IMF	International Monetary Fund
INDC	Intended Nationally Determined Contributions
IPC	Integrated Phase Classification
IPCC	Intergovernmental Panel on Climate Change
ISTVS	IGAD Sheikh Technical Veterinary School and Reference Center
KMF	Kismayo Meat Factory
KSA	Kingdom of Saudi Arabia
LPA	Livestock Professional Association
LSD	Lumpy Skin Disease
MoLFR	Ministry of Livestock, Forestry, and Range
NAPA	National Adaptation Programme of Action to Climate Change
NDP	National Development Plan
NGO	Nongovernmental Organization
OIE	World Organization for Animal Health
PENHA	Pastoral and Environmental Network in the Horn of Africa



PET	Pictorial Evaluation Tool
PIMS	Promoting Inclusive Markets in Somalia
PPP	Public-Private Partnership
PPR	<i>Peste des Petits Ruminants</i>
RVF	Rift Valley Fever
SATG	Somali Agriculture Technical Group
SDG	Sustainable Development Goal
SELF	Small Entrepreneurs Loan Facility
SFD	Saudi Fund for Development
SGP	Sheep and Goat Pox
SLA	Somali Leather Agency
SME	Small and Medium Enterprise
SNU	Somali National University
SPS	Sanitary and Phytosanitary (Standards)
SWALIM	Somalia Water and Land Information Management
TAD	Transboundary Animal Disease
UAE	United Arab Emirates
UN	United Nations
UNDP	United Nations Development Programme
UNFCCC	United Nations Framework Convention on Climate Change
UNFPA	United Nations Population Fund
UNICEF	United Nations Children's Fund
USAID	United States Agency for International Development
WFP	World Food Programme
WTO	World Trade Organization



PREFACE

In mid-2018, the Ministry of Livestock, Forestry and Range (MoLFR) of the Federal Government of Somalia (FGS) began preparation of the Somalia Livestock Sector Development Strategy (LSDS), with technical assistance from the World Bank and the Food and Agriculture Organization (FAO) of the United Nations. Under the leadership of MoLFR, an extensive consultative and participatory approach involving all Federal Member States (FMSs), other relevant stakeholders (public, private and civil society) and teams of specialists was adopted during all phases of the LSDS preparation. Consultation workshops were held with MoLFR and Benadir (October 14th -15th, 2018), South-West (16th -17th), Somaliland (20th -21th), Puntland (23th -24th), Galmudug (28th -29th), Hirshabelle (30th -31th) and Jubaland (November 3th -4th, 2018), while a validation workshop was conducted in Mogadishu with all stakeholders (June 17th -19th, 2019). This LSDS therefore fully captures the livestock sector's main challenges and opportunities as well as a common vision for its development by 2030 among all stakeholders.

The economic and social development of Somalia lies in its ability to exploit sustainably its natural resource base, which is increasingly buffeted by a fragile natural environment and climate change, while at the same time fostering shared growth and lifting the health and educational status of its people. This requires the public sector to adopt enabling and effective policies, strategies, institutions, laws and regulations and to provide key public goods (including essential infrastructure and services) that can increase private investment across the productive sectors. Livestock remains the backbone of the Somali economy in terms of production value, value added, and foreign currency earnings, and the vast majority of Somalia's population is dependent on livestock and its products for their livelihoods and their food and nutrition security. The failure to develop the country's livestock sector, including strengthening its resilience to climate change and ensuring the quality and safety of its products would, thus, would greatly constrain the country's prospects for economic growth and recovery.

The FGS, led by H.E. Prime Minister Hassan Ali Khayre, and the MoLFR are committed to improving the livelihoods of millions of pastoralists and agro-pastoralists, traders, processors and rural communities across Somalia who are dependent on the livestock sector. It welcomes the collaboration of its domestic and external partners to maximize the available technical and financial support to realize its potential as a major contributor to the country's sustainable, inclusive, and equitable socio-economic development and poverty reduction. An imperative to achieve these goals will be to establish partnerships with international organizations and academic institutions, including for North-South, South-South and triangular cooperation on scientific and research projects and programs, which can help put this sector's strategy and other innovative and creative ideas into action, deliver the desired results and impacts for all the sector's stakeholders, producers and consumers alike.

MoLFR would like to thank the officials of all governmental and non-governmental organizations, the technical teams at the federal and state levels, and all other relevant stakeholders (public, private and civil society) who participated in formulating the Somalia LSDS. Furthermore, the FGS wishes to express its deep gratitude to the joint team of the World Bank and the FAO for their strong dedication, effective technical contribution, and sustained efforts in coordinating inputs from many Somali and external experts, while respecting and supporting the leadership and ownership of the process by MoLFR. A special appreciation is also extended to the international community and donors for their continuous support and partnership towards the development and sustainable growth of Somalia's livestock sector, its economy, and its development objectives.

Hon. Hussein Mohamud Sheikh-Hussein, MP.
Minister, Livestock Forestry and Range
Mogadishu, July 2019



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The Ministry would like to express its utmost gratitude to His Excellency Prime Minister Hassan Ali Khaire of the Federal Government of Somalia (FGS), who resolutely introduced the Economic Development Roadmap initiative, known as “*Tubta*” in the Somali language, of which MoLFR is a member. The Somalia Livestock Sector Development Strategy (LSDS) is a key part of the planned activities by MoLFR under this initiative and its preparation within a period of only one year is a miraculous achievement. It is a great pleasure, an honor and a privilege that Somalia LSDS will be implemented under the auspices of Hon. Hussein Mohamud Sh. Hussein (MP) the Minister, Livestock, Forestry & Range of the Federal Government of Somalia. As envisioned, the implementation of this LSDS holds strong scope to drive sustainable and resilient growth of Somalia’s livestock sector and to make an important contribution to the country’s socio-economic recovery, poverty reduction gains and long-term development.

Core contributors from MoLFR were Mohamed Omar Nur (Director General), Dr. Abdirahman Nur Kailie (Senior Livestock Sector Advisor), Shire Ahmed Musse (Senior Natural Resource Management and Resilience Advisor), Dr. Abdulahi Araye Addow (Director Animal Production) and Dr. Hashi Osman Mohamed (Director Forestry and Range), Abdulkani R. Barrow (Senior National Coordinator, Joint Charcoal Reduction Programme (PROSCAL)), and Dr. Mohamed Mukhtar Aden (Senior Livestock Sector Advisor)

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EXECUTIVE SUMMARY

A Bold Vision to Tackle Constraints and Harness Potential of the Largest Economic Sector

Somalia's livestock sector has proved remarkably resilient amidst headwinds of fragile governance, social unrest and recurrent droughts. Despite absent or very weak public services, an increasingly narrow and fragile natural resource base, and an arid and semi-arid climate that has become drier, more extreme, and more variable in recent decades, the sector has until recent years posted strong growth, reaching 62 percent of GDP at its peak in 2013-2016 from 37 percent pre-war. Even at the height of a severe drought in 2017, the sector remained the largest contributor (43 percent) to GDP, employment and livelihoods.

The livestock sector remains key to economic growth and poverty reduction, but its potential is hampered by structural weaknesses and by acute exposure to downside risks. The sector's high vulnerability to natural and economic shocks was only recently evidenced by an estimated \$2 billion in damages and losses suffered during the 2016/17 severe drought period and a drastic drop by 2018 of \$183 million in live animal exports after Saudi Arabia and the UAE imposed an import ban on health grounds in December 2016 (from \$430 million in 2016; the drop would be even larger if compared to the peak \$494 million of 2015). Major production constraints related to poor natural resource management, health, nutrition, and genetic resources are compounded by other structural and institutional weaknesses that impede sustainable growth, diversification and value addition.

Nonetheless, despite current challenges, there is strong scope for the sector's recovery and growth in the medium- and long-terms. Sizeable areas of the country are suitable for grazing, browsing, and fodder production. At an estimated 56 million heads in 2018, the country has one of the largest animal populations in the world (ranking 1st for camels and 6th for goats in 2015). Production of milk and meat has grown steadily to satisfy a growing domestic demand in its urban centers, supported by large remittance inflows. And its close proximity to, and favorable consumer preferences in, Gulf markets provide Somalia with a competitive advantage for the export of live animals and animal products.

Success in tackling the sector's constraints and harnessing its potential will require a clear vision of the priority policy reforms and interventions needed to alleviate the most binding constraints and concerted action across federal and state levels to mobilize and guide public and private investment, with strong, coordinated support by its development partners. The shared vision by the Federal Government and Federal Member States and by the key stakeholders is for the livestock sector to make a major contribution by 2030 to: i) improve livelihoods of the majority of the rural population by strengthening the sector's climate resilience and increasing incomes from livestock products; (ii) enhance the nutrition and food security of its people via an expanded supply of high-quality and hygienic animal proteins; and iii) fuel stronger economic growth via more efficient, inclusive and environmentally sustainable livestock production and expanded value-adding processing.

The vision and its objectives, principles, and operational priorities as articulated above were first outlined in the Livestock Policy 2015-2020, embedded in the NDP 2017-2019, and were subsequently reconfirmed by key federal, state and private stakeholders during broad-based stakeholder consultations during October-November 2018. If such a vision is achieved, the sector will not only provide a major contribution to the country's overall economic development, but will also help cement peace and security, alleviate poverty and malnutrition, and enhance health outcomes in both rural and urban areas. Among its key operational goals are:

- *Preserving Animal Resources* - Safeguarding the diversity and optimizing the use of existing animal resources is critical for both resilience and growth.



- *Improving the management of the livestock industry* – Also critical is to improve productivity, including for the rapidly expanding dairy, meat and poultry farming subsectors;
- *Ensuring the health and quality of live animals and livestock products in domestic and international markets* – Stronger food safety is needed to protect consumers, to regain lost export markets for live animals, and to expand into new ones, including for livestock products;
- *Increasing the processing of livestock products* – The sector’s long-term growth is dependent on higher value addition, as the current trajectory of increasing animal populations, supplying domestic and export markets with live animals, and minimal processing has reached its limit;
- *Building public and private service and research institutions, and human capacity* – Public and private investment in finding solutions to current and emerging problems is currently lacking, but is necessary for both resilience and sustainable growth of the sector; and
- *Ensuring security throughout the country* – Without enhanced security in many areas of southern and central regions of Somalia, rehabilitation and development interventions and new business ventures would remain challenging and costly, thus attracting limited donor funding and private investment.

To inform and guide needed action, this Livestock Sector Development Strategy (LSDS) provides a coherent, nationwide framework for the medium- and long-term development and growth of the sector, reflecting the priorities and needs of its key stakeholders beyond those related to short-term recovery from the recent 2016/2017 drought. It is largely informed and benefits from extensive consultations during 2018 and 2019 with public and private stakeholders who shared their vision, challenges, and priority recommendations. The LSDS will also provide a timely contribution to the National Development Plan II that will guide in 2020-2024 public and private investments, as well as external assistance from external donors and development agencies.

Institutional Strengthening to Support Public and Private Interventions

This LSDS supports a co-management framework based on collaboration among public and private institutions. This is needed to fill knowledge gaps, improve the management of the rangelands, deliver core animal health services, rebuild and expand production and trade infrastructure, and promote investment in processing and value-addition. To render PPP arrangements effective, efficient and accountable, however, the roles and responsibilities of public and private agents must be more clearly defined and coordinated. Furthermore, a consultative and participatory approach requires that livestock stakeholders have a voice in policy formulation and project design and implementation through producer/pastoralist associations, civil society organizations and/or other sector interest groups.

The livestock sector falls under the responsibility of the Federal Ministry of Livestock, Forestry and Range (MoLFR), with the engagement of the livestock ministries of the Federal Member States (FMSs). All ministries, however, have major capacity and resource gaps, which inhibit their ability to deliver even the most basic services. Furthermore, budget allocations and physical facilities are very limited, with most staff lacking updated training and needed specializations. The most recently established states in southcentral Somalia, in particular, lack any physical infrastructure and receive virtually no budgetary resources and external funding, with most staff working as unpaid volunteers.

The capacity needs of the public sector to effectively support the sustainable growth of production and trade by the private sector are extensive. They start with lack of pay for virtually all ministry staff in the recently established FMS, who are volunteering their time and operating without any vehicles and other supplies. They extend to policy formulation and review, planning, coordination and implementation of emergency and development programs and projects, drafting and enforcement of laws and regulations, and data collection and analyses. The list continues with



research and extension; monitoring and compliance of, and service provision to, private businesses; and strengthening of public veterinary services. Furthermore, the MoFLR requires technical and financial support from external development partners to establish, with engagement of the FMS line ministries, forest and range reservation lands and sustainable range land management; to rehabilitate, improve and expand water infrastructures and, under public-private partnerships, wholesale and retail market facilities. The relevant ministries need to ensure quality higher education in veterinary science, animal husbandry, and food processing. The FGS, moreover, needs to provide adequate security to facilitate access to finance for the private sector so that it is able to invest in value added processing of livestock products. All stakeholders need also training along the main input and product value chains (feed and fodder, milk, meat, hides and skins, and poultry).

Particular attention in the design and implementation of this strategy has been given to women and youth, whose engagement in livestock trading and marketing has been traditionally at the local and subsistence levels and whose equitable benefits and participation in the future development of the sector should be ensured. The live animal export value chain provides at present the bulk of the employment opportunities, including for traders, both small- and large-scale, brokers, trekkers, feedlot operators, lorry drivers, loaders and others providing ancillary services. However, large employment (and income) opportunities, especially for women and youth, lie in the collection, processing and distribution of livestock products (outputs, such as dairy and meat, and inputs such as fodder).

Modernizing Traditional Systems for Production and Marketing of Live Animals

Both resilience to climate shocks and expansion of livestock production are constrained by several factors linked to the nature of traditional production systems. The key constraints include: (i) degradation and deforestation of the rangelands; (ii) shortages of water, largely due to insufficient rainfall harvesting facilities; (iii) high incidence of diseases and weak systems for the monitoring and response to disease threats; and (iv) inadequate availability of natural feed and commercial fodder. Another set of institutional and structural constraints and gaps is related to the lack of and/or inconsistent implementation of regulations and policies, and insufficient knowledge and data about animal populations, offtake rates, incidence of diseases, and domestic consumption, among others. Somalia conducted its last national census of livestock in 1975, and subsequent animal population and offtake estimates are largely based on limited surveys and extrapolations. Thus, reliable information, including a comprehensive animal population survey, is needed.

There is still no consensus in the scientific literature over whether the Horn of Africa will experience more or less rainfall in coming decades. However, all studies and projections by climate scientists agree on a warmer future, with increased variability and frequency of extreme rainfall events, with related cycles of intensified floods and drought. The current large livestock population estimated at 56 million, compared to historical pre-war levels of less than 40 million, moreover, has been putting a major strain on limited natural resources such as rangeland, vegetation cover, and water. Moreover, the increasing frequency of severe droughts has meant that rangelands have less time to rebound and recover before the next one. Recommended priority actions are:

- Large scale investments in watershed management and infrastructure that would mitigate the impact of extreme cycles of rainfall, floods and drought will be critical for the resilience of Somali livelihoods, including those dependent on livestock.
- Improving the natural rangelands resource base is critical to sustaining the production and productivity of livestock, in parallel with improving livestock productivity and value addition.



- Efforts to improve land tenure management are also required to deal with grazing rights, the fast-expanding private enclosures on previously communal rangelands, and commercial crop- and grasses-producing areas
- Ensure secure migration routes to maintain mobility of livestock both within the country and cross-border.

While local breeds are well adapted to the country's harsh geographical and arid climate, poor breeding practices are a factor contributing to their low live weight and productivity, e.g. in milk and meat yields. Experiences from similar countries, such as Ethiopia, however, shows that the extensive pastoral production systems in challenging arid and semi-arid lands pose a big challenge for the feasibility, adaptability and sustainability of improved exotic breeds and hybrids. Such experiences instead clearly demonstrate the benefits of community-based breeding programs, which should remain the main core strategy to improve productivity while maintaining adaptive traits, though cross-breeding is recommended for cattle.

Despite widespread livestock ownership, most Somalis have a diet composed more of cereals (home grown sorghum and maize and imported rice and wheat) and sugar and oil (also imported). This is complemented by consumption of animal products, mostly fresh milk and meat (including from poultry), which provide the most important source of protein. Within even pure pastoral households highly dependent on markets and basic food imports, Somalia is very vulnerable to market disruptions caused by conflict or by severe global market price hikes in commodities. A flourishing livestock export sector is thus crucial in financing the substantial annual import bill required to ensure food security for many Somalis. Disruptions in exports, therefore, have negative effects on pastoral households' food security as they rely on animal sales to access food markets.

Somalia's close proximity to various export markets and favourable consumer preferences in the Gulf markets provide Somalia with a competitive advantage that can support growth of exports of live animals as well as of meat, hides and skins, and possibly other livestock products. To exploit this potential advantage, however, Somalia needs to address successfully the vexing issue of the reliability and credibility of its health certification system for exports, and to diversify and reduce both its export product concentration and its heavy reliance on a few export markets, which makes the live-animal export sector acutely vulnerable to exogenous risks. These markets include Saudi Arabia and the U.A.E. for shoats, Yemen and Oman for cattle and shoats, Kenya for cattle, and Egypt for camels.

A deeper understanding of the dynamics and decision-making processes behind the recurring import bans is required, as are urgently needed interventions to address export ban and related market risks, including:

- Reassessing the relationship between public and private veterinary services and quarantine stations and the responsibilities of each stakeholder;
- Using a unified Animal Health Certificate issued only by FGS public veterinarians, as already agreed among the FGS and the FMs and with Saudi Arabia, and as mandated by law;
- Analyzing the cost of regulatory compliance and enforcement (for example the three-week quarantine and health certificate emission) for both public and private sector players;
- Mainstream standard methods and procedures (SMPs) for export quarantines, already validated by IGAD member states.

Strengthening Animal Health, Welfare and Nutrition

While the private sector and livestock professional associations (LPAs), accredited by the MoLFR and funded by development partners, have largely taken up the provision of critical veterinary



services, such efforts remain insufficient to address the serious risk of animal diseases and consequent import bans. Moreover, animal welfare issues arise during loading, transporting and unloading of livestock, not so much because of mistreatment by their handlers, but mainly because of the inadequacy of available facilities and services, such as sanitation, nutrition, water, and veterinary care, both in rural areas and along transit routes. Even though the FGS adopted in 2016 a National Veterinary Law, there is need for a more comprehensive legislative and regulatory framework and for effective institutional arrangements for its implementation and enforcement.

To improve animal health as well as to meet international standards and requirements of importing countries, this LSDS prioritizes the following interventions:

- Apply the OIE-PVS Pathway Tool to assess, monitor and improve veterinary services;
- Establish a national agency for recording disease information and outbreaks reports and for conducting epidemiological studies;
- Mainstream livestock health data in the OIE/AU-IBAR-managed information systems and within the framework of the One Health Framework;
- Improve the administrative structure of the public veterinary services, and its collaboration with LPAs, entrepreneurs and producers' cooperatives;
- Make qualified manpower available in the field and as advisors to federal and state ministries;
- Support the development of community-managed health services;
- Monitor and promote proper drug quality and use to minimize the risk of antimicrobial resistance; and
- Study animal welfare issues and train stakeholders in relevant international standards.

Ensuring year-round affordable good quality feed is important to improve animal quality, to mitigate lack of pasture and browse during seasonal or prolonged severe droughts, and to reduce the pressure of a large animal population on increasingly scarce natural resources. The feed subsector, however, has always been and is to this day characterized by a traditional low-input grazing and browsing system, which is subject to seasonal shortages of pasture and very limited commercial feed/fodder production. Other major challenges are degradation, deforestation and continuous overgrazing driving massive-scale soil erosion, desertification, and reduction in edible biomass on the rangelands, due to weaknesses of government agencies tasked to manage the country's natural resources (especially since the demise of the pre-war National Range Agency) and the erosion of traditional rangeland management institutions. Other constraints and gaps are the uncontrolled expansion of enclosures on previously open rangeland, the low levels of technical knowledge on fodder production and storage, poor access to fodder seeds and other production inputs, inadequate availability of water, storage, and transport infrastructure, and in southern Somalia, unsettled land tenure issues.

The 2016-2017 major prolonged drought that led to high mortality of livestock across the country, however, has stoked interests among investors and traders in producing, commercializing and storing improved feed and fodder from grasses, crops residues (mainly sorghum and maize stalks), and oil seeds. Development opportunities in this subsector can rely on much increased demand for fodder throughout the year by expanding commercial livestock rearing operations, field testing by some research institutes (e.g. SATG) of promising better grasses, and existing large-scale commercial production of cereals and oil seeds in some regions. Crop production could easily be doubled to reach again pre-war levels, if necessary investments in the rehabilitation of irrigation and flood control infrastructure will be made and in promoting the widespread adoption of smart agricultural techniques in rainfed cultivation.

To address the major constraints and exploit available opportunities, the subsector would benefit from the following priority interventions related to water management and fodder supply:



- Rehabilitate the existing water reservoirs and increase the number of water catchment facilities, both along the two major rivers and on the rangelands;
- Conduct a feed and fodder inventory and balance assessment, to assess availability based on supply estimates and feed needs based on the estimates from the expected animal population survey;
- Study potential feed production capacity, linkages of feed and fodder value chains with relevant subsectors, access to production areas and markets, and transport linkages between them and develop a livestock feed and rangeland strategy;
- Analyze feed and fodder value chain linkages with all relevant subsectors, access to production areas and markets and transport linkages between them
- Consider market support schemes to stimulate and facilitate inclusion of distant communities;
- Support technical training on feed and fodder production;
- Support private sector and cooperative investment in feed and fodder production; and
- Re-establish a rangeland management agency and enforce pre-war and/or more recent policies and laws (and traditional systems) to rehabilitate and manage the rangelands.

Ensuring Food Safety and Promoting Value Addition in Livestock Products

The main constraints that have kept milk production and the processed dairy sector from reaching its potential are both institutional and structural. However, demand for fresh milk and other processed dairy products will continue to increase in all states beyond the natural population growth, which alone will provide over a million new potential consumers in the next decade. Major growth constraints include most of those affecting negatively the health and welfare of live animals as well as the poor food safety practices and poor environment for private sector investment. Promoting more varied dairy and by-products of meat production and consumption, including from poultry, would have positive nutritional outcomes for the country's population, besides being an additional source of income for Somali pastoralists and agro-pastoralists.

Dairy-subsector stakeholders see strong prospects for value-addition via processing (pasteurization and yogurt and cheese production, among others), if supported by an appropriate public policy and robust private investment. With regards to the prospects for dairy processing, two recent studies by Oxfam and the FAO-Investment Centre found that most consumers perceive fresh milk to be of superior quality and hence prefer that to any other dairy product on the market. Thus, they concluded that value addition via processing of milk may not be profitable, as demand for pasteurized milk and other dairy products represents a small niche market segment, as most buyers are unwilling or unable to pay for processing of any kind. The governments' and stakeholders' more optimistic view, however, is rooted in the current poor and inefficient state of the subsector, the large number of available milk-producing animals, the commercialization trend that has already started, and the high, growing and more sophisticated domestic demand linked to rapid urbanization, higher incomes, and diaspora returnees. This view is also buttressed by international comparisons and by trade statistics, in particular, the large bill in recent years for imported dairy products, ranging between US\$60 million and \$82 million annually, mostly for milk powder and long-life packaged milk.

Despite many challenges, a market-oriented peri-urban camel and cattle production system has been emerging around many towns across Somalia, where the sale of raw milk is already a very important economic activity. The growth opportunities for milk producers in the peri-urban context are based and depend upon mainly the remaining availability of fertile rangelands near many peri-urban areas, opportunities to deploy solar and wind power to reduce energy costs, especially for cold chains, and improved livestock management and feeding practices that can raise productivity by reducing seasonal fluctuations without expanding herd size.



In addition to improving subsector specific knowledge, there is a need to strengthen the subsector's organisation and management and to draw private investment for its expansion and modernization. The immediate priority interventions proposed from the literature and the stakeholders are:

- Improve veterinary and extension services to control the spread of animal diseases, improve the health of the peri-urban herds, and meet hygienic standards for milk handling and transport;
- Strengthen government oversight and auditing of milk production, handling and transport facilities, to ensure food safety and that quality standards are applied effectively;
- Support the establishment of a Dairy Board to represent peri-urban investors and serve as an entry point for discussion and private sector participation in policy dialogue;
- Adopt a two-pronged approach to the sector's development: promote safe raw fresh milk production and marketing for the rural population and processed dairy products, including yogurt made from camel milk, for urban and diaspora consumers;
- Promote and facilitate credit access for, and private sector investment in, dairy processing (including cold chain infrastructure, technology, and business management); and
- Foster widespread adoption, including by facilitating access to credit, of dairy processing and preservation techniques—to support expanded supply of products with a long shelf-life.

The meat value chain, like that for dairy, is also still almost exclusively focused on supplying the local market. Most slaughterhouses and meat markets are largely unhygienic, in various states of disrepair, and lack drainage systems. Many small and medium-size towns have no slaughterhouses at all. While there are signs of incipient commercialization, current processing is minimal, with only three meat processing companies operating in the country. The oldest in Mogadishu (SoMeat) has two modern slaughterhouses and exports frozen cattle carcasses to Oman (and in small quantities also to Qatar) in refrigerated containers, with plans to invest in the machinery required for producing and exporting cuts. The other two operations on the outskirts of Hargeisa (Hilibsan) and Galkayo (Beira Food Factory) to process and sell meat in local markets in a hygienic way, have been recently established but have plans to expand, including into meat canning.

Unlike for dairy, however, there are strong opportunities for expanding meat exports to Gulf States and in the broader East African region, where demand is projected to increase. In addition, there are prospects of selling meat to AMISOM troops in the country. Meat export prospects, however, need to be further analyzed vis-à-vis other countries like Pakistan, whose exports have already displaced Somali meat in the UAE market. Furthermore, an expansion of frozen meat exports via sea beyond Mogadishu and Berbera, the only two container ports in the country, looks unlikely and that of chilled meat beyond major cities with international passenger flights may not be cost-effective, due to the high cost of operating pure cargo aircrafts.

Based on the above issues as outlined, the following priority interventions are thus recommended for the meat subsector's development:

- Approve and implement a federal Meat Inspection Act;
- Support public and private investments and PPP arrangements for the establishment and management of modern slaughterhouses and cold chains in all urban centers, after clarifying roles and responsibilities of investors and supervising agencies;
- Improve integration of backyard producers of goat meat with slaughterhouses and cold chains;
- Study the constraints and opportunities for developing a thriving offal industry; and
- Offer training on proper food (meat) handling practices for stakeholders across the value chain;



Hides and skins, cured mostly to sun-dried or dry-salted stages with only some processed to wet-blue stage, albeit of very poor quality, barely register today in the trade statistics because of the extremely low price they fetch. They are nonetheless still exported, suggesting potential international demand for a revival of exports from Somalia that would provide significant opportunities to livelihoods, especially in rural areas, as was indeed the case before the war. To harness this potential, it would be necessary to implement the following priority recommendations:

- Undertake studies that look at the economics of production and marketing of hides and skins and assessing requirements and preferences in existing and potential export markets;
- Collect statistical information about the subsector (starting under the planned livestock population survey) and about the constraints to attracting foreign investment from the importing countries;
- Formulate an enabling policy and regulatory framework for a future tanning industry;
- Formulate subsector and product quality standards that are based on international requirements;
- Support and strengthen organizational capacity of hides and skins trading associations;
- Encourage private investment in technology upgrading of existing and new slaughterhouses and tanneries, including via tax incentives and rebates; and
- Expand access to finance for traders and processors in hides and skins trade.

The poultry subsector remains underdeveloped, with poor linkages between producers, market and consumers, and minimal external donor support. However, because of an expanded commercial production and related consumption, poultry has become increasingly important to Somalia's food security and protein availability. Despite poor knowledge of the sector and myriad growth bottlenecks, government and private stakeholders believe that the available high market demand underlies a strong potential for expanding and upgrading the poultry subsector.

The recommended priority interventions for the development of the poultry subsector, so that it can contribute to strengthening food security and complement traditional Somali livestock, are mainly focused on modernizing and professionalizing the entire value chain, as follows:

- Create an enabling legal and regulatory environment that encourages private investment;
- Explore PPP arrangements for the development of a local chicken feed mill industry;
- Promote investments in hatcheries, if it is determined that there is sufficient demand;
- Provide extension services on poultry-specific animal health services; and
- Develop appropriate policies, regulations and guidelines for hygiene, production and packaging standards.

Beekeeping remains a small-scale side business for most beekeepers, who have only a weak understanding of modern production, handling and safety techniques, with honey consumption mostly limited to local markets. There are, however, several socio-economic benefits of beekeeping, especially for rural women. Unlike other agricultural enterprises, beekeeping is less affected by climate-change and honey sales greatly reinforce women's income sources during drought periods when there are few alternatives. Small producers and traders use income from honey sales to purchase food and invest in education and other household items, thus diversifying their livelihoods options. Furthermore, honey is a popular product, widely believed to possess medicinal properties and thus locally in high demand. Prices for raw honey in recent years have been very high, thus profitable for beekeepers. Stakeholders' recommendations for the subsector's development are to:

- provide extension services on production and handling practices and on market opportunities;



- facilitate their access to modern production and protective equipment;
- support the establishment by the private sector of honey refineries, to add value; and
- launch an awareness campaign to educate the public on beekeeping as a supplementary source of income and livelihood, and of its benefits in pollinating trees and crops.

Financial and Human Resources for the Sector's Development

Few financial services, with the exception of money transfers and some microcredit, are available to Somali pastoralists, due to their inability to pledge collateral. Livestock insurance is not available at all. Furthermore, traditionally women in pastoral and agro-pastoral communities have faced even greater barriers than men in accessing finance. There is a need to increase financial access throughout livestock value chains, including via enhancing financial literacy, formal financial credit services, and innovative digital financial services provision. Moreover, adoption of risk transfer instruments like livestock insurance, as for example is already the case in Kenya, would be a useful tool for managing drought shocks in pastoral areas.

Mostly private technical secondary schools and higher education institutions in veterinary sciences and animal husbandry lack regulations and standards usually set by governmental authorities responsible for education at the local, state or federal level. They also lack the infrastructure, financial resources and systemic governmental and institutional checks and balances that ensure effective institutional operations. As a result, the quality of teaching and learning is weak and inadequate to meet the educational demands of Somalia's livestock sector. The sector's educational institutions, moreover, have the added disadvantage of not attracting female students.

Key recommendations for strengthening educational policy, regulations and institutions and the capacity of both public and private technical schools and universities to meet the needs of the livestock sector include:

- Enact national education legislation and regulations to improve the quality of programs offered, including those in veterinary science and animal husbandry, by standardizing the academic curriculum and establishing higher-education commissions at federal and state levels;
- Enhance monitoring and evaluation of performance and enforcement of regulations by the education and livestock ministries over the sector's educational institutions;
- Build capacity of faculty of veterinary and animal husbandry educational institutions to improve the quality of teaching and learning.

External Development Assistance: Challenges and Lessons Learned

External assistance during the past decade has been highly concentrated on Somaliland, Puntland, and the federal government, with only one recognized project covering southcentral Somalia. Most interventions have aimed to strengthen animal health systems, the policy and regulatory environment, and private sector development. Activities to accomplish these goals included training sessions, workshops, knowledge and technology sharing, bringing together stakeholders, capacity building, creation of cooperatives, and trade facilitation. These objectives were and are indeed still relevant to the development needs of the country, the livestock sector, and the targeted beneficiaries. However, some major binding constraints have remained unaddressed. For instance, while many projects focused on animal health or market linkages, they did not address environmental issues such as degradation and deterioration of pastures and rangelands, even though these issues were widely recognized across projects. Similarly, few (none in southcentral Somalia) projects focused on rehabilitating watering facilities or fostering commercial feed supplies and markets.



External development partners have generally expressed a high degree of confidence in the sustainability of the outcomes of their interventions. However, the sustainability of project benefits remains unclear, given the many daunting implementation issues that they faced, including major construction delays (in the case of slaughterhouses and fish processing plants) and outright failures (as in the case of cold chains) in ensuring hand-overs and operations of facilities that they built in the northern regions, as well as the lack of rigorous impact evaluations and slow progress on domestic resource mobilization. Implementation challenges were related to poor education/skills availability; weak institutional capacity; narrow project design; limited advance planning for hand-over of project assets to local communities or to the private sector and related management issues; poor donor coordination, beneficiary involvement and poor sector governance; dilapidated infrastructure; and insecurity.

Key lessons learned from donors and development agencies' self-assessments and independent evaluations are summarized below:

- There needs always be a major skills and capacity building component in all projects. While long procurement delays and slow delivery should be expected, using local procurement, sourcing in-country expertise, and engaging private and community stakeholders and local authorities in project design and implementation would enhance capacity building, ownership and chances for project success.
- Because of insecurity in some areas affecting government officials and foreign personnel, partnering with private sector and local community organizations is necessary for project implementation.
- A collaborative approach among different institutions that can leverage different expertise and strengths and enhance technical coordination is also critical for project success.
- Value chains often require good understanding of social dynamics, as industry stakeholders may value their relationships more than pure economic gains.
- Selection of project sites must be transparent, evidence-based and technical criteria-driven.
- Strong awareness is needed by regulators and other value chain actors of SPS and quality standards prescribed by the OIE and/or required by importing countries, to avoid trade restrictions.
- A pooled and programmatic approach that layers different activities, has longer timeframes, and provides flexibility is essential in fragile and complex environments such as Somalia.
- Shocks and disasters are highly likely and thus their risk should be addressed at project design stage.
- Linkages with other sectors and ministries should be considered, as multi-sectoral approaches to strengthen resilience and disaster management are more effective than just a one-sector approach.
- Suitable technology that is geographically, environmentally, climatically, economically and culturally appropriate should be used to ensure resilience, reduce risk and increase sustainability.
- Local animal species, while less productive, are better adapted to survive the conditions of the local environment; thus, improving herd genetics with foreign breeds can only be successful if required inputs are available and is better undertaken by relying on community-based breed improvement programs.

Conclusions and Next Steps

As summarized above and elaborated in the report in further detail, the livestock sector remains key to economic growth and poverty reduction. However, the sector's performance and prospects are undermined by considerable structural weaknesses, severe exposure to downside risks, and vulnerability to climate and market shocks. The main constraints for the sector's development are:



- Weak institutional and policy framework, including limited and largely informal organization of the livestock value chain; lack of coordination and integration among its actors; weak human capacity; minimal or zero budgetary resources; lack of enforcement of existing rules and regulations; and poor linkages and ineffective negotiations with main trading partners;
- Dilapidated water and transport and inefficient energy infrastructure (e.g. insufficient and irregular and inadequately harvested rainfall and broken pavement) causing water shortages and higher transport and marketing costs than necessary, all of which affect negatively livestock health and survival, availability of natural fodder, delivery of quality animals to markets, and expansion of dairy production and processing;
- Lack of access to good quality pasture, due to land degradation and reduction of productive land, which is in turn the result of deforestation, soil erosion, invasion of unpalatable plant species, spread of private enclosures, and climate change bringing increasingly frequent recurrence of drought and floods;
- Poor veterinary health services and transboundary animal disease control, with occasional flare-up of epidemics, other livestock diseases, and pests, compounded by limited availability of quality veterinary education and technical training facilities in pastoral areas;
- Overreliance on only a few export markets and on live animal exports;
- Financial constraints – with very low government budgetary allocations to the livestock sector and private investment in processing for value addition, partly due to poor access to credit and no access to insurance, and lack of incentives and encouragement by governments; and
- Persistent insecurity in large parts of southcentral Somalia.

Despite the daunting challenges of overcoming these constraints, there is in fact strong scope for the sector's recovery and growth in the medium- and long-term, if managed properly, in both live animals and livestock products. This LSDS reflects the key sector needs and strategic development priorities of the FGS, its FMS, and the sector's key stakeholders, beyond those related to emergencies and short-term recovery from shocks. The key goals this LSDS has endorsed are to:

- Strengthen the production and marketing systems for live animals, via improving environmental stewardship of the country's rangelands, including their adaptation to an increasingly threatening climate change; health services, including the credibility of health certifications; and feed and fodder availability;
- Develop the many value addition opportunities available in livestock products, via promoting investments in processing and quality improvements, while ensuring food safety.

A comprehensive and ambitious Master Plan for the sector's development, however, needs to be developed in order to operationalize this LSDS' priority and medium-term recommendations. Also needed are quantifiable targets for the achievement of the LSDS objectives, as well as specific interventions (policy and institutional changes, technical assistance and investment projects), with estimated costs. proposed timeline and clear roles of government and external development partners. These interventions should clarify and address the following areas:

- Overall sector governance and institutional reform, including (i) assignment of responsibilities among all levels of governments; (ii) capacity building; (iii) broad participation, including by women, youth and disadvantaged groups, in policy formulation, institutional reforms, project design and implementation; and (iv) effective aid coordination and partnership with international donors and development agencies;
- Animal production and marketing systems, including effective rangeland management, land tenure, and climate change adaptation;
- Animal health services (and credible health certifications for exports), nutrition, and welfare;
- Production and marketing of dairy, meat, hides and skins (for exports), poultry and beekeeping products, to promote value added and ensure food security and safety;



- Access to modern financial services, including insurance; and
- Human resource development, including capacity building for the public and private stakeholders and for technical and higher education in veterinary, food sciences and animal husbandry sciences.

Under the recently approved “Water for Agro-pastoral Productivity and Resilience (WAPR), or Biyoole Project, the World Bank can financial support to the MoLFR’s for the preparation of such a Master Plan and related project proposals. However, to attract external funding to carry out an ambitious program of interventions, the FGS and FMS also need to clarify effective implementation mechanisms and a monitoring and evaluation framework that can assess the relevance of the objectives, effectiveness of interventions (outputs and outcomes), the performance of implementing agencies, domestic and external. To carry out all necessary next steps, and also to implement much-needed animal population survey and feed inventory and balance, the MoLFR is seeking technical assistance from its development partners during the second half of 2019 and through 2020. This will also ensure adequate input for the most important sector of the Somali economy into the National Development Plan II (2020-2022), currently under preparation.



1.0 INTRODUCTION

More than two decades after the start of a civil war and the collapse of the central government, which destroyed most of the country's physical and socio-economic infrastructure, Somalia turned a page when a peaceful transfer of power from a transitional government in 2012 generated domestic political momentum and a new Federal Government of Somalia (FGS) obtained widespread international recognition in 2013. Four Federal Member States (FMS) have been established since in southern Somalia, Jubaland, Southwest, Galmudug and Hirshabelle. They complement two other longer-established states in the northern regions, **Puntland and Somaliland.**

To date, businesses operate flexibly across all state borders, but there is a need for further improvement in the working relationship and coordination between the FGS and FMS towards creating a clearer and more favourable environment for regulatory and economic governance, including that related to interstate trade and exports, especially of livestock. Government at all levels, moreover, has remained primarily focused on addressing still formidable security and humanitarian challenges, to the detriment of providing essential oversight, services and infrastructure for inclusive and sustainable economic growth and development. Widespread environmental degradation—fuelled by demographic pressures, unsustainable exploitation of groundwater, rangelands and forests, and breakdown of customary and deep-rooted clan-based land management systems—and global climate change also threaten the viability of traditional livelihoods and the country's food security. Further, continued conflict in the southern and central regions of the country remains a threat to Somalia's overall political stability.

While Somalia's livestock sector faced and still faces formidable challenges, it has proved remarkably resilient amidst headwinds of social unrest and recurrent droughts, and the sector remains key to economic growth and poverty reduction. In contrast to a crop sector whose production is today only about half its pre-war peak, the livestock sector experienced surprisingly strong growth, including impressive export growth, in recent decades through the mid-2010s, largely due to substantial investments in infrastructure and service quality improvements by the Diaspora, foreign private companies and donor-funded programs. With almost half of the population still living in rural areas and engaged in some way in the livestock sector, the livestock sector remains the largest contributor to Somali GDP and the biggest source of foreign exchange after remittances.

The sector's performance and prospects, however, are undermined by considerable structural deficiencies and by exposure to downside risks. Production constraints related to poor resource management, climate change adaptation, animal disease control, poor nutrition, and genetic resources are compounded by other structural and institutional weaknesses that impede value added diversification and expose the sector to myriad risks. The sector's vulnerability and fragility is evidenced by an estimated \$2 billion in damages and losses suffered during the most recent 2016/17 severe drought period and a drastic drop between 2016 and 2018 of \$183 million (from \$430 million in 2016) in live animal exports after Saudi Arabia and the UAE imposed an import ban on health grounds at the end of 2016.

Despite current challenges, there is in fact strong scope for the sector's recovery and growth in the medium- and long-term, if managed properly. Sizeable areas of the country are suitable for grazing, browsing, and fodder production. At an estimated 56 million heads in 2018, the country has one of the largest livestock populations in the world (and the largest camel population). Its close proximity



to and favourable consumer preferences in Gulf markets provide Somalia with a competitive advantage that can support export growth of live animals and animal products. Production of milk and meat, moreover, has grown steadily since the late 1980s to satisfy a growing urban demand in the country supported by large remittance inflows. Strengthening the sector's resilience to increasingly frequent and severe weather shocks and improving its growth performance will not only provide a major contribution to the country's overall economic development, but will also foster peace and security, alleviate poverty and malnutrition, and enhance health outcomes in both rural and urban areas.

Success in tackling the sector's constraints and harnessing its potential, however, requires a clear vision and strategy for the priority policy reforms and interventions needed across federal and state levels, with strong, coordinated support by its development partners, to alleviate the binding constraints and mobilize and guide public and private investment. In 2004, the Food and Agricultural Organization of the United Nations (FAO), the World Bank and the European Union (EU) collaborated with senior officials of the then Somalia Transitional National Government (TNG) in the development of a background document for a livestock sector strategy. Due to overwhelming concerns with political and security challenges in southern Somalia and the lack of external financing for development activities, the TNG did not adopt nor implement any development strategy or plan.

This Livestock Sector Development Strategy (LSDS) provides an updated, coherent, nationwide vision and framework for the sector's medium- and long-term development, reflecting the priorities and needs of its key stakeholders beyond those related to the sector's short-term recovery from the recent 2016/2017 drought. It is based on a comprehensive literature review and two additional original studies on livestock-related higher education and on the peri-urban dairy value chain. Extensive consultations with public and private stakeholders on their vision, challenges, and priority recommendations were held via a series of state-level 2-day workshops conducted in October-November 2018 throughout the country (see Main Annex 1 and Online Annexes) and a 3-day validation workshop in June 2019 in Mogadishu (see Main Annex 2). The LSDS will also provide a timely contribution to the just-launched National Development Plan II that will guide in 2020-2024 public and private investments, as well as external assistance from external donors and development agencies.

This document is organized as follows. It starts by summarizing the country's geography, its overall economic and trade performance, and its environmental and social context in Chapter 2. The livestock sector's historical economic and social performance is discussed in Chapter 3, followed by a brief preview of its main development challenges. Chapter 4 presents the overall vision for the sector's long-term development, including the roles of its public and private stakeholders and the capacity needs of federal and state livestock ministries. Chapter 5 discusses in more detail constraints, prospects, opportunities, and recommended interventions from the literature, from governments' policy and planning documents and from stakeholders for the development of its live animals' production and marketing systems, as well as two critical inputs, namely animal health and nutrition (water and feed). Chapter 6 similarly addresses the main product subsectors, that is, dairy, meat, hides and skins, poultry, and beekeeping. Chapter 7 discusses and presents challenges and recommendations related to cross-cutting sectoral issues, including lessons from external assistance. Chapter 8 concludes by discussing the next steps necessary to operationalize this sector's development strategy.



2.0 COUNTRY BACKGROUND

2.1 Climate and Geography¹

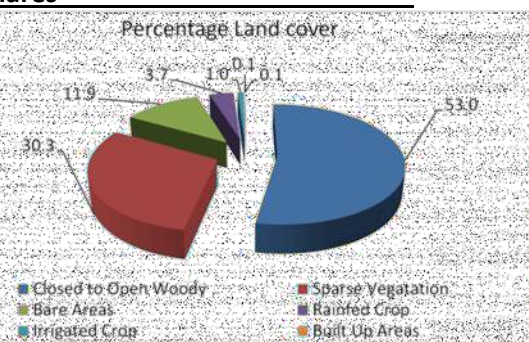
Somalia's climate profile is generally arid and semi-arid with bi-modal rainfall. Very small, usually elevated areas have an annual average rainfall of 500–600 mm, but most of the country has an average rainfall that is only 100–200 mm, with rain mostly falling in isolated and heavy storms. The wettest regions have typically 40–60 rainy days each year with daily rainfall of the order of 5–15 mm (Kammer, 1986). Open water evaporation usually far exceeds rainfall and is in the range 1,600–2,400 mm per year in the south of the country. The bimodal rainfall pattern has two rainy seasons throughout most of the country (less pronounced in the northern mountain region) known locally as the 'Gu' (April to June) and the 'Deyr' (October to November). These result from the north-south oscillation of the Inter-Tropical Convergence Zone (ITCZ) and its related frontal systems.

Droughts occur regularly at intervals of 2–3 years in the 'Deyr' and 8–10 years in both seasons (UNDP, 1987). The temporal patterns of high rainfall variability over Somalia is the main direct cause behind such extreme events impacting the country, including floods (UNDP/ICPAC, 2013). The occurrences of these extreme events happen when the Indian and Pacific Ocean experiences anomalous sea temperatures and circulation anomalies during El Niño/La Niña, together with IOD events. These include the recent droughts of 2000, 2004, 2008 and 2010–2011; and floods of 1997/1998. Furthermore, analysis of global data from 1901–2005 shows temperatures have increased by 1.0°C over the course of the century, while inter-annual analyses of more recent national data shows that mean air temperatures in Somalia remain high throughout the year, ranging from 15–25°C in the northern mountains to 25–35°C in the southern regions.

As a result, most of the country's landmass covering an area of 638,000 square kilometers in the Horn of Africa is arid or desert, but also with a large semi-arid zone, and in southern Somalia a sizable humid zone. Nomadic pastoralism takes place mostly in the arid zone, agro-pastoralism in the semi-arid zone, and irrigated crop cultivation in the humid zone (see Figures 2.1 and 2.2), with 53 percent of Somalia recently classified as rangeland (SWALIM 2014), compared to 45 percent in 1979 (World Bank 1979). Another 30 percent is covered by sparse vegetation and less than 5 percent is suitable for either rainfed and irrigated cultivation (see Table 2.1 and for more details of land use systems also Figure SA1.1 and Table SA1.1 in Statistical Annex 1).

Table 2.1: Somalia Land Cover / Land Use, 2014 – Areas Size and Shares

CLASS	USER LABEL	Area (ha)	Area km ²
Natural Woody Vegetation Closed to Open	NVCO	33843064	338430.6
Natural Woody Vegetation Sparse or Herbaceous	NVSH	19320323	193203.2
Bare Areas	BA	76030749	76030.75
Rainfed Crop	RC	23619971	23619.97
Irrigated Crop	IC	62805034	6280.503
Built Up Areas	BU	60815869	608.1587
Water Bodies	WB	32317731	323.1773
TOT:		63849643	638496



Source: Swalim

¹ This section is based on FAO 2004, Somalia 2017, and on SWALIM's assessment of land cover and uses as summarized on its website.

Figure 2.1: Somalia Land Cover / Land Use, 2014

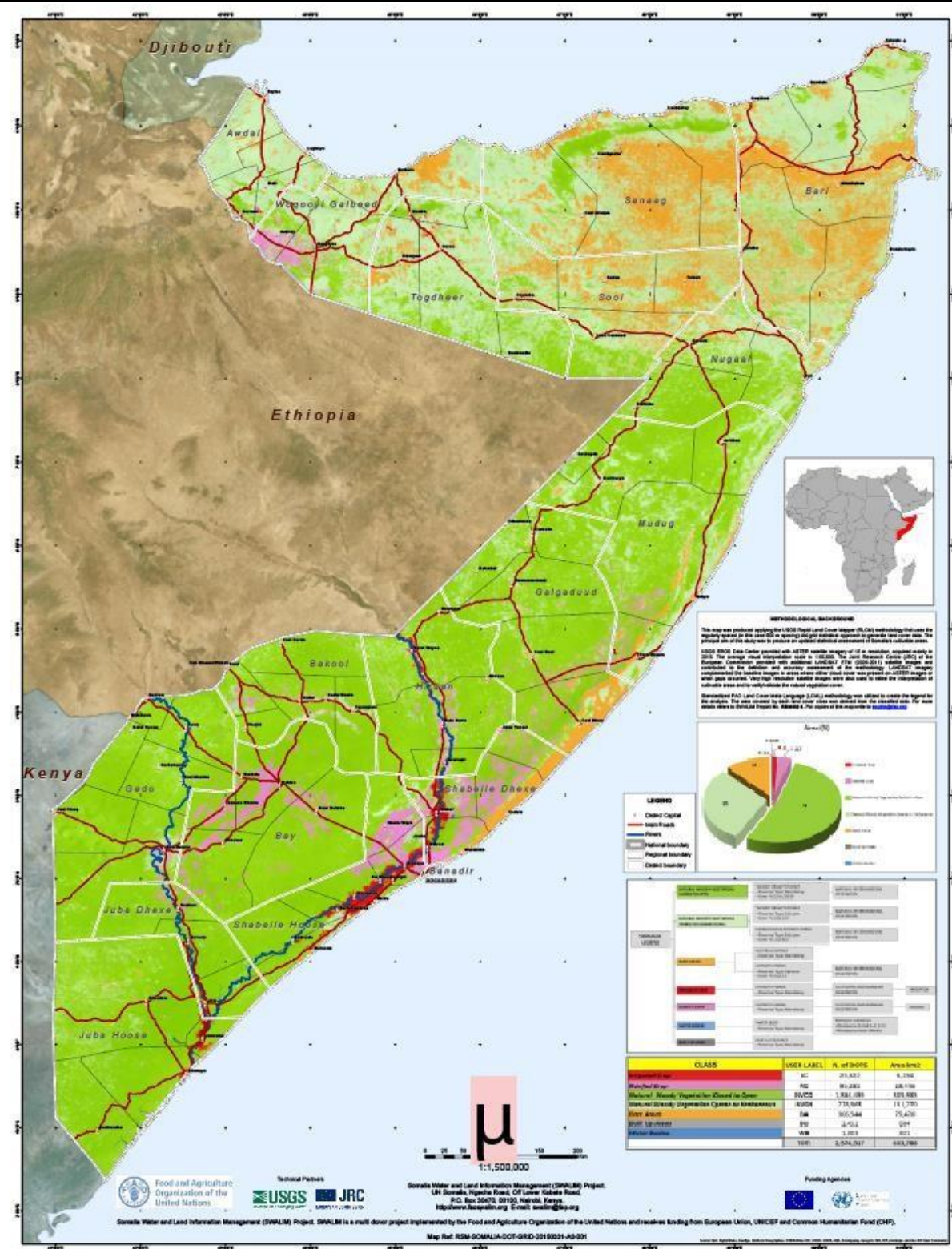
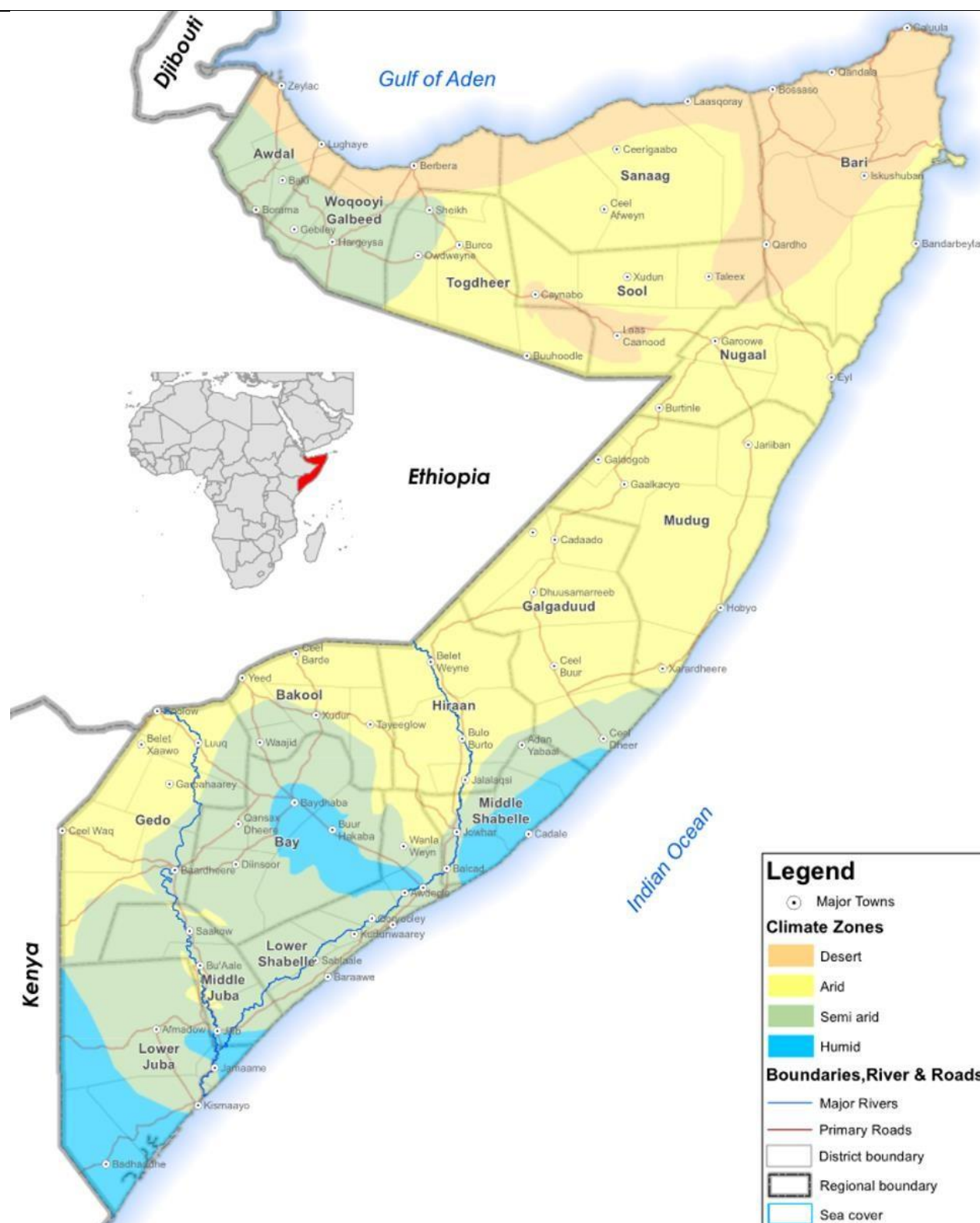


Figure 2.2: Somalia Climate Zones, 2018



Source: SWALIM

By 2015, according to the FAO, what was left was a total forest cover estimated at only about 10 percent of the country's land area, compared to 62 percent through 1980.² Already by the end of the 1980s, virtually all the floodplain forests had been cleared for irrigated agriculture production.

² See SDG Indicator 15.1.1 - Forest area (as a percentage of total land area) at <http://www.fao.org/sustainable-development-goals/indicators/1511/en/> for 2015 and Openshow 1982 for an historical baseline.



Between 1990 and 2009, Somalia lost an average of 76,750 hectares, or 0.93 percent of its forests a year (FAO 2010). All the remaining forests, which include juniper trees in the northern Golis mountain range, tropical vegetation along the Juba and Shabelle rivers and on their adjacent floodplains and mangroves along its coast, are facing increased pressure of commercial exploitation. The related progressive destruction of plant life has already impaired animal habitats and reduced forage, affecting Somalia's wildlife and its greatest food resource, its livestock.

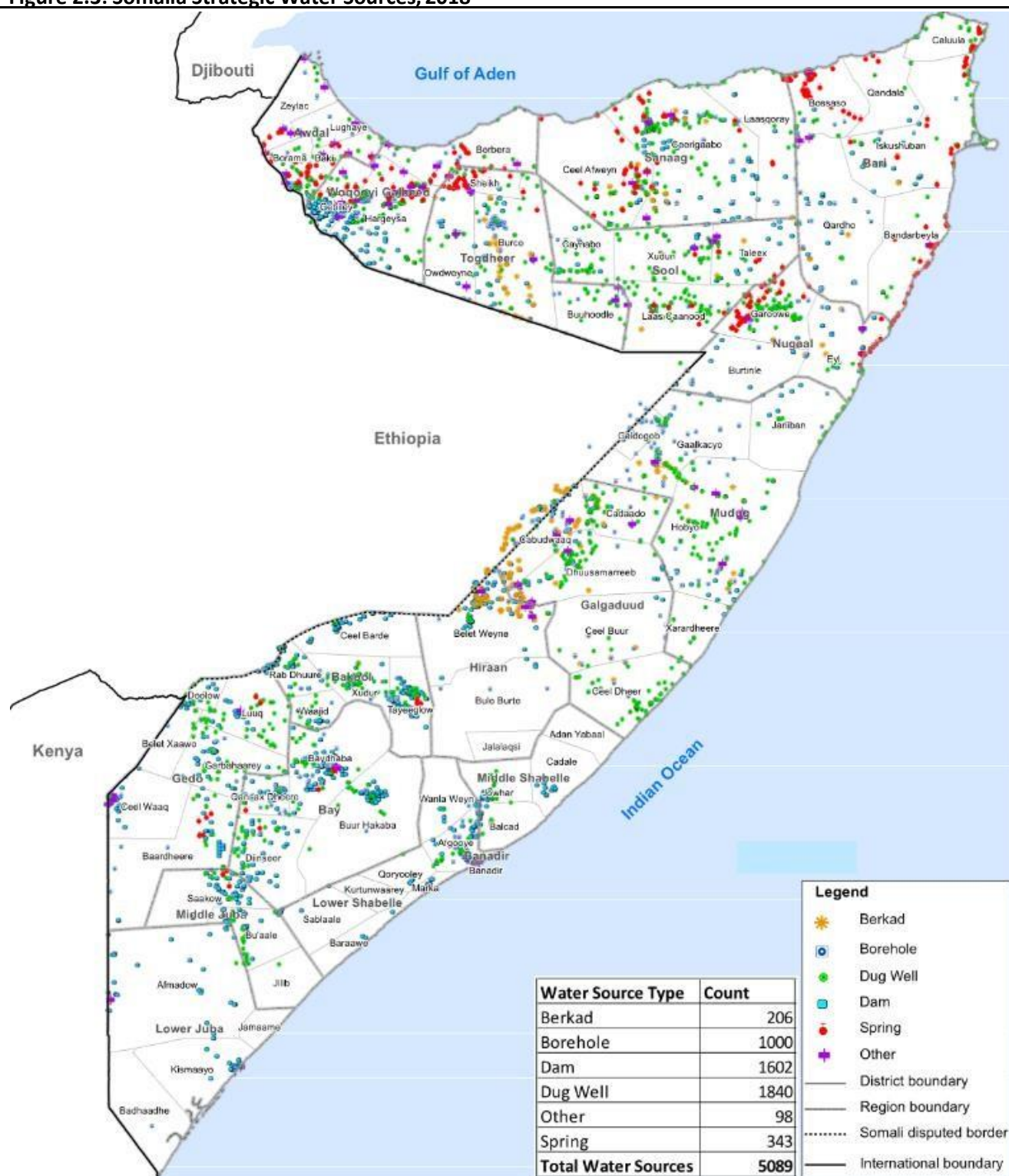
Somalia's terrain consists mainly of plateaus, plains, and highlands. However, in the far north, the Karkaar Mountains lie at varying distances from the coast of the Gulf of Aden. In the north, a maritime plain that parallels the Gulf of Aden coast is covered with *guban*, a semi-arid Scrub well suited for grazing livestock in the rainy season. The mountain ranges descend to the south, forming an elevated plateau without perennial rivers, which include shallow valleys with mostly dry watercourses and is called the Ogo. The Ogo in the central part form the Mudug plain while the eastern part gently undulates towards the Indian Ocean. This plateau merges in the Haud to the south forming broad undulating plains best for grazing by the nomads, despite having inadequate rainfall for a greater part of the year.

Livelihoods in Somalia are highly dependent on the ecosystem, but climate change, the spread of invasive species and past scale of charcoal production for exports has resulted in environmental degradation in most parts of the country. The Horn of Africa is one of only two biodiversity hotspots that are mostly arid and semi-arid, holding more endemic reptiles than any other region in Africa. Unfortunately, this region is also one of the most degraded in the world, with only about 5 percent of its original habitat remaining, caused largely by overgrazing and charcoal production with inadequate resource management efforts. Invasive species such as *Prosopis juliflora* are also an increasing threat to species extinction and habitat destruction. Many were initially introduced to East Africa for the stabilization of dune systems and for providing fuel wood after prolonged droughts in the 1970s. In many areas, however, species have hybridized to an extent that current varieties have lost most of their valuable woody attributes and have aggressively suppressed native shrub and tree vegetation.

In 2019, SWALIM mapped more than 5,089 water points across the country (see Figure 2.3). From previous surveys, it estimated that only 2,200 are functional and perennial under normal non-drought conditions. Of these functional sources, only around 500 are improved ones, mainly deep boreholes, which in many other African countries commonly provide water for domestic use, livestock, and irrigation. In Somalia, however, groundwater sources are technically demanding to identify and exploit because aquifers are deep (more than half the boreholes are over 130 m deep, with some over 400 m) and water within aquifers is often of low quality (salty or hard) that makes it unsuitable as drinking water or for irrigation.³

³ See <http://fmt.faoso.net/imms/fmt/maps/website/227>

Figure 2.3: Somalia Strategic Water Sources, 2018



Source: SWALIM

The vegetation in Somalia is still varied, although much less so than three decades ago, as inappropriate land use has led to the original vegetation cover being heavily degraded, especially



in northern Somalia, and in various places it has been entirely destroyed, according to SWALIM⁴ and to Odouri et. al 2007. Current vegetation cover ranges from the forests of the Golis mountains in the North to the bush land in the Lag Badana ecosystem of the South. Patches of mangroves are found in Zeylac, Berbera and Calula on the northern coastline and in Kismayo on the southern coast. Expansive grasslands are found in Puntland and sparsely covered sand dunes cover a band of several kilometers along the coast. **Somalia's distribution of vegetation, visually illustrated in Figure SA1.1 in Statistical Annex 1, is as follows:**

- The coastal plains vegetation consists mainly of herbaceous plants. The vegetation in this region extends to the feet and slopes of the Golis Mountains. Close to the mountains, the predominant vegetation is sparse bushy *Acacia*, *Balanites aegyptiaca*, and *Commiphora* associations including *Boswellia* species.
- In the hinterland plateaus, vegetation is dominated by open shrubs and woody plants of *Acacia bussei*, *Acacia etbaica*, *Boscia spp.*, *Cadaba spp.*, and *Acacia mellifera*. Some herbaceous plants (mainly *Chrysopogon aucheri* and *Sporobolus spp.*) can also be found here.
- In the northern Golis mountain range the vegetation consists of evergreen trees of *Juniperus procera* and *Juniperus excelsa* and open shrubs of *Buxus hilderbrandtii*, *Dodonea viscosa* and *Terminalia brownii*, etc. In the past, the Golis Forest remained largely uninhabited and protected while today it is experiencing increased exploitation pressure. There are now even human settlements in the forest.
- In the central plains, vegetation varies from extensive grassland along the fixed dune areas to shrubby bushland with scattered trees in the west toward the Ethiopian border. Plants and bushes include *Andropogon kelleri*, *Chrysopogon aucheri*, *Sporobolus ruspolianus*, *Indigofera ruspolii*, *Acacia spp.*, *Commiphora spp.*, *Cordeauxia edulis*, *Delonix elata*, *Terminalia orbicularis* and *Dobera glabra* etc.
- In the southern regions (especially in the floodplain), the vegetation type is mainly low deciduous bushland of *Acacia spp.* which extends to the coastal dunes.
- Mangrove swamp communities are also situated at the tidal estuaries of the seasonal rivers towards the Indian Ocean coast and Gulf of Aden. They areas include Bushbush, Caanoole and Lag Badanaa. At certain points along the River Juba there are sections of riparian forest.

2.2 The Economy⁵

The Somalia population can be divided into five diverse livelihood types: pastoralists, agro-pastoralists, fishing and coastal communities, urban population, and internally displaced persons (IDPs, also mostly located at the outskirts of major urban centres). According to UNFPA estimates from its 2014 nationwide population survey (UNFPA 2015), about 49 percent of the population still lives in rural areas, with 26 percent (or 3.2 million) of the total population classified as pure pastoralists and about 23 percent (or 2.8 million) as agro-pastoralists, depending on settled crop and livestock farming. Alternative estimates for direct employment by the ILO for 2014 (ILO 2015) suggest that about 46 percent of all employed persons in the country are working in agricultural activities, with about 25 percent employed in agro-pastoral activities, 9 percent in nomadic herding, 4 percent in fishing, 7 percent in other rural activities. The rest or about 54 percent of employed people work in urban-based governments and the private sector (the latter includes marketing of livestock products), according to the ILO.

⁴ Downloaded as of March 15, 2019 from <http://faoswalim.org/land/land-cover> for both text and Figures 2.1 and 2.2. See also Odouri et. al 2007 for more detailed description of main land cover types and vegetation units and their locations.

⁵ This section is largely based on the analysis of economic trends presented in recent World Bank reports, covering the broad agricultural sector, including livestock (World Bank 2018) and the whole economy (World Bank 2019b, forthcoming).



Almost three decades after the start of the Somali civil war and the eventual collapse of the central government in December 1990, the country's economic growth is still weak and mainly consumption-driven, enabled by large remittance and aid inflows. Between 2013 and 2017, real GDP has grown an average of 2.5 percent p.a., while population grew by 2.9 percent p.a. The result has been an annual contraction in per capita incomes of 0.4 percent during the same period—significantly below fragile, sub-Saharan, and low-income country averages—leaving per capita GDP at just over \$500, making Somalia among the poorest countries in the world.

In the past two to three decades through the mid-2010s some major structural changes stand out in Somalia's economy and in its agricultural production and trade. Many pre-war sub-sectors such as crop production, agro-processing, other manufacturing, utilities and government services shrunk, increasingly buffeted by the country's degraded and fragile natural environment, more extreme and frequent cycles of drought and floods related to an intensifying climate change, lack of research and extension service, insecurity and the consequent deterioration of flood control, irrigation, water facilities, and transport infrastructure in the south-central regions. Just before the most recent 2016-2017 drought, cereal production was about half of what it was pre-war and banana production an even smaller fraction, with sesame the only crop experiencing growth in production and in exports.

Other sectors, unshackled by the stifling pre-war government monopolies and regulations, managed to thrive through the mid-2010s, including livestock, telecommunications, construction and retail trade. The livestock sector's performance however experienced a precipitous drop after 2016. Its recent contribution to the country's gross domestic product (GDP) remains speculative, due the lack of reliable and consistent estimates across subsectors for production, farmgate prices, and input use. The World Bank estimated for this report that its share rose significantly from its historical, pre-war level of 43.7 percent in 1980 and 37.2 percent in 1986-88 to 61.5 percent in 2013-2016. However, because of the double whammy of the 2016-2017 drought on the animal population, health, and productivity, and of the December 2016 livestock import ban by Saudi Arabia and the U.A.E., the livestock sector's GDP share is estimated to have fallen to 43.4 percent in 2017 (see Table 2.2).

Construction, telecommunications and money transfer services have been the key growth sectors in recent years, but due to a poor enabling environment agro-processing, manufacturing and other services have all suffered from weak growth, even though the country has experienced rapid urbanization during the past two decades. The growth of agro-processing and manufacturing, in particular, is limited by an adverse investment environment – Somalia is ranked at the bottom of the WBG's Doing Business Survey. This is largely because the government, especially in the southcentral regions, has not yet been able to establish strong security and public sector management frameworks, to ramp up its capacity and muster the needed resources for the provision of essential public services, and to attract large scale private investment.

Somalia's agro-processing sector, in particular, was negligible before the civil war and remains even more limited today. Before the outbreak of the civil war, almost all of the agro-processing industries, including for dairy products, meat, and hides and skins, were owned or heavily controlled by the government. Few of these were fully developed and well managed. All of them, however, were completely destroyed soon after the start of the civil war. Since the 1990s, a few new privately owned agro-processing industries have emerged, with high-potential opportunities for future expansion, job creation, and contributing to a higher and sustained economic growth, as the northern regions re-established security and self-governance institutions in the subsequent years



and the regions in the south-central part of the country have become more secure in recent years (see World Bank 2018, including the related online Annex listing surveyed agro-processing ventures in operations). Nonetheless, in all regions of the country, and despite its enormous potential, agro-processing, including livestock processing, was minimal even before the civil war and is even smaller today. In addition to the generally poor enabling environment noted above, other factors play a role in stifling the growth of agro-processing, including limited access to financing, high energy costs, and domestic consumers' preferences for fresh products (e.g., produce, raw milk) over processed ones.

Table 2.2: Agriculture's Share of Somalia's GDP, 2010s vs. 1980s

	1980 (National Accounts) /1	1986-88 (National Accounts) /1	2013-16 /2	2017 /2	2018 /2
Agriculture	64.4%	62.0%	n.a.	n.a.	n.a.
Livestock and crops	61.9%	55.3%	72.2%	48.0%	n.a.
Livestock and livestock products	43.7%	37.2%	61.5%	43.4%	n.a.
Live animals exports \3	3.5%	2.7%	7.5%	3.0%	3.3%
Livestock products	40.2%	34.5%	54.0%	40.3%	n.a.
Dairy (milk) \4	28.4%	13.9%	42.9%	33.0%	n.a.
Meat for local consumption \5	9.5%	19.8%	11.0%	7.2%	8.9%
Meat for exports \6	0.0%	0.0%	0.1%	0.1%	n.a.
Hides & Skins\6	0.8%	0.2%	0.1%	0.0%	n.a.
Others (poultry & beekeeping)	1.4%	0.6%	n.a.	n.a.	n.a.
Crop production \4	18.3%	18.1%	11.3%	4.6%	n.a.
Other	2.4%	6.7%	n.a.	n.a.	n.a.
Forestry \7	2.1%	6.1%	1.5%	n.a.	n.a.
Fishing	0.3%	0.6%	n.a.	n.a.	n.a.

Source: As elaborated by the World Bank, sourced from Somalia 1989, FAOSTAT, U.N. COMTRADE, FSNAU IDS and Somalia 2018 (DINA).

Notes:

1\ 1980s shares are nominal production values divided by nominal GDP, from the official National Accounts 1977-1988 (Somalia 1989; and Statistical Annex in World Bank 1991)

2\ 2010s shares are production values estimated as in notes 3\ through 6\ below divided by official nominal GDP in US\$, as of June 2019

3\ 2013-16 export values are from U.N. COMTRADE (partner countries' reporting); 2017 and 2018 export value is FSNAU volume exports multiplied by the average nationwide export prices of FSNAU disaggregated export prices (recorded monthly in the main Somali markets for camels, cattle and shoats)

4\ 2013-16 and 2017 dairy and crop shares are the production values as estimated in the Drought Impact Needs Assessment or DINA (Somalia 2018), divided by official nominal GDP in US\$ as of May 2019.

5\ 2014-18 annual meat production volumes for local consumption were estimated by applying the ratio of FAOSTAT meat production per each livestock category for 2011-2013 over the average FAOSTAT animal population to the new 2014-2018 animal population as estimated by FSNAU & MLFR. Annual nationwide prices necessary to derive production values are averages of the monthly, disaggregated main markets prices found in the FSNAU IDS database.

6\ For these categories, 2013-2018 production values are assumed to be equal to export values as reported in U.N. COMTRADE (partner countries' reporting)

7\ Among forestry products, only frankincense production value was estimated under the DINA for the (pre-drought) 2013-2015 years.

Export performance through the mid-2010s (before the recent 2016-2017 drought) has been impressive, with annual exports climbing every year since the late 2000s to a peak in 2015 of US\$701 million (\$644 million in 2016) and an average annual value in the early 2010s about fivefold higher than before the civil war. This was mostly due to strong growth of livestock exports, whose peak value of \$546 million in 2015 was almost 10 times higher than in the late 1980s. Overall crop exports, on the contrary, have been in recent years a fraction of what they were before the civil war. As a result of such record performance of livestock exports and the collapse of banana exports,



once a close second item by value, the livestock sector accounted in the peak 2015 year for about 78 percent of total export earnings, a much higher share than the 53 percent registered in the pre-war years (livestock exports are discussed further and in more detail in chapter 3).

Table 2.3: Top Exports, 1981-2018 (US\$, millions; annual averages)

Annual averages in US\$ millions; SITC-3 categories in brackets

Product Description	1981-85	1986-90	1991-95	1996-00	2001-05	2006-10	2011-14	2015	2016	2017	2018
Livestock and Livestock Products	92.4	61.1	73.4	83.6	92.2	153.8	432.8	546.2	466.4	221.3	<i>n.a</i>
Animals, live, chiefly for food (00)	88.4	55.7	66.6	72.9	76.7	126.3	377.1	493.9	429.9	215.3	246.8
Animals, live, n.e.s., incl. for zoo & breeding (9410)	0.0	0.0	4.2	6.5	0.6	5.2	42.8	38.7	30.1	<i>n.a</i>	<i>n.a</i>
Livestock Products (21, 01, 02, 0616-honey)	4.0	5.4	2.6	4.3	14.9	22.4	12.8	13.6	6.4	5.9	<i>n.a</i>
of which: Hides & Skins (21)	3.9	5.3	2.4	2.3	6.2	8.7	8.6	4.0	2.3	2.7	<i>n.a</i>
Forestry	4.5	4.9	4.8	9.7	19.0	33.9	30.4	20.4	31.3	46.7	<i>n.a</i>
of which: Gums and resins, etc. (2922)	4.4	4.9	3.7	2.8	2.9	2.9	5.4	20.4	30.6	46.7	<i>n.a</i>
of which: Cork and wood (24)	0.1	0.0	1.1	6.9	16.1	31.0	25.0	0.0	0.5	0.0	<i>n.a</i>
Crop Agriculture	15.9	32.0	5.6	10.7	11.0	11.9	51.0	39.4	47.8	23.7	<i>n.a</i>
Sesame seeds (2225)	0.0	0.5	0.1	0.8	1.8	6.0	40.0	34.2	26.8	5.6	<i>n.a</i>
Bananas (0573)	13.8	28.1	4.3	6.6	0.0	0.0	0.0	0.6	0.0	0.0	<i>n.a</i>
Citrus Fruits (0571+0572)	0.2	0.1	0.1	0.2	0.3	0.6	4.6	4.0	5.4	17.2	<i>n.a</i>
Other (residual, incl.04,06,07,08,40,42,43; minus ho	2.0	3.2	1.1	3.1	8.9	5.3	6.4	0.5	15.5	1.0	<i>n.a</i>
03-Fishery	2.1	9.0	10.1	8.1	18.3	7.6	3.6	27.8	25.5	26.9	<i>n.a</i>
Chemicals, Manufactures, Machinery, Transport (3+5+6-	4.1	4.3	7.6	7.3	14.6	20.3	16.4	15.8	13.8	13.2	<i>n.a</i>
Other (residual)	1.3	4.5	1.9	0.4	18.6	56.3	28.4	50.8	59.4	1.4	<i>n.a</i>
TOTAL EXPORTS	120.3	115.7	103.4	120.0	173.8	283.9	562.6	700.5	644.2	333.2	<i>n.a</i>
of which: Livestock and Livestock Products	76.8%	52.8%	70.9%	69.7%	53.1%	54.2%	76.9%	78.0%	72.4%	66.4%	<i>n.a</i>

Source: COMTRADE, as of June 27, 2019; FSNAU, for live animals in 2017 and 2018 (FSNAU does not differentiate between animals for food vs zoo and breeding).

Source: UN COMTRADE (Somalia's partner countries data), accessed via WITS on June 27, 2019; except for FSNAU for live animals in 2017 and 2018 (FSNAU does not differentiate between animals for food vs zoo and breeding).

Note: 2017 estimates are still preliminary, as some countries (including Saudi Arabia and Bahrain) have yet to report to COMTRADE.

Qualitative studies and anecdotal evidence strongly suggest that Somalia has also become a major informal supplier to large parts of the Horn of Africa of cattle and charcoal and a re-exporter of sugar, pasta, and electronics, among other commodities subject to high import tariffs in neighbouring countries. This formal and informal trade expansion was facilitated by the combination of the growth of the telecommunication sector and of the remittance companies, which supported a deeper domestic, cross-border, and international market integration despite the highly insecure conditions in the southcentral regions of Somalia. The country has become a de facto duty-free zone and a critical part of a regional trade network spanning from Dubai—a critical supply hub for Somali businessmen—to Ethiopia and Kenya.

There has also been a steady and large increase in food imports, including livestock products, since the late 1980s, reaching around US\$1 billion this decade from an annual average of only US\$72 million in the late 1980s. The combination of increased domestic food demand, mostly for cereals, sugar, and other preparations, and for vegetables, including *qat* (mild narcotic leaves), driven by rapid population growth and urbanization and largely supported by remittances, and of the collapse of domestic crop production are two key factors that have led to a massive increase in food imports. Imports of live animals and animal products have also increased, to about \$76-81 million in 2016-2017 from around \$7 million before the civil war (see Table 2.4).



Table 2.4: Top Imports, 1981-2017 (US\$, millions; annual averages)⁶

Annual averages in US\$ millions; SITC-3 categories in brackets

Product Description	1981-85	1986-90	1991-95	1996-00	2001-05	2006-10	2011-14	2015	2016	2017*
Crop Agriculture	90.0	64.3	68.5	92.3	134.4	376.5	968.0	1196.2	853.7	973.4
Cereals and cereal preparations (04)	63.0	40.3	32.8	38.1	51.8	132.7	297.6	314.0	334.4	380.4
Sugar, sugar preparations (06; minus 0616-honey)	4.5	3.2	17.9	37.6	55.9	93.8	160.6	188.7	307.2	274.5
Vegetables and fruits (05)	2.6	2.9	6.7	2.0	3.4	94.2	387.7	558.5	52.3	32.4
Coffee, tea, cocoa, spices, manufactures thereof (07)	2.2	1.0	1.7	3.5	2.6	4.1	2.3	3.4	4.5	31.4
Oil seeds (22)	0.0	0.0	0.0	0.1	0.0	1.7	0.2	0.2	2.2	1.4
Other (residual, i.e. 08, 09, 4; minus 41)	17.7	16.9	9.4	11.1	20.7	49.9	119.7	131.5	154.3	254.2
Live Animals and Products (00,01,02,41,0616-honey,9410)	16.5	7.1	3.0	1.7	7.9	28.2	160.0	239.1	76.0	80.9
of which: Dairy products (02; minus 025)	15.9	5.9	2.4	1.7	4.7	16.2	62.0	82.0	66.0	72.1
of which: Meat and Meat Preparations (01)	0.5	1.2	0.5	0.0	1.9	1.2	1.9	8.2	7.4	7.0
Forestry (24 and 2922)	1.6	2.8	0.2	0.1	0.3	1.0	8.3	6.4	10.7	19.3
Fish, crustaceans, molluscs, preparations thereof (03)	0.2	0.0	0.2	0.1	0.6	2.9	8.2	13.9	10.7	2.9
Beverages and tobacco (1)	5.5	4.7	9.7	16.4	34.9	43.3	61.7	25.5	21.0	43.3
Crude materials,inedible,except fuels (2; minus 21,22,24,29)	1.4	1.0	2.1	12.5	26.6	76.3	10.2	5.1	5.9	52.7
of which: Crude animal & vegetable materials,n.e.s. (2)	0.2	0.2	1.7	11.8	25.4	75.5	8.1	2.0	1.3	49.1
Chemicals, Manufactures, Machinery & Transport (3+5+6+7)	182.5	199.1	60.4	43.9	83.7	191.0	459.8	854.1	1,044.4	1,100.1
Other (residual, i.e. 9 minus 9410)	4.0	4.3	4.0	0.6	4.9	6.2	22.6	28.5	13.8	14.4
TOTAL IMPORTS	301.8	283.3	148.1	167.7	293.3	725.3	1,698.9	2,369.0	2,036.2	2,286.9
of which: Food Imports (0,11,22,4; minus 8, 4314)	107.7	72.1	73.7	94.6	143.8	410.0	1,094.7	965.3	268.1	1,053.5

Source: UN COMTRADE (Somalia's partner countries data), as of March 2019.

Note: 2017 estimates are still preliminary, as some countries (including Saudia Arabia and Bahrain) have yet to report to COMTRADE.

Remittances are the major source of external finance for consumption and investment in Somalia.

Somali migrants and refugees outside Somalia doubled between 1990 and 2017 to total more than 2 million. During 2015–2017, Somali diaspora sent home about an official US\$1.3 billion per year, but remittances may be significantly larger when considering unrecorded flows. Remittances represent 20 percent of GDP, about the same amount as grants to Somalia, and more than three times foreign direct investment (FDI). Remittances may be counter cycle as relatives and friends often send more during economic downturns, disasters, conflicts, or other negative shocks (World Bank 2019a).

2.3 Poverty and Vulnerability: High Overall, Including among Pastoralists⁷

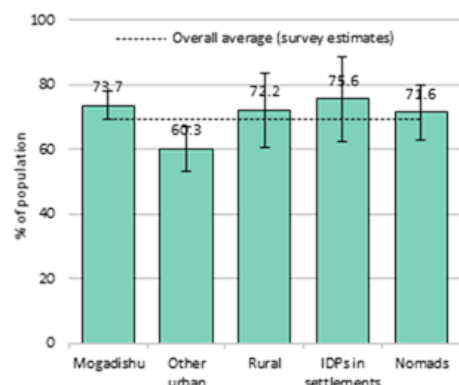
According to a recent study of poverty and vulnerability, Somalia is one of the poorest countries in Sub-Saharan Africa, and in the absence of appropriate poverty-reduction policies its vulnerability will impede its economic and social development. Nearly 7 of 10 Somalis live in poverty, the 6th highest rate in the region, behind only the Democratic Republic of Congo, Burundi and South Sudan. The overall incidence of poverty of 69 percent is 19 percentage points higher than the unweighted average of low-income Sub-Saharan African countries of 51 percent in 2017. Moreover, a significant group of non-poor are vulnerable to falling into poverty, as about 10 percent of the population is within 20 percent in terms of total daily consumption above the poverty line. Poverty incidence is highest in IDP settlements and in Mogadishu, but it is also high at 72.2 percent for the rural

⁶ As for total exports, there is a very large and unexplained divergence between import totals based on product-by-product and country-by-country statistics captured by COMTRADE (partner countries) and import totals based on aggregate flows captured by the IMF (also based on partner country reporting). For instance, total imports, including non-food, nonagricultural imports, from Somalia were US\$2,143 million in 2014, according to the UN COMTRADE database, but US\$3,482 million according to the balance-of-payments statistics in the latest IMF report (IMF 2016, see Table 4a on page 23).

⁷ This section is based on an analysis of poverty and vulnerability in Somalia, based on the second wave of the Somali High Frequency Survey, implemented in late 2017/early 18. The survey achieved greater geographical and population coverage, including of nomads and of many households in insecure areas, compared to Wave 1 of the Somali High Frequency Survey (SHFS) conducted in 2016 covering Puntland, Somaliland and Banadir only and the Somaliland Household Survey (SLHS) carried out in 2013. Wave 2 targeted almost 6,400 households distributed among rural and urban areas all regions except Middle Juba.

population (which includes agro-pastoralists, in addition to pure farmers and other rural dwellers) and for nomads at 71.6 percent, with both ratios being higher than for other urban areas (see Figure 2.5).

Figure 2.5: Poverty is high for all and highest in rural areas and IDP settlements



Source: World Bank 2019a.

Every nomadic household owns at least one goat, sheep, camel, donkey, cattle or chicken, and they tend to own more than non-nomadic households (World Bank 2019). Every nomadic household owns some type livestock, compared to 53 percent of the rural households, 35 percent of IDPs in settlements, 23 percent in other urban areas and 19 percent in Mogadishu. Chickens are the only type of livestock owned by a higher share of rural (including agro-pastoral) than of nomadic households; 18 vs. 11 percent respectively (Figure 2.6a). Among owners, nomadic households also own a larger number of livestock than non-nomadic households; nearly twice for cattle, sheep, goat and camels (Figure 2.6b). Wealth in the form of livestock represents an advantage against other populations living in IDP settlements, urban and rural areas. Yet, relying primarily on livelihood based on livestock makes them more vulnerable to climate-related shocks.

Figure 2.6a: Livestock Ownership

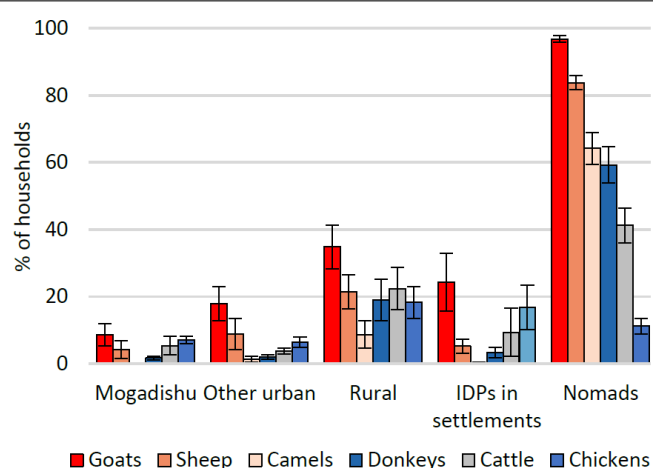
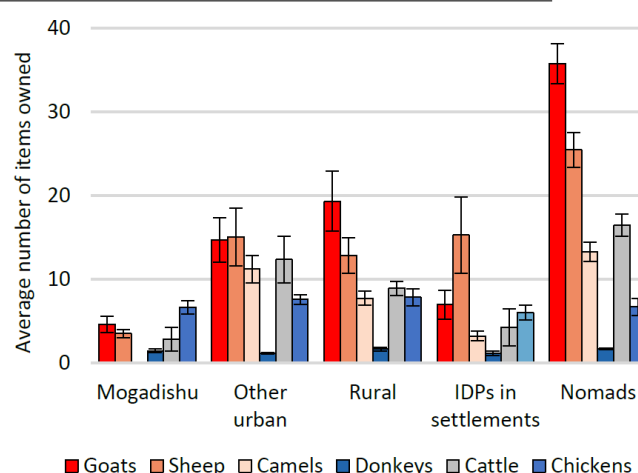


Figure 2.6b: Number of Livestock Owned



Source: World Bank 2019a.

The latest IPC tabulation by the FSNAU for the period April-June 2019 shows that 44.2 percent of the pastoral and agro-pastoral population is classified as either stressed (IPC-2, 29.1 percent), in crisis (IPC-3, 14.2 percent) or in emergency phase (IPC-4, 0.9 percent). Somaliland appears the most affected, with IPC-coverage rates ranging between 50-88 percent across its regions for its



pastoralists and agro-pastoralists. In Puntland's regions, IPC-rates range between 50-70 percent. The impact of the current drought is less in all other regions. In Galmudug's regions, IPC rates range from 21 percent in the coastal cowpea agro-pastoral belt to 70 percent in the Hawd and Addun pastoral areas. IPC coverage rates in Hirshabelle's regions are also relatively low, in the 10-50 percent range, except for a peak of 70 percent in its Hawd pastoral area. The range in South-West's regions is similar, at 10-45 percent, while Jubaland's regions appear the least affected so far, at 10-20 percent.

Though information on households' herd numbers and composition over time remains weak and preliminary, recent reports indicate that in many areas pastoralists have become poorer, with more households shifting from middle to poor wealth groups and from poor to very poor wealth groups (Bishop, et al., 2008; World Bank 2019). Recurrent drought episodes, often combined with insecurity, have led to pastoralists losing large numbers of animals and many migrating to (peri) urban areas. For pastoralists, loss of animals is closely related to destitution and marginalization from community life.

According to the most recent household survey of poverty and vulnerability in Somalia, access to services is limited particularly for rural residents, IDPs in settlements, and nomads (World Bank 2019). Improved water & sanitation is critical for health, school performance, and productivity, but only 5 of 10 households have access to improved sanitation, and 8 of 10 to improved water sources. Only 5 of 10 households have electricity. Access to services is somewhat higher in urban areas. Poor households are less likely to have access to improved sanitation and electricity. Markets and health clinics are far—more than 30 minutes walking distance—for 34 to 40 percent of Somali households and for most nomads. Over-age school enrollment is common, with stark geographical and gender disparities, and with nomads and girls facing the biggest challenges. Improving service provision—especially education—is crucial to improve human capital and reduce inequality that disproportionately affects girls/women, IDPs, and rural and nomadic households.

Somali households are vulnerable to shocks such as natural disasters and epidemics, as well as to household-level shocks such as injury, death, or unemployment. Shocks contribute to extreme poverty and vulnerability, constraining economic opportunities and livelihoods, damaging assets, and limiting access to farms, fishing, and pastoralist routes. The persistent cycle of shocks increases Somalis' vulnerability to future shocks as there is limited public and private insurance. The threat of climate change also increases the risk of displacement by reducing the viability of livelihoods that support rural (and urban) populations in Somalia.

Jobs play an important role in fragile environments, but Somalia suffers from acute unemployment and underemployment. Unemployment rates are considerably higher among women and Somalis under 30 years of age, who represent 70 percent of the population. Jobs not only contribute directly to poverty reduction, productivity, and economic growth but also facilitate social cohesion and reduce the risk of violence. However, according to the International Labour Organization (ILO), only 54 percent of the active population (ages 15 to 64) in Somalia is unemployed. The unemployment rate among youth ages 14 to 29 years is 67 percent—among the highest rates in the world (UNDP 2015). Women fare worse (74 percent) compared to men (61 percent)⁸. Due to high fertility rates (6.2 births per woman in 2010-2015), the youth population is

⁸ Gender inequality is alarmingly high in Somalia at 0.776 out of a value of 1 (complete inequality), with Somalia at the fourth lowest position globally on the Gender Inequality Index, if internationally comparable data were available. Women suffer severe exclusion and inequality in all dimensions of the index—health, employment, and labor market participation (UNDP 2015).



arguably more vulnerable than any other to extremism and criminal behavior and the numbers will continue to swell.

Job-creation, especially for youth, is fundamental to sustainable poverty reduction, vulnerability mitigation, and conflict avoidance. Somalia has a large youth bulge, so youth must be able to find jobs to contribute to economic growth. Policies to encourage business and entrepreneurship to create jobs are needed to avoid idle youth from resorting to conflict. Furthermore, enhancing access to domestic markets can increase inclusivity, spur economic activity, and accelerate poverty reduction.

Investment in resilience is also needed to prevent livelihood loss for vulnerable rural households, especially due to likely future droughts. A consumption shock of the same magnitude as the 2016/17 drought is estimated to increase rural poverty from 65 to 76 percent. Investing in resilience and rural market access would help these households avoid livelihood loss. Measures may include agricultural insurance, enabling households to diversify income, and improving access to roads and clean water.



3.0 LIVESTOCK SECTOR: STRUCTURE AND PERFORMANCE

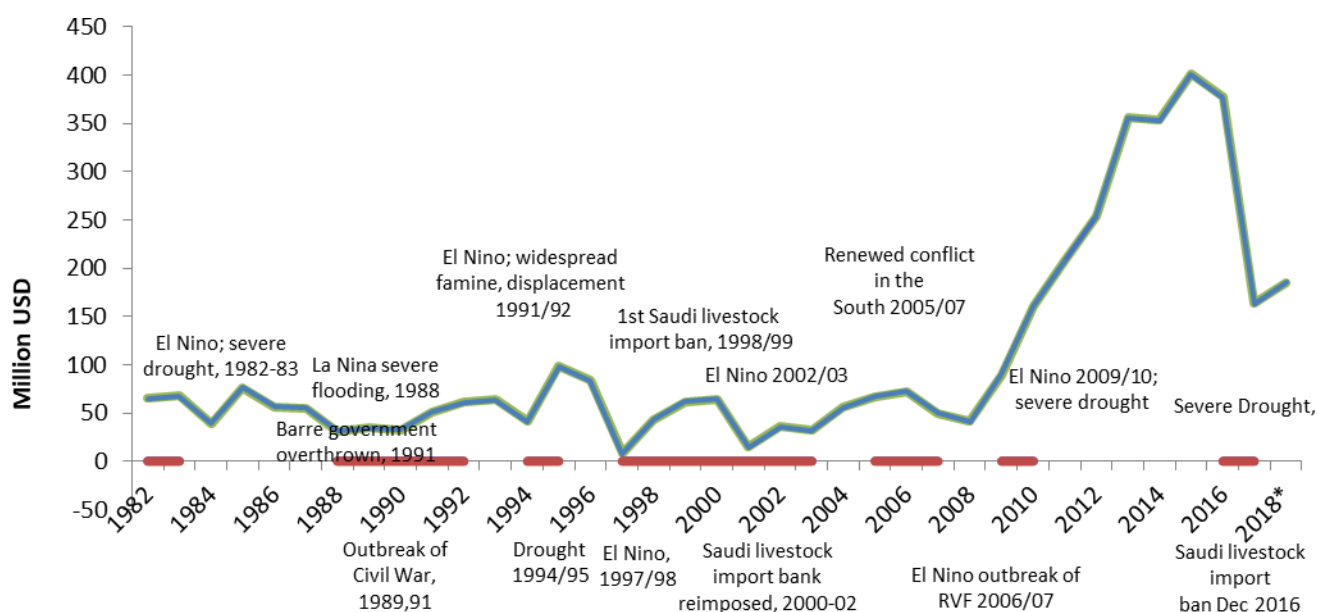
3.1 The Natural Environment⁹

Severe environmental degradation of the country's fragile natural environment, including of its rangelands and forested areas, and intensifying climate change, with more extreme and frequent cycles of floods and drought (eight major droughts since the early 1980s, see Fig. 3.1), have affected negatively in the last four decades the performance of all agricultural sectors, including the livestock sector. In addition to crop failures in agro-pastoral areas, the impact of recurrent droughts includes decreased water availability, reduced pasture availability, accelerated land degradation, impaired health, resilience and productivity of livestock, lower livestock prices and household incomes, and pastoral migration in search of pasture and water, which can increase and exacerbate conflicts over scarce resources.

The cumulative impact of various failed rainy seasons since 2016 hit pastoralists, farmers and the internally displaced particularly hard and led to a very serious humanitarian crisis whose human, social, and economic effects still linger to this day in a few areas of the country. After coming back from the brink of famine in mid-2017, food security and nutrition (and livestock welfare and numbers) have been on a steady path to recovery. However, they remain fragile, with weak and late rains so far through May 2019. Current humanitarian needs in early 2019 are greater than the pre-crisis levels of three years ago, with acute malnutrition still persisting in many areas and the real prospect of another drought-related crisis through 2019. Humanitarian action has prevented worse outcomes, but sustained assistance for emergency nutrition needs and resilience-building efforts remain critical in preventing further deterioration (FAO, FSNAU, FEWS-NET 2019).

In the recent 2016-2017 drought, massive livestock and other damages and losses were experienced following three consecutive poor seasons of erratic and below average rainfall (Somalia 2018). Insufficient rain and water availability towards the end of 2016 severely reduced water and pasture availability for livestock. The drought stretched the government's capacity to handle the devastating effects of the drought on pastoralists. This also increased the risk and vulnerability of livestock-dependent households, with the poor losing up to 60 percent of their total stock in some of the most severely affected areas. Somalia suffered over US\$1.6 billion in losses and damages in the livestock sector – around 50 percent of the total drought damage and losses – during the drought period. In addition, an estimated USD 400 million revenue loss from live animal exports in 2017 from the import ban imposed by Saudi Arabia and the UAE brought the overall damage and loss for the sector to over US\$2 billion. Following the drought, the sector and the country faced numerous challenges including but not limited to: absence of fodder reserves and alternatives, the absence of allocated budget by Government to the sector, and weak government institutional capacity to cope with such large-scale drought impact. The overall effect of recurring severe droughts, moreover, is on limiting regeneration of pasture and water availability for both animals and plant growth. Somalia therefore remains one of the most drought vulnerable country in the Horn of Africa, even more so because feed and forage production and supply chains are not well-developed and pre-war flood-control and irrigation infrastructure are in disrepair.

⁹ This section is largely based on section 4.2 of a recent World Bank-FAO joint report analyzing the impact of climate change on the broad agricultural sector, including livestock (World Bank 2018).

Figure 3.1: Impacts of Climate and Other Shocks on Shoats Exports since 1982


Source: Shoats export values from U.N. COMTRADE through 2016 and World Bank elaboration of FSNAU disaggregated estimates of export volumes and prices for 2017-2108.

3.2 Livestock Production Systems

Livestock remains the traditional repository of household wealth in Somalia and is an important source of livelihood and resilience for a large part of the country's population. In fact, self-consumption of livestock (meat) and livestock products (milk and ghee) as well as their market sales is critical for the food security of pastoral and agro-pastoral households, with market sales important also for the rest of the Somali population. While well adapted to the harsh geographical and climatic environment, Somali sheep and goats (shoats), camels and cattle are characterized by poor weights and milk yields, largely as a result of young animals with larger frames and body weights being selected for slaughter or live export rather than breeding.

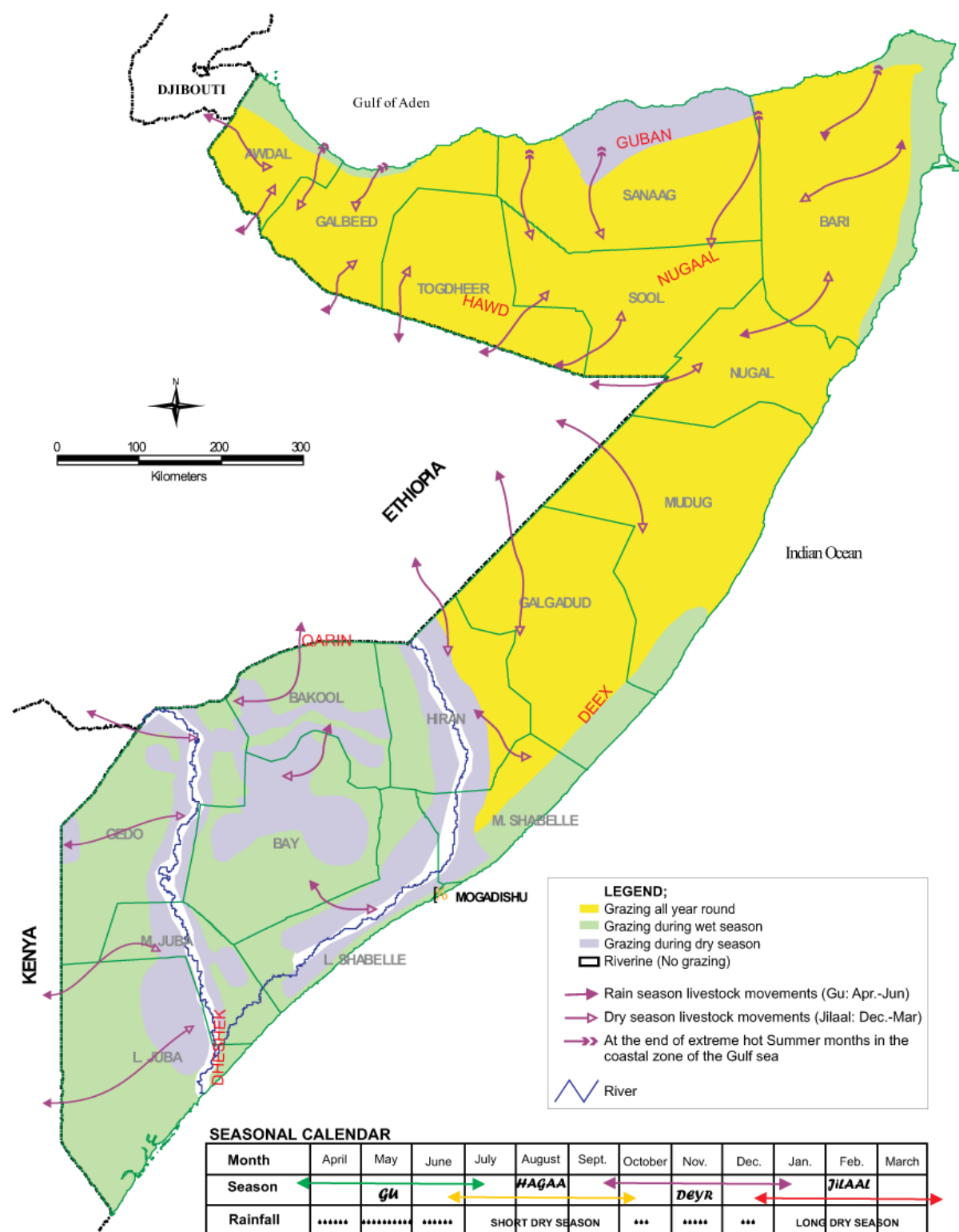
The main production systems in Somalia are nomadic pastoralism and agro-pastoralism (settled mixed farming), with little or no integration with crops in the first system, which relies heavily on seasonal migration in search of pasture (see Figure 3.2). There is also a smaller peri-urban dairy farming system, which mainly relies on purchased supplements and grain and crop by-products as fodder. Commercial livestock-rearing operations with dairy animals (mostly camels) have been on the rise in peri-urban settings, ranging from small to very large (3,500 animals for a camel dairy farm in Benadir), supplying nearby urban markets with fresh raw milk.

- Pure pastoralists, at 3,186,965 or about 26 percent of the total population in 2014 (UNFPA 2014), are mostly nomadic, poor, and are found throughout all rural areas of Somalia, but predominantly in the arid open rangelands of northern and central Somalia (tending to their sheep and goats, cattle, and camels). They also dominate along the Ethiopian and Kenyan borders in southern Somalia (tending in addition also to cattle), using little or no external inputs, including feed and health services. This group suffers from a poverty incidence estimated for 2018 at 71.6 percent (World Bank 2019). This predominant production system is characterized by low levels of inputs and productivity and relies largely on (i) natural pastures on the country's vast rangelands, which are covered by grass, herbs, shrubs and sparse trees and (ii) palatable



fresh crop residues for fodder. This is coupled with high mobility by both pastoralists and agro-pastoralists in search of forage.

Figure 3.2: Seasonal Migration of Livestock



Source: FSNAU 2019

- Agro-pastoralists, at 2,806,787 or about 22.8 percent of the population in 2014 (UNFPA 2014), depend on both settled crop production and livestock rearing, and live mostly in the inter-riverine regions of Bay, Bakool, western Hiran, and eastern Gedo in south-central Somalia. They also predominate in certain areas of southwestern and northwestern regions with underground water and high annual precipitation. A small proportion are riverine crop growers, some of



whom also keep animals on their farm, but most of whom keep no livestock. The poverty incidence among agro-pastoralist is also believed to be high, around the 72.3 percent level which has been estimated by the World Bank (2019) for rural households (to include also pure farmers and other rural dwellers).

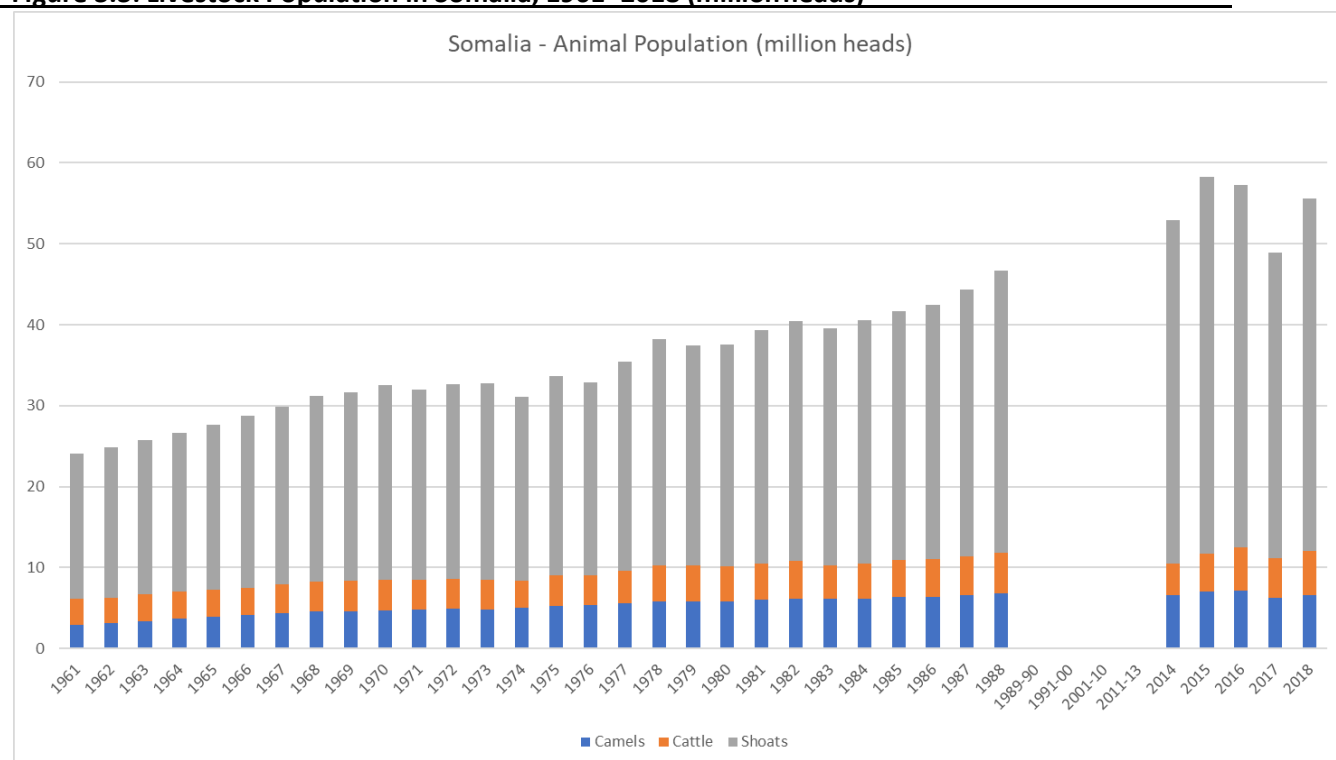
3.3 Livestock Population, Distribution and Composition

Somali livestock are of a broad range of species and types with some breeds that are well adapted to a nomadic way of life, limited feed resources and intermittent water supply. Because Somali people live in several countries other than geographical Somalia, some of their traditional livestock breeds are also found in Djibouti, Ethiopia and Kenya. See Main Annex 1 for more details about Somalia's livestock genetic resources.

The last livestock census in Somalia was undertaken in 1975. Until the late 1980s, livestock population figures showing a total of about 40 million were estimated by government agencies from reliable surveys supported by externally funded nationwide projects and can be found in both the official national accounts (Somalia 1989) and in FAOSTAT. Since then, however, animal population estimates have been largely based on projections and extrapolations by external agencies with very limited information and access to many parts of the country. One such point estimate suggested a livestock population of 39 million in 1998 (see FAO, World Bank and EU 2004). All sources nonetheless concord that there was a sharp decline in the 1990s, largely because of the civil war, followed by a solid recovery only interrupted by a sharp decrease in small stock in 2007-2008.

More recent baseline livestock estimates for 2014 imply a remarkable resilience by the sector until the latest severe drought that affected during 2016 and 2017 the entire country, though at different times and with different intensities.¹⁰ As illustrated in Figure 3.3, these baseline estimates point to a sharply increased animal population in 2014 (baseline) compared to historical time series estimates from FAOSTAT as well as to other point estimates from various source: 52.9 million total heads in 2014, including 28.7 million goats, 13.6 million sheep, 6.6 million camels, and 3.9 million heads of cattle (Somalia 2018; also in World Bank-FAO 2018, section 5.7, pp 56-58). See also Main Annex 2 for an overview of available livestock statistics and Table SA2.1, Statistical Annex 2 for a regional breakdown of the baseline animal population estimates for 2014.

¹⁰ These estimates were originally proposed by the FSNAU, applying historical household profiles which included livestock holdings (FSNAU-FEWS 2016) in many districts of Somalia to district-disaggregated official human population estimates for 2014. They were discussed with and agreed to by federal and state ministry officials and FAO-Somalia and World Bank experts at the DINA workshop held in October 2017 in Mogadishu, and even though based on heroic assumptions and a weak methodology, they remain the best available baseline estimates until a reliable comprehensive animal population census or survey will be conducted. For goats these estimates were much larger and for camels and cattle somewhat smaller than previous estimates by other sources, and their regional distribution was also different. The most recent livestock figures estimated independently in 2012 by the Somaliland government and including the northcentral regions (whose control is in dispute with Puntland) are roughly consistent with the regional FSNAU-MoLFR estimates shown in Table SA2.1, Statistical Annex 2. They show a steady increase in all types of livestock between 1998 and 2009, with almost 90 percent of its livestock being shoats (World Bank-FAO 2018).

Figure 3.3: Livestock Population in Somalia, 1961–2018 (million heads)


Source: FAOSTAT through 1988 (same as official National Accounts); Somalia 2018 (DINA) for 2014; MoLFR and FSNAU for 2015–2018.

Animal mortality due to the latest severe drought in 2016/2017, estimated at 6.4 million heads and valued at over \$350 million (Somalia 2018), contributed to bringing the total animal population in the country down to less than 49 million by 2017, according to the most recent FSNAU-MoLFR estimates. The animal population, however, is estimated to have recovered to almost 56 million heads in 2018, after more normal rainfall patterns resumed during the second half of 2017 (see Table 3.1). Statistical Annex Tables SA.2.2 through 2.4 show the animal population estimates by the FSNAU-MoLFR for 2015, 2016 and 2017, which factor in higher drought-related mortality and lower natality rates compared to normal years (see also Somalia 2018). In addition, they also take into account the impact of the import ban imposed by Saudi Arabia and the UAE in December 2016, implying a lower offtake for export since then.¹¹

Table 3.1: Animal Population, 2018

Region	Total Camel	Total Cattle	Total Goats	Total Sheep	Total Livestock Population
Lower Juba	370,297	711,130	1,135,906	830,700	3,048,033
Middle Juba	190,149	622,124	781,905	659,508	2,253,686
Gedo	545,980	302,851	882,262	346,382	2,077,474
Bay	430,587	1,206,575	2,286,970	242,310	4,166,442
Bakol	725,686	566,859	2,788,372	783,605	4,864,522
Lower Shabelle	511,114	1,088,774	2,409,516	945,386	4,954,789
Benadir	n.a	n.a	n.a	n.a	n.a

¹¹ Estimates for 2015 to 2018 are based on reported livestock ownership and herd dynamics among poor households, from FSNAU twice-a-year surveys among poor households. Estimates for 2015 to 2018 for better-off and middle-wealth households are based on an assumed fraction (25% and 50%, respectively) of the growth rate (plus or minus) among poor households



Middle Shabelle	210,502	395,858	2,325,460	1,037,903	3,969,723
Hiran	912,460	420,856	2,719,780	939,966	4,993,061
Galgadud	293,947	35,822	1,305,587	530,146	2,165,501
Mudug	324,593	13,305	1,883,372	777,236	2,998,507
Nugal	311,267	0	2,040,739	1,125,306	3,477,311
Bari	67,435	0	1,472,581	672,792	2,212,808
Sool	186,099	0	899,617	668,414	1,754,130
Sanaag	174,091	0	1,656,482	1,084,301	2,914,875
Togdheer	445,999	4,503	1,601,514	477,478	2,529,494
North-West (Woqooyi Galbeed)	535,393	96,567	2,360,891	968,418	3,961,269
Awdal	411,568	65,696	1,965,468	893,303	3,336,034
Total All Regions	6,647,164	5,530,921	30,516,421	12,983,154	55,677,660

Source: FSNAU and MoFLR, May 2019

Note: These estimates exclude the animal population in Benadir region (in and around Mogadishu), which is believed to be sizeable.

Even after considering the animal mortality from the recent drought, Somalia remains the country with the largest camel population in the world, more than twice the camel population of Kenya and Ethiopia combined (Mahmoud 2013 and, for 2015, FAOSTAT). Somalia also ranked 6th in the world and 1st in Africa for goat population in 2015, if the 31.8 million estimated by the MoFLR and FSNAU (as opposed to the roughly only one third recorded by FAOSTAT) are compared to those by FAOSTAT for all other countries. Livestock herd sizes and composition vary across Somali production systems and regions, as illustrated in Table 3.1. Camels are found in all agro-climatic zones of the country. Cattle are mostly concentrated in the southern regions, particularly in the Juba and Shabelle river valleys. Large numbers of sheep are found in the northern and central regions, whereas goats are predominantly found in the northern regions, especially in Puntland, and in the southern regions. Average herd sizes are large to very large in the nomadic pastoral system and medium size in the mainly transhumant agro-pastoral system. In contrast, mixed crop/livestock farming households may have only a few animals. Different species are generally held together, to make the most efficient use of range vegetation, because camels and goats are primarily browsers while cattle and sheep are largely grazers. Combining different types of livestock also provides pastoral groups with a wide array of animal products. Herd diversification is an important risk management strategy for protecting against diseases, drought, and lack of drinking water or forage (Dahl 1981).

3.4 Health and Disease¹²

Animal diseases are regularly reported in the country, with the most significant being of transboundary nature. Among the most notable are *Peste des Petits Ruminants* (PPR),^{12F13} Contagious Caprine Pleuropneumonia (CCPP), Contagious Bovine Pleuropneumonia (CBPP) occurring mostly in the southern areas, Foot and Mouth Disease (FMD), Sheep and Goat Pox (SGP), Camel pox, and Lumpy Skin Disease (LSD). Other commonly found diseases are Anthrax, Blackleg, Brucellosis, Hemorrhagic Septicemia, Trypanosomiasis, Salmonellosis, Clostridium, and ecto- and endoparasites all of which affect both production and reproduction of animals kept by pastoralists and

¹² This section is mainly based on FAO et al. 2004 and World Bank-FAO 2018, and updated by FAO-Somalia.

¹³ During vaccination campaigns spearheaded by the FAO since 2012, an overall individual animal PPR sero-prevalence of 62 percent has been observed, with a low of 39 percent in the northwestern region and a high of 74 percent in southern Somalia, much higher than the 5.5-37.6 percent range found in 2001 baseline surveys. So far, there have been no studies to determine PPR mortality and morbidity rates. (Njue 2018).



agropastoralists as well as of the wildlife population.¹⁴ Fowl pox is also commonly observed disease affecting poultry. Due to the aggregation of a large number of livestock in common grazing areas, watering points, and markets and along trading routes, and to shared grazing on the rangelands with wildlife, the likelihood of disease transmission remains high. Climate variability also has had a negative impact on livestock and human health through vector-borne diseases. Higher average temperatures have supported the expansion of vector populations into cooler areas at higher altitude (Thornton et al. 2007).

Rift Valley Fever (RVF) remains at the forefront of many of the problems that the livestock export trade has suffered after its presence was first suspected in 1998.¹⁵ Outbreaks in East Africa have occurred periodically at intervals ranging from 2 to more than 30 years. The Somali areas bordering with Kenya and the north eastern region of Kenya went through a massive epizootic of RVF in 1997/98 facilitated by extensive flooding occurring in that region. RVF is otherwise enzootic throughout Sub-Saharan Africa and Madagascar and reached the Arabian Peninsula in 2000. Permanent irrigation in the western coastal zones of Yemen and Saudi Arabia will contribute to the risk of the disease becoming endemic in these areas (FAO et al. 2004). The FAO, however, believes that RVF is currently not endemic in Somalia (unlike northern Kenya) and thus has not conducted mass vaccinations campaigns against this disease, leaving quarantine stations to administer vaccines against RVF to the animals in their care depending on the requirements of the importing country.

The presence of transboundary animal diseases (TADs), the absence or weakness in available veterinary services, and the constant mobility of animals within the country and across the Horn of Africa region pose a serious risk of disease transmission and subsequent trade bans by importing countries. Recognizing the regional dimension of addressing the animal health challenge in each of IGAD member states, a regional animal health strategy for 2017-2022 was prepared with funding by the World Bank (ICPALD 2017). Its strategic pillars are: (i) strengthening national veterinary institutions; (ii) strengthening disease surveillance and reporting systems and regional sharing of data; (iii) identify priority disease control measures; (iv) enhancing regional and national emergency preparedness; and (v) instituting sustainable mechanisms for regular and timely coordination between member states.

In May 2019, furthermore, Somalia and Ethiopia agreed on a Memorandum of Understanding (MOU), facilitated by IGAD/ICPALD for cross-border cooperation and coordination on animal health and sanitary measures. The areas of cooperation, to be backed by commitment of financial, staffing and physical resources, that are identified in the MOU include: (i) control of trans-boundary animal diseases; (ii) mapping of livestock routes, markets and natural resources (water and pasture), including control of invasive weeds; (iii) enforcement of regulations on veterinary drugs and vaccines use; (iv) facilitation of livestock trade through quality control and certification; (v) access to livestock-related infrastructure including diagnostic laboratories, cold chains, holding grounds and quarantines; (vi) joint promotion of livestock identification and traceability; (vii) joint capacity building of stakeholders; (viii) promotion of cross-border networking among stakeholders.

¹⁴ For an historical perspective and more details about the major diseases affecting Somali livestock, see pp. 13-14 of the 2004 "Towards a Livestock Sector Strategy - final report" (FAO et al. 2004).

¹⁵ RVF is a viral disease that is maintained and transmitted by mosquitoes whose eggs are resistant to desiccation. The disease persists at low levels and can remain undetected until favorable climatic conditions, including higher rainfall with flooding, create a breeding explosion in its vector, which is the *Aedes (Neomelanimonium)* group of mosquitoes, followed by amplification of the virus via other species of mosquito.



In the absence of a strong public sector to take the lead in protecting animal health, many critical services are provided by the private sector, often but not always with funding from development partners. These services include mass vaccination, vector control and supportive treatment campaigns, training, and institutional capacity development. A central role in delivering animal health services is played by three private, accredited livestock professional associations (LPAs) in south central Somalia and two in Puntland and Somaliland. The LPAs are regularly mobilized to deliver targeted services such as vaccinations, treatments, and disease investigation and disease detection, generally with external funding (see Table 3.2). They also manage the veterinarian drug stores in the regions and, with support of the FAO, train community-based animal health workers (CAHWs) in their remote local communities on reporting and treating basic livestock diseases, veterinary drug administration, basic surgical procedures and reporting to regional veterinary offices on suspected TAD cases.

Table 3.2: Sheep and Goats Vaccinated in 2012-2018 for Main Animal Diseases

PPR SGP & CCPP VACCINATION SUMMARY			
YEAR	VACCINE	NUMBER OF ANIMALS VACCINATED	COST OF VACCINATION CAMPAIGN (US\$)
2012	PPR	19,666,847	3,730,960
2012	CCPP	7,476,391	1,418,332
2013	PPR&SGP	11,814,414	4,006,128
2014	CCPP	11,860,193	4,021,650
2014	PPR&SGP	11,873,321	3,859,810
2015	PPR&SGP	13,821,212	4,543,138
2016	CCPP	14,006,923	5,000,889
2017	PPR&SGP	14,362,091	4,748,725
2018	CCPP	14,027,352	5,008,183
2018/2019	PPR&SGP	14,300,485	5,120,253
Total		133,209,229	36,709,342

Source: FAO-Somalia.

Substantial investments have been made in recent years by international development agencies such as the FAO to improve animal health and welfare and also some by foreign private investors in quarantine stations to meet the requirements of the importing countries. However, no comprehensive evaluation or monitoring of the Performance of Veterinary Services (PVS), according to the OIE-PVS Pathway tool, has been done. The FAO, through the accredited LPAs and under the supervision, coordination and monitoring of MoLFR, has carried out mass vaccination, treatment and vector control campaigns covering in the seven years to end-2018 more than 133 million shoats (see Table 3.2) and about 4.3 million households, at a total cost of about US\$37 million.¹⁶ Other development partners have also conducted other vaccination campaigns, albeit on a smaller scale, and various treatment programs.

Seven quarantine stations for the export of live animals have also been built, each with their own isolation areas for sick animals: three in Berbera, Somaliland, with a combined maximum capacity

¹⁶ From year to year, different animals are vaccinated, including young animals (usually less than three months of age) that were not vaccinated during the previous round.



to hold up to 230,400 animals at any one time (211,000 for shoats and 19,400 for camels), and two others in Bosaso, Puntland, that can hold more than 367,000 heads of livestock. One of the quarantine stations in Berbera and one in Bosaso are owned by a Saudi company (SEIVQMC), while the others are owned by Somali business people. A new, locally owned private quarantine facility was established in Jazeera, near Mogadishu, in 2011, with a holding capacity of 250,000 animals; and another was opened by a private company in Kismayo in 2017, whose capacity is 13,500 animals, and handling so far mostly cattle. All these facilities are equipped with laboratories to perform various tests before health certification, including for PPR and brucellosis (also known as undulant fever, a highly contagious disease), according to the specific requirements of the importing countries.

Through the mid-2010s, such investment in a system of vaccinations campaigns, health pre-inspections, quarantine stations, and livestock health certifications at the tail end of the market chain had dramatically enhanced the confidence of importers in the safety and quality of livestock from Somalia and had led to the lifting of the livestock import ban by the Saudi government and to reductions in livestock handling time and overall marketing costs (Godiah et al. 2015; Khadijah and Kabue 2012). The northern quarantine stations have on site federally-accredited but state government-employed and privately paid veterinarians, who issue health certificates of compliance with importing country requirements. Animals are also clinically inspected in some regional markets, such as in Galkayo, Puntland, and Belet Weyn and Jowhar, Hirshabelle, before healthy animals are trucked to the quarantine stations near the export ports.¹⁷

This system, however, was unable to prevent the shipment to Saudi Arabia from Mogadishu Al-Jazeera quarantine of animals that tested positive in December 2016 to RVF. The December 2016 RVF incident led Saudi Arabia and the UAE to re-impose an import ban on live animals from all Somali ports. Based on available evidence, the FAO and the MoLFR believe the animals had been shipped off to sea from the Jazeera quarantine station (near Mogadishu) by unscrupulous traders with the complicity of inexperienced government and private veterinarians. The animals were not held under quarantine neither to the end of the two-week incubation period nor to the end of the required holding period following the administration of vaccines against RVF, resulting in a false-positive result at the port of entry. Another shipment of 27,000 shoats from Bosaso under the sacrificial animal exemption was rejected by Saudi Arabia in the months preceding the 2018 Haj, also following positive tests to RVF antibodies at destination. This second incident derailed until recently the provisional agreements on lifting the import ban with the Ministry of Water, Environment and Agriculture of the Kingdom of Saudi Arabia (KSA), which included a request by MoLFR to the World Organization for Animal Health (OIE) to evaluate the standards and practices of all quarantine stations in Somalia, including their laboratories. As part of renewed negotiations, it was agreed in February 2019 that MoLFR shall produce a federally-mandated uniform health certificate form that will accompany all animals exported from all quarantine stations, including those in Berbera, Somaliland.

At stake are three critical issues, related to weak capacity, conflicts of interest, and ineffective controls that results in poor credibility among key importing countries of the system at Somali ports for testing for RVF and other TADs and for preventing tainted shipments. Weak capacity is

¹⁷ Before September 2009, exports from Somaliland to Saudi Arabia transited through the livestock quarantine facilities at the port of Djibouti. This not only increased the marketing costs for Somaliland traders, but also constrained export volumes to the handling capacity of the facilities at the port of Djibouti. While the findings by Khadijah and Kabue (2012) cover the case of Somaliland, the same applies to exports from Puntland.



exemplified by the poor conditions of the public laboratories and the lack of accreditation and supervision of the private laboratories at the quarantine stations. Another major issue affecting credibility is that health certificates are issued by privately paid veterinary officers, a practice that gives rise to conflicts of interest. Although government-accredited and formally government-employed, these veterinarians are not independent from the private companies owning and managing the quarantine stations and paying their salaries (in addition to government-mandated fees). The third issue relates to the general institutional weaknesses affecting export controls by the authorities, coupled with the impatience among Somali exporters regarding the long waiting period of three weeks in the quarantine stations and reluctance to shoulder the associated livestock maintenance costs. All these factors have led and can still lead to rushed shipments too soon after vaccinations against RVF and FMD, in contravention of importing country requirements.

Other weaknesses of the current animal health system are that all vaccination campaigns are donor driven and paid; meaning vaccinations only take place when external funds are available.

Also, existing vaccination campaigns only target TADs. For other important diseases, such as anthrax, haemorrhagic septicaemia, blackleg, etc, vaccines are not widely available in the country and are imported directly, with no assurance of their quality and certifications, by the private sector for use mainly at the quarantine stations. There have also been challenges related to the timing of the administration of vaccines, specifically for live vaccines, such as for RVF, and for which it is difficult to distinguish between a recently vaccinated animal and an infected animal. The unregulated nature of the animal health care system in Somalia makes it difficult to regulate vaccinations, for example by ensuring that animals are not vaccinated too close to the point of export, and to inform receiving countries of the vaccination campaigns and notify them that tests might show false positive results.

Widespread availability of low-quality veterinary drugs is another challenge. The bulk of veterinary drug trade is undertaken by the private sector outside the purview of the Department of Veterinary Services in the MoLFR. Despite the recent adoption of the Somali Veterinary Code, there still is no veterinary drug quality assurance program and, in any case, enforcement of licensing procedures and monitoring of veterinary drug use would be difficult due to current capacity constraints. Most livestock owners often purchase veterinary drugs solely according to price, rather than from veterinarians, CAHWs, or agro-vet stores. They also lack awareness on their proper use, leading to under-dosage, over-dosage and residues in animal products meat and milk. This uncontrolled use poses a high risk of bringing about also in Somalia antimicrobial resistance (AMR), which has become a global concern around the world, to current antibiotic, antifungal, antiprotozoal, and antiviral drugs. With a largely unregulated market, the use of expired, substandard, and fake drugs cannot be ruled out.



Decades of donor-funded animal health projects/interventions, including distribution in the northern regions of free veterinary drugs, has also eroded livestock owners' willingness to pay for good-quality drugs, making the development of a private sector-driven drug supply system very challenging. Moreover, many veterinary workers and professionals across Somalia are aging, with few youthful ones having been trained in the last decades. As discussed in Chapter 4, IGAD Sheikh Technical Veterinary School in Sheikh, Somaliland and a few private universities in Somalia offer training programs in animal health and veterinary science, but their output quality and graduation numbers still fall far short of the needs of the country's largest economic and export sector. Due to this lack of domestic capacity, many of the livestock exporting companies use veterinarians from outside Somalia, who are not registered and accredited by MoLFR and whose qualifications and experience on animal inspection and disease investigations are unknown to either federal or state authorities.

3.5 Feed Resources and Animal Nutrition

There is a high and growing demand for quality fodder along trading routes, in market centers, and at terminal ports, even though the predominant pastoral production system still relies on natural pastures on the country's vast rangelands. However, little or no commercial fodder is currently available (see Box 3.1), and storage practices of hay production from native grasses is so poor that its nutritive value drops rapidly equivalent to that of cereal straws soon after grain harvest. In the agro-pastoral and settled mixed farming systems there is medium to high integration with crops and even very high integration in the flood plain areas where fodder can be grown. In addition, fodder is produced seasonally in some of the rain fed areas with some of the rangeland areas in northern Somalia being increasingly enclosed illegally, against the traditional practices, in order to grow fodder. Fodder is also grown under irrigation along the riverine areas using mechanical pumps and within flooded basins in the plains. There is a proliferation of small-scale irrigation in peri-urban, coastal areas and dry river beds areas based on groundwater extraction from boreholes as well as in dry river beds using mechanical and solar-powered pumps, where cut-and-carry livestock feeding is being practiced. A study undertaken in the border area of Ethiopia, Kenya and Somalia (Nyangaga, et al., 2009) found that agro-pastoralists use part of their crops, mostly residues, to feed their livestock. They then sell the rest to peri-urban livestock keepers in and around towns as well as pastoral communities living further away from the rivers when they move closer to their farms during the dry season.

Because of the wide catchment area, most animals destined for export are trucked large distances from the main livestock markets to the Somali export ports, with rest stops along major trekking routes, generating a high and growing market demand for fodder at such rest stops, in market centers and at terminal ports. To meet the substantial water and fodder requirements for this trade and also to overcome natural feed shortages in seasonal droughts, the biomass currently used as fodder are mostly crop residues such as stalks that is mainly made up of maize and sorghum straws, that are generally of low nutritional value. Others less commonly marketed fodder sources because of their higher prices are green maize stovers, grass hay, pulse and cereal milling by-products (brans), and oilseed cakes, except during severe droughts when the very survival of the animals is at stake. Fish waste is another valuable by-product but still under-exploited as a source of feed, particularly suited to the poultry sector. Successful pilots have also been undertaken, for example by FAO, to produce feed from the pods of the invasive *Prosopis Juliflora* (mesquite or, in Somali, *garaanwa*).



Box 3.1: Promising Commercial Venture in Seed Production for Animal Fodder

CSET is a private seed company established in October 2018 in Mogadishu, focusing on the production, processing and sale of various kinds of quality seeds for animal fodder and human consumption, and aiming to become a leading company in the agro-processing subsector. Such a business venture is a promising sign that the private sector is willing to step in to meet the growing demand for commercial quality feed and fodder, spurred by the recurring severe droughts experienced during the most recent decade.

The company offers five different varieties of sorghum seeds specifically intended for providing two harvests (50-60 days each) per growing season and maximizing the yield from the green stalks, rather than maximizing the yield of the sorghum head for human consumption that requires a full season to maturity (110-120 days). For instance, the height of their Deeqsom (IRAT204) variety is 1.6 meters, with a protein content of 24 percent, a potential yield of 1.25 kg/ha and is medium-resistant to diseases and insects, including the Fall Army Worm. Its first harvest, fully sold to livestock owners, was in June 2019 from their demonstration farm under irrigation in the Lower Shabelle region and the company is planning another demonstration farm for the next season under rainfed conditions in the Bay region.

Source: interview with CSNET management in June 2019 and company marketing brochure.

The demand for fodder and consequently also its prices peak when livestock exports soar before and during the main Islamic celebrations and during the two regular dry seasons that Somalia experiences every year. Demand is also high at all times near towns where the number of urban consumers of milk and meat has been increasing, generating greater demand for fodder from dairy and other animal farms in peri-urban areas. Nomadic pastoralists have also increasingly demanded fodder as the rangeland pasture is declining, more so at times of prolonged severe droughts. Prices of fodder are highest during the dry season when pasture is scarce and fodder production is generally low. A 10-ton truckload of dry grasses (hay) harvested from areas around Burao in Somaliland sells for as low as US\$50 in the wet season but can go up to US\$350 in the dry season. Around Dollow in the Gedo Region, when large populations of pastoralists start migrating with their animals to the riverine areas due to availability of crop residues produced under irrigation, fodder prices increase from less than US\$0.20 around May/June per bunch (equivalent to a single handful or three stalks of fresh Sudan grass or maize) to US\$0.50 two to three months later, when drought peaks (FAO, 2017).

Despite the higher cost which allow many poor pastoralists to purchase only the small amounts necessary to ensure the survival of their animals until the next rains, fodder availability offers a good coping strategy during dry season and even prolonged droughts. The increased demand for and availability of quality fodder in the riverine areas around Dollow, in fact, provides much needed support for breeding and lactating animals, for increased milk yields and for consumption by the local communities, as well as an opportunity for women to engage in the marketing of fodder as an alternative source of income.

3.6 Marketing of Live Animals

Similar to livestock production, Somalia's domestic livestock marketing and export trade is still mostly guided by traditional institutions and practices, including kinship and ethnic groupings, identity, and religious practices, which serve as alternatives to formal contracting. Livestock marketing supplies animals for slaughter, primarily for domestic meat consumption (exports of



frozen beef carcasses remain negligible), and for live animal exports, mostly to Gulf countries. In fact, the Somali livestock sector sets itself apart by the degree to which its production systems are geared to supply a thriving export market. Livestock is exported year-round, but with a considerable peak in exports during the seasonal Haj, with Saudi Arabia being the main importer of live animals, responsible for almost 65 percent of all livestock exports from Somalia in 2015, followed by other Gulf countries.

The accurateness and reliability of available offtake estimates are questionable but available figures appear to be in line with those for pastoral and agro-pastoral production systems elsewhere in sub-Saharan Africa.¹⁸ Estimates quoted in previous sector strategy and needs assessment documents for Somalia show annual offtake percentages of 27.3% for sheep, 23.3% for goats, 11.3% for cattle, and 1.6% for camels (in FAO et al. 2004, also in World Bank 2007).¹⁹ A more recent study using 2013 data obtained from various federal and state livestock and planning ministries presented the following offtake rates, as shown in Table 3.3 (ICPALD, 2015), which shows a large variation for cattle and camels between the northern and southern regions of the country.

Table 3.3: Livestock offtake rates (for slaughters and for export of live animals)

Percent or %	Northern Regions	Southern Regions
Cattle	9.4	4.9
Camel	11.4-14.5	1.9
Goats	23.0	23.0
Sheep	23.0	23.0

Source: ICPALD, 2015.

In the absence of government programs and institutions that protect animal health, many critical services (e.g., extension, mass vaccinations, quarantine stations, and health certifications) have been provided by the private sector and by local veterinary associations funded from development partners. These interventions and services supported a strong export performance for live animals through 2016, but they remain insufficient to address the serious risk of various animal diseases in the country. Such efforts led to large reductions in handling time and marketing costs, and through 2016 dramatically enhanced the confidence of importers in the safety of Somali livestock through 2016. In fact, the substantial public and private investment in vaccination campaigns, quality improvements, and in quarantine stations have supported an impressive growth in live animal exports through the mid-2010s, with export volumes and values in 2015-2016 considerably higher than before the

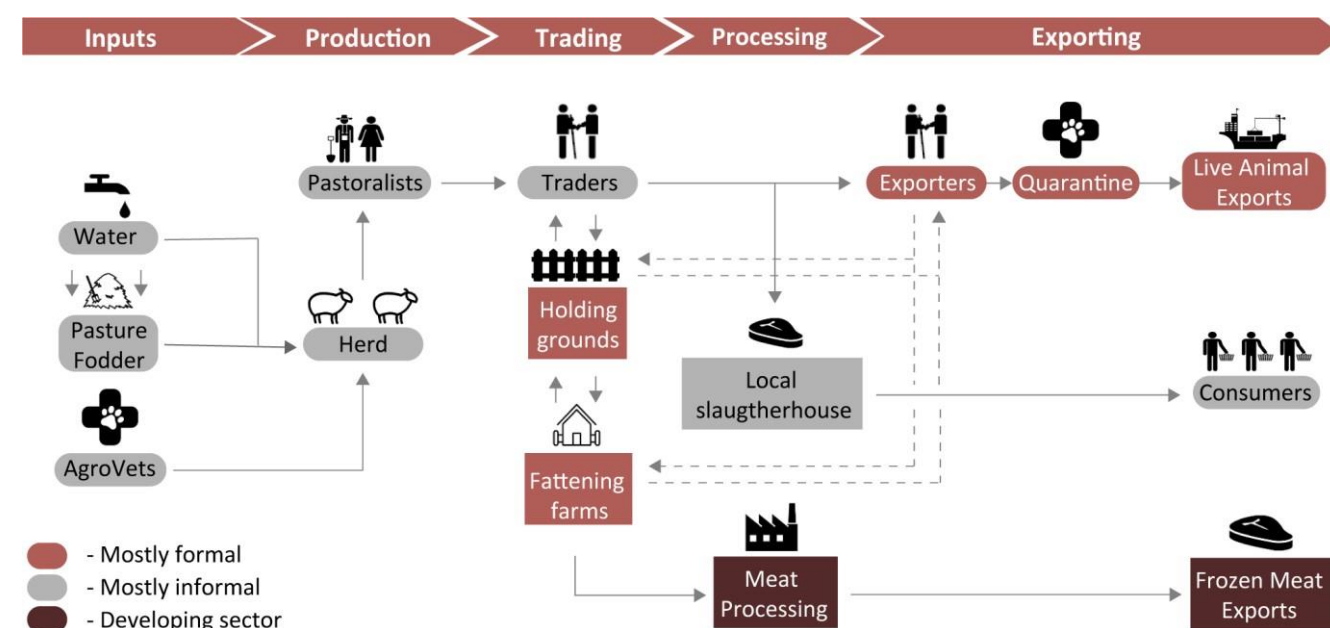
¹⁸ FAO in its review of Cattle and Small Ruminant Production Systems in sub-Saharan Africa (2002) notes that mean offtake rates for cattle in pastoral production systems in arid and semi-arid areas are 11,7% and 12,3% respectively, against 10,2% for agro-pastoralists in semi-arid areas; for sheep these rates are 22,6%, 20,6, 5 and 15,6% and for goats 30,2%, 17,2 and 16,7% respectively.

¹⁹ The Somalia Joint Needs Assessment (2006) further notes that about 33% of the off-take is accounted for by household consumption, 17% by other domestic consumers, and around 50% by exports. Average livestock mortality is below 5%.

civil war and providing about three-quarters of total exports value, as shown in Table 2.3 (World Bank and FAO, 2018). They did not manage to prevent, however, the latest import ban imposed by Saudi Arabia and the UAE in December 2016, with the only exception of 1.2 million sacrificial shoats allowed in annually (during the Haji season). For the sector's historical performance, refer to Tables 2.3 and 3.3 and Figure 3.3. Furthermore, for a visual representation of the main features of the sector's export performance see Figure 3.2.

On the Arabian Peninsula, there is a strong preference for Somali livestock and meat, both because of similar and familiar halal handling requirements and because of similar genetic characteristics between Somali and Gulf region's shoats. This latter factor ensures similar body shapes, and taste and cooking properties of the meat, unlike with animals from Australia, New Zealand, Europe, and the Americas (FAO 2013). While sales of Somali cattle dominate many of the larger and smaller markets in the Horn of Africa, particularly the southward routes and markets on the border of Kenya and Ethiopia, its small stock, including a large share originating from Ethiopia, dominates northward trading routes to the Gulf States, with a noteworthy peak during the Haji season.

Figure 3.4 - Live Animal Exports and Meat Value Chain in Somalia

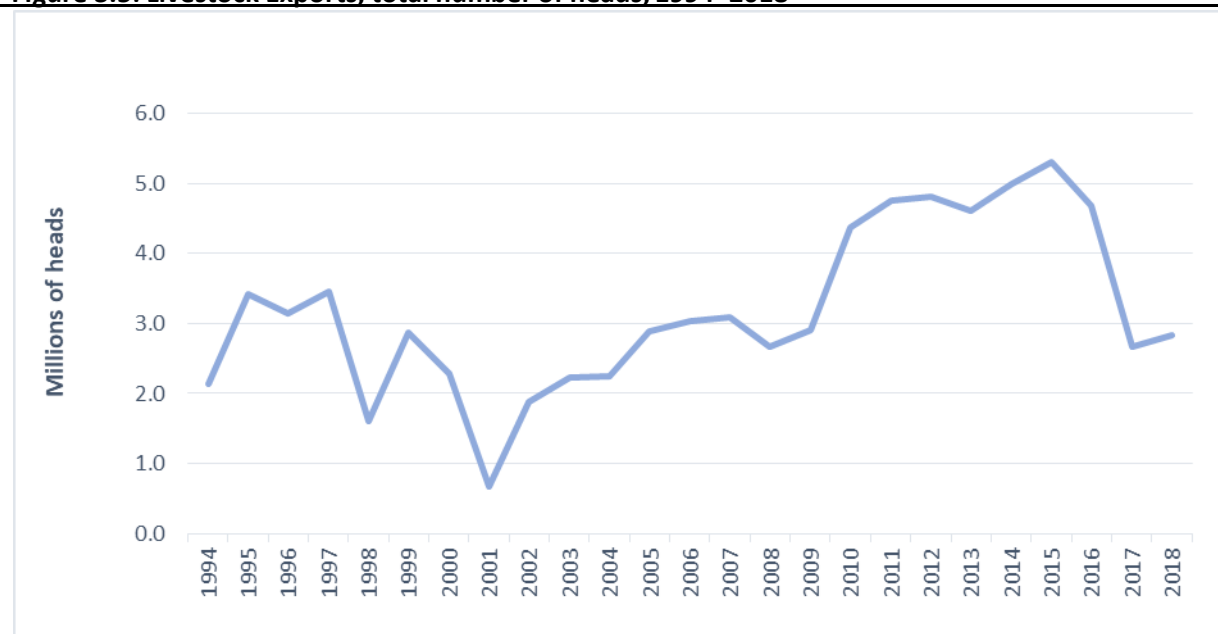


Recorded exports in 2017 and 2018 are dramatically lower than at their peak in the mid-2010s, as shown in Figure 3.5 and Table 3.4, and concentrated on a few destination markets. They are currently limited mainly to Oman throughout the year, with 1.2 million sacrificial animals shipped to Saudi Arabia from the northern ports during the Haji period, and to small occasional shipments of camels to Egypt. This is due both to the lingering impact of the latest severe drought in 2016-2017 and the import ban on the ground of suspected RVF imposed by Saudi Arabia and UAE in December 2016. Efforts to lift this ban have proven ineffective so far, only partly because of another shipment to Saudi Arabia during 2018 was found to include animals affected by positive RVF antibodies, but mainly because of a chronic failure to meet international standards and requirements of importing

countries and related lack of credibility of the official veterinary certifications issued by the quarantine stations in the northern ports.

Most of the livestock is exported through the northern ports of Berbera and Bosaso transiting through Beletweyn, Galkayo, and Burao. These towns are important collection markets for most of the pastoral livelihood zones of Somaliland and Puntland, as well as from the southern and central rangelands and from across the border in Ethiopia's and Kenya's Somali regions. From Burao the highway leads straight to Berbera, Somaliland's main port, from where animals are shipped to Saudi Arabia and Yemen, as Berbera is closest to the Yemeni ports across the Gulf of Aden from where many animals also transit into Saudi Arabia. Bosaso is the second most important port for exporting animals, mainly to Oman and the UAE. Exports of cattle and camels, the latter dominated by exports to Egypt, also take place from the northern ports, though numbers are significantly lower than those of shoats. Beletweyn is the main connection hub between the southern regions and the main northern export markets. For both Somaliland and Puntland several studies are available that describe the live animal export value chain. Livestock exports from southern ports came to a complete stop following the civil war and exports of shoats resumed from Mogadishu only in 2013 and of cattle from Kismayo only in 2017. Camel exports from Mogadishu to Egypt have also recently started.

Figure 3.5: Livestock Exports, total number of heads, 1994–2018



Source: FSNAU (IDS online database).

At the peak in 2015, a record of 5.3 million live animals were exported from Somalia, composed of 4.9 million shoats, 295,000 cattle, and 72,000 camels, 6.3 percent above the previous year's level. The value of exports of live animals for food reported by Somalia's trading partners to the U.N. COMTRADE database rose in parallel from around US\$56 million annually (on average) in the late 1980s to an estimated US\$377 million annually during 2011–2014, and a peak of US\$494 million in 2015. The additional value of live animals exported for breeding purposes also jumped from virtually nil before the civil war to a peak of US\$81 million in 2014, though it fell back to \$30 million in 2016.

Eexports of camels and cattle have also seen a spectacular increase, though with wide annual variations, since the 1990s, with their number of heads increasing from a range of 16,000-65,000



to around 140,000 at their peak in 2011 (according to FSNAU), mainly to Egypt. This may have been due to political tensions in Sudan, the traditional camel supplier for Egypt, and between Sudan and Egypt. The unfortunate plight of Sudan and the new demand from Egypt has opened an upcoming market opportunity for Somalia. Berbera being the larger and more modern port, its export figures for both camels and cattle are higher than for Bosaso. However, since 2015 there has been a marked increase in cattle exported from Bosaso and a decrease from Berbera.

The most recent drought spell, which started in 2015 in the northern regions and extended to the south central regions in the second half of 2016, already started reducing during 2016 both the animal population and, due to the very poor health of surviving animals, also livestock exports, which dropped from 5.3 million in 2015 by almost 13 percent to 4.7 million heads in 2016. The most recent import ban imposed by Saudi Arabia and the UAE in December 2016 has taken a further massive toll, at least in the short term, with the total number of animals exported from all four major Somali ports collapsing to 2.7 million during 2017 and 2.8 million during 2018 (according to FSNAU-IDS database). Of these about 1.2 million or about 40 percent were sacrificial shoats absorbed by Saudi Arabia just before and during the Haj and shipped from the northern ports of Berbera and Bosaso. The rest were destined mostly to Oman and Yemen.

Table 3.4: Livestock Sector Exports, 1981-2017

Annual averages in US\$ millions; SITC-3 categories in brackets

Product Description	1981-85	1986-90	1991-95	1996-00	2001-05	2006-10	2011-14	2015	2016	2017	2018
Animals, live, chiefly for food (00)	88.4	55.7	66.6	72.9	76.7	126.3	377.1	493.9	429.9	215.3	246.8
Sheep & Goats (0012)	64.7	42.0	63.2	52.3	40.9	82.7	292.4	400.9	376.9	163.6	185.0
Cattle (0011)	15.7	0.4	3.4	20.3	35.7	40.8	79.4	93.0	52.9	46.9	59.4
Other (residual, incl. 0014, 0015, 0019)	4.4	3.3	0.0	0.3	0.1	0.0	1.0	0.0	0.1	4.8	2.5
Animals, live, n.e.s., incl. for zoo & breeding (9410)	0.0	0.0	4.2	6.5	0.6	5.2	42.8	38.7	30.1	<i>n.a</i>	<i>n.a</i>
Livestock Products	4.0	5.2	2.6	4.3	14.9	18.1	11.9	13.6	5.9	5.9	<i>n.a</i>
Hides, skins and furskins (21)	3.9	5.1	2.4	2.3	6.2	8.6	8.6	4.0	2.3	2.7	<i>n.a</i>
Bovine & equine hides, raw (211 & 2112)	0.2	0.2	0.4	0.2	0.9	1.2	2.1	1.0	0.4	0.0	<i>n.a</i>
Goat & kid skins, raw (2114)	2.2	2.8	1.1	0.9	2.6	0.8	0.0	0.0	0.0	0.0	<i>n.a</i>
Sheep & lamb skins, raw (2116+2117)	1.4	2.0	0.8	1.0	2.2	3.4	2.2	1.0	0.4	0.7	<i>n.a</i>
Other (waste, leather, furskin; 2119, 2120)	0.0	0.1	0.1	0.1	0.6	3.3	4.3	2.0	1.5	2.1	<i>n.a</i>
Meat and Meat Preparations (01)	0.1	0.0	0.1	1.7	7.5	9.0	3.1	9.6	3.6	3.6	<i>n.a</i>
Meat of bovine animals (0111)	0.0	0.0	0.0	0.0	0.1	0.3	0.0	0.1	0.1	0.1	<i>n.a</i>
Meat of sheep and goats (0112)	0.0	0.0	0.0	1.6	7.2	8.3	2.9	9.4	3.2	3.4	<i>n.a</i>
Other, incl. offals (residual of 01)	0.0	0.0	0.0	0.0	0.2	0.5	0.2	0.1	0.3	0.1	<i>n.a</i>
Dairy products and birds' eggs (02)	0.0	0.0	0.1	0.3	1.2	0.4	0.2	0.0	0.0	0.0	<i>n.a</i>
Milk & cream (0223 & 0224)	0.0	0.0	0.1	0.3	1.1	0.3	0.1	0.0	0.0	0.0	<i>n.a</i>
Other (residual, incl. 0230, 0240, 0251)	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.0	0.0	0.0	<i>n.a</i>
Natural Honey (0616)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	<i>n.a</i>
TOTAL LIVESTOCK and LIVESTOCK PRODUCTS	92.4	60.9	73.4	83.6	92.2	149.5	431.8	546.2	465.9	221.3	<i>n.a</i>
of which: Live Animals (00+9410)	96%	92%	96%	95%	84%	88%	97%	98%	99%	97%	<i>n.a</i>
of which: Livestock Products	4%	8%	4%	5%	16%	12%	3%	2%	1%	3%	<i>n.a</i>
of which: Hides & Skins	97%	99%	93%	53%	42%	48%	72%	29%	39%	46%	<i>n.a</i>
of which: Meat and Meat Preparations	2%	1%	2%	39%	50%	50%	26%	71%	61%	61%	<i>n.a</i>

Source: UN COMTRADE (Somalia's partner countries data), accessed via WITS on June 28, 2019; except for FSNAU for live animals in 2017 and 2018 (note: FSNAU does not differentiate between animals for food vs zoo and breeding).

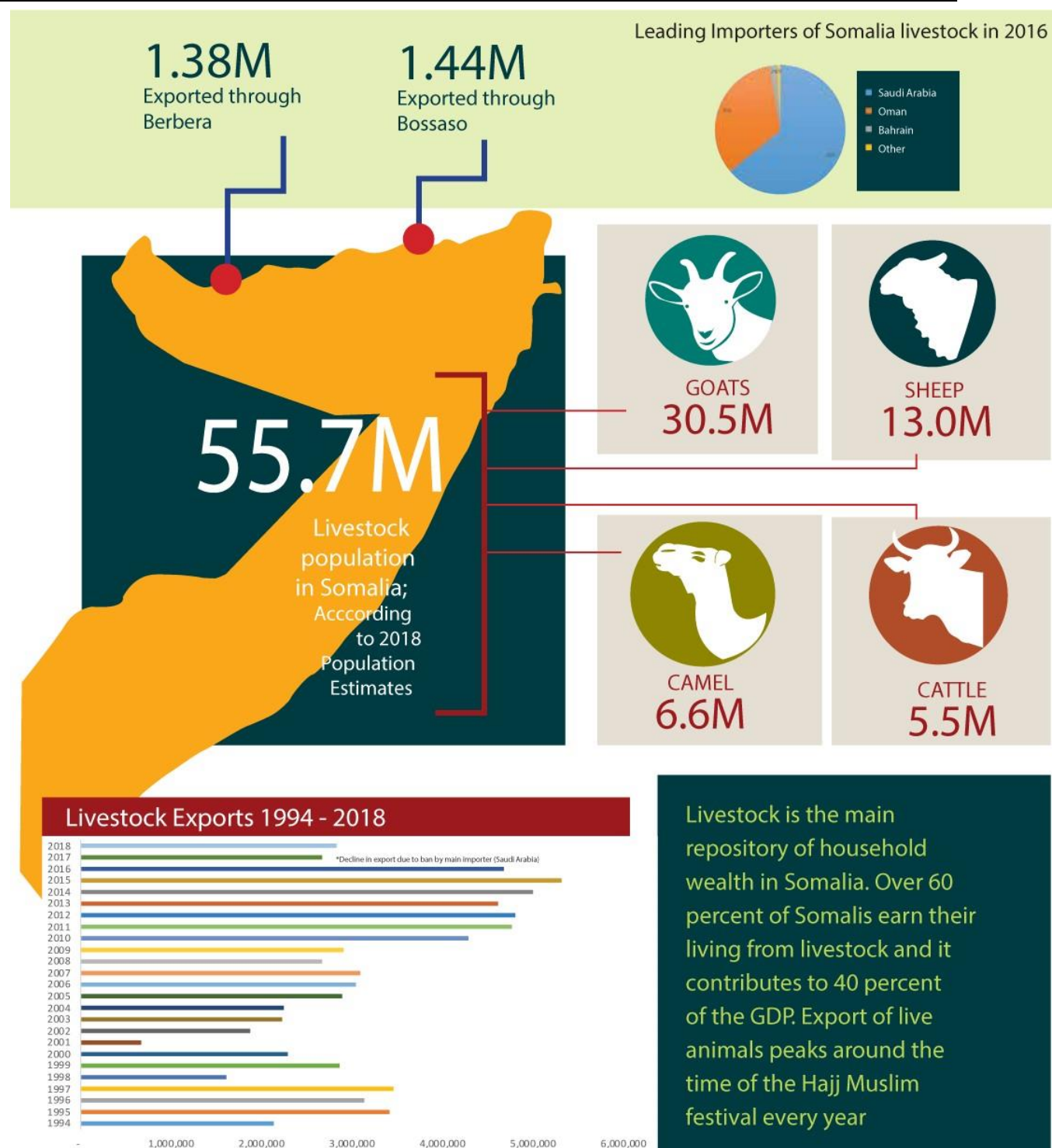
Note: 2017 estimates are still preliminary, as some countries (including Saudi Arabia and Bahrain) have yet to report to COMTRADE.

Compared to the more formal northern trading routes, less information is available on the southern cross-border trading routes for livestock, which are largely unrecorded, including cattle exports from Jubaland into Kenya and Ethiopia and shoats from Kenya's and Ethiopia's border areas to be re-exported through Somalia's northern ports. Though numbers on formal and informal livestock cross-border trade are unknown, they are believed to be substantial, as for Somali pastoralists clan territories transgress the Somali border for almost its entire length, from northeast



Djibouti to northeast Kenya (Bishop & Catley, 2008). Before the collapse of the Somali state there was a common cross-border route with livestock moving from Ethiopia and northern Kenya into Somalia for exporting through the port of Mogadishu. Collapse of the state caused a reversal in the direction of livestock trade, leading to an outflow of livestock, particularly camels and cattle from Somalia to Kenyan and Ethiopian markets (Mahmoud, 2013). However, in recent years shoats from northern Kenya have been and are still been transported all the way to and exported from the northern ports.

Figure 3.6: Livestock Exports Infographic, 2018



Source: FAO 2019

Table 3.5: Top 8 Destinations for Livestock Sector Exports, 1981-2016

Annual averages in percent of total exports



Country	1981-85	1986-90	1991-95	1996-00	2001-05	2006-10	2011-14	2015	2016
Oman	0.4%	0.1%	0.9%	7.3%	7.6%	23.4%	26.8%	27.2%	33.3%
Egypt, Arab Rep.	1.0%	1.2%	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Kuwait	0.0%	0.0%	0.3%	0.8%	0.3%	1.4%	0.1%	0.0%	0.0%
Saudi Arabia	95.8%	95.5%	77.8%	44.4%	0.0%	21.9%	54.5%	56.4%	64.2%
Bahrain	0.0%	0.0%	0.1%	0.5%	6.5%	1.4%	2.1%	0.9%	1.8%
United Arab Emirates	0.0%	0.0%	9.8%	13.5%	56.3%	9.4%	7.6%	13.8%	0.6%
Qatar	0.1%	0.9%	0.7%	0.3%	0.4%	0.7%	0.1%	0.0%	0.2%
Yemen	2.7%	0.0%	10.3%	33.1%	28.8%	41.7%	8.7%	1.6%	0.0%

Source: COMTRADE, accessed via WITS in March 2019.

Livestock sector exports, furthermore, are characterized by highly concentrated product and destination patterns. Most exports are of live animals, with only very limited exports of meat (virtually all to Oman) and hides and skins. Moreover, following the lifting in 2009 of the prior import ban by Saudi Arabia imposed in 2001 and the opening of modern quarantine stations in Somaliland and Puntland managed by a Saudi company, over 98 percent of live animals exports (by value) went before the most recent drought and import bans to only four countries in the Arabian Peninsula (see Table 3.5).

Table 3.6: Livestock Sector Imports, 1981-2017

Annual averages in US\$ millions; SITC-3 categories in brackets

Product Description	1981-85	1986-90	1991-95	1996-00	2001-05	2006-10	2011-14	2015	2016	2017*
Livestock and Livestock Products	16.5	7.1	3.0	1.7	7.9	28.2	160.0	239.1	76.0	80.8
Livestock	0.0	0.0	0.0	0.0	1.2	10.6	95.4	148.2	0.4	0.0
Live Animals Chiefly for Food (00)	0.0	0.0	0.0	0.0	1.2	9.4	93.1	147.7	0.4	0.0
Sheep & Goats (0012)	0.0	0.0	0.0	0.0	0.0	0.1	3.9	40.4	0.3	0.0
Cattle (0011)	0.0	0.0	0.0	0.0	1.2	9.0	89.2	107.3	0.0	0.0
Other animals, for food	0.0	0.0	0.0	0.0	0.0	0.3	0.1	0.0	0.0	0.0
Animals, live, n.e.s., for zoo & breeding (9410)	0.0	0.0	0.0	0.0	0.0	1.2	2.3	0.4	0.0	0.0
Livestock Products (21+01+dairy+other)	16.5	7.1	3.0	1.7	6.7	17.6	64.6	90.9	75.6	80.8
Dairy products	15.9	5.9	2.4	1.7	4.7	15.9	60.3	82.0	66.0	72.1
Milk & cream, fresh (0223)	0.0	0.0	0.0	0.0	0.1	1.0	4.4	5.3	4.9	3.1
Milk & cream, prsrd or cncntrtd (0224)	10.3	4.7	2.2	1.6	4.6	14.7	54.9	76.4	60.8	68.7
Butter and ghee (0230)	5.5	1.1	0.1	0.0	0.0	0.2	0.9	0.2	0.1	0.1
Cheese and curd (0240)	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.2	0.3
Meat and Meat Preparations (01)	0.5	1.2	0.5	0.0	1.9	1.2	1.9	8.2	7.4	6.9
Meat of bovine animals (0111)	0.0	0.0	0.1	0.0	0.0	0.3	0.7	4.1	2.7	1.9
Meat of swine (0113)	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.1	0.1	0.3
Poultry, dead & edible offals (0114)	0.0	0.0	0.1	0.0	1.8	0.6	0.5	3.4	2.4	2.6
Other (residual, if any different from zero)	0.5	1.2	0.2	0.0	0.1	0.2	0.7	0.6	2.1	2.1
Birds' eggs (0251+0252)	0.0	0.0	0.0	0.0	0.0	0.1	0.6	0.5	1.9	1.7
Natural honey (0616)	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.3	0.0
Other (residual, incl. 21-hides&skins & leather)	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0
Animal Feed (08)	0.0	0.0	0.0	0.0	0.1	0.7	0.4	0.8	2.2	2.7
of which: Hay & fodder, green/dry (0811)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.2	0.9
Medicinal and pharmaceutical products (SITC 54)	5.1	4.8	3.5	6.8	7.7	18.5	33.8	50.9	51.0	71.0
Provitamins & vitamins (5411)	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.1
Antibiotics (5413)	0.0	0.1	0.0	0.0	0.0	0.1	0.1	0.1	0.2	0.2
Vegetab.alkaloids (5414)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Hormones (5415)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Glycosides; glands & extracts (5416)	0.1	0.3	0.1	0.1	0.2	1.4	1.3	1.2	2.5	6.3
Medicaments (including veterinary; 5417)	4.9	4.3	2.8	6.4	7.2	16.4	31.4	48.9	47.4	63.4
Other pharmaceutical goods (5419)	0.1	0.2	0.6	0.3	0.3	0.6	0.9	0.7	0.9	1.1

Source: COMTRADE, accessed via WITS in March 2018

Regarding the imports of livestock, Table 3.6 offers a more detailed illustration of the top livestock-related import items, compared to the aggregate import estimates for the overall economy shown in Table 2.4 in Chapter 2.2. Historically and also for the most recent years 2016-2017, Somalia does not import livestock in



any sizeable quantity, whether for food or for breeding. Sizeable cattle imports, however, for about \$42-46 million were recorded (as exports by Somalia’s trading partners) during 2011, which was a severe drought year and 2012, when presumably Somali pastoralists were focused on rebuilding their cattle herds. Furthermore, during 2013-2015, there was an even higher and extremely unusual level of cattle imports, in the \$107-\$132 million range; in 2015, there was also a large import of about \$40 million of shoats: for these three years, this spike of livestock imports remain unexplained, though it might be related to a statistical oddity, when neighbouring countries recorded (some) cross-border trade which usually goes unrecorded.

Table 3.7: Gross Production Values for Somalia’s Livestock, 1986-2018

Annual average in US\$ million

	1986-88 /1	2013-16 /2	2017 /2	2018 /2
Livestock and livestock products	606	4,083	3,091	n.a
Live animal exports /3	104	495	215	247
Livestock products	502	3,587	2,876	n.a
Dairy (milk) 4/	227	2,848	2,353	n.a
Meat for local consumption 5/	262	730	516	665
Meat exports 6/	227	2,848	2,353	n.a
Hide & skins exports 6/	3	5	3	n.a
Others (poultry & beekeeping)	10	n.a	n.a	n.a

Source: As elaborated by the World Bank, sourced from Somalia 1989, FAOSTAT, U.N. COMTRADE, FSNAU IDS and Somalia 2018 (DINA).

Notes:

1\ As in Table 2.2

2\ As in Table 2.2

3\ As in Table 2.2

4\ As in Table 2.2

5\ As in Table 2.2

6\ As in Table 2.2

Regarding livestock products and key inputs to the livestock sector, the large bill starting in the late 2000s and ranging since 2011 from \$60 million to \$82 million for dairy imports, of which 90 percent is for “preserved or concentrated” milk, that is, long-life processed milk and milk powder, stands out as the most obvious missed import substitution opportunity. Since 2015, there are also sizeable meat imports around US\$7-8 million, less than half of which is explained by poultry meat imports (mostly frozen). Imports of animal feed are almost nil, which is further evidence of the minimal level of modern, commercial operations of the Somali livestock production and trading systems, as well as of the feed subsector. On the contrary, imports of “medicaments, including for veterinary use” have been growing and have hovered during 2015-2017 around \$49-63 million annually. Import of vitamins, antibiotics, and other supplements (e.g. minerals) and pharmaceutical products for either people or animals barely register in the international trade statistics.

3.7 Production and Consumption Trends of Livestock Products

As shown in Table 3.6, the production (and domestic consumption) of the main livestock products, milk and meat experienced substantial growth through the mid-2010s (World Bank/FAO 2018), but suffered severely from the impact of the recent drought spells during 2016-2017. The aggregate drought-related loss just from lower milk production and reduced body weight for exported or slaughtered animals has been estimated at a staggering \$1.2 billion (Somalia 2018). The production of other items like hides and skins, which used to be important for export and for



domestic tanning before the civil war, poultry, which has grown in the post-war period from almost nil before, and honey are still small today, though providing alternative and/or complementary sources of livelihood to some (smaller) population groups throughout the country.

3.7.1 The Dairy Subsector

Pastoral dairy production is one of the most valuable goods for both rural and urban populations, used for both household consumption and commercialization. It is one of the key sources of income and employment generation for women and youth. It is also vital for child nutrition. The dairy value chain includes individual producers selling directly at nearby markets (villages/towns), as well as specialist milk traders (mainly women) who receive or collect milk in bulk from wide catchments of producing areas and then market them in the towns (see Figure 3.7). Further to this, in recent years, commercially oriented herds of dairy camels and cows are reared quasi-permanently in confined areas within a short range to urban centers and marketed to hotels and restaurants in the cities. Solely on the basis of the projected growth of 2.9 percent for the human population, there will be more than half a million new potential consumers in 2030 compared to 2019, but demand is believed to grow further due to income gains and urbanization (Oxfam 2012 & 2018; FAO 2019; Hussein 2019).

Camel milk, which is not typically processed beyond cooling and packaging in plastic pouches for street-side sale, is the most popular and fetches the highest price in the market, as the demand is hardly met (demand for camel milk can surge by up to 22% during fasting months like Ramadan). Camel milk is composed of lactose, fat, and protein in roughly the same proportion as bovine milk. However, their relative composition, distribution, and the molecular structures of the milk components are different and as a result camel milk is difficult to process into products and is thus mainly suitable for drinking as fresh or sour milk.²⁰ If refrigerated the shelf life of camel milk can be as long as 7 days after which the milk become sour and still consumed as such. Quality concerns such as bacteria levels in primary processed items like sweets and curd were often not attributed high priority, and it was suggested that consumers have low or no awareness of food safety issues (Oxfam, 2018). Notwithstanding the popularity of camel milk, cow and goat milk are also in demand and often home-processed into ghee for self-consumption and local sale.

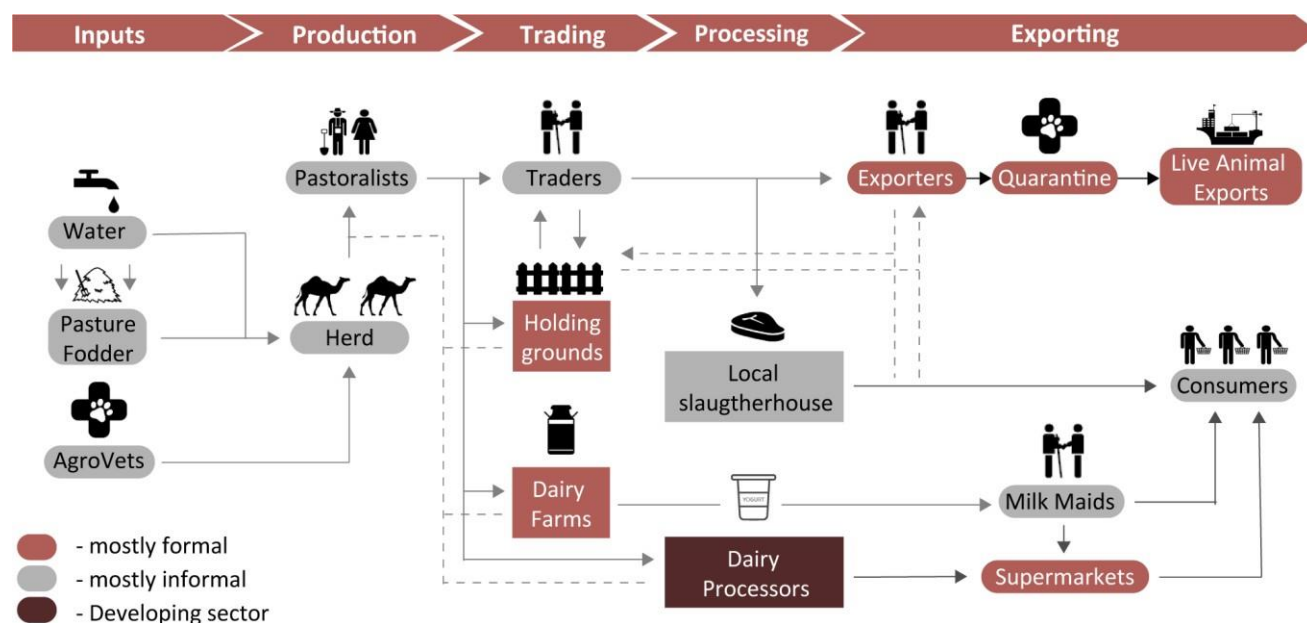
Agro-pastoral and pastoral fresh milk production still accounts for most of the milk sold in city markets and will continue to do so in the foreseeable future. Over-dependence on degraded pastures leaves the agro-pastoral and pastoral milk producers struggling to maintain current low milk production levels. Feed supplementation of lactating animals is widely practiced, but is constrained by lack of skills and high costs and limited availability of quality animal feeds (USAID, 2017). According to stakeholders, lack of finance, knowledge and trained technicians is the main impediment for the expansion of storage and cold chain facilities and processing of milk, enabling the large-scale importation of long-life milk powder and packaged milk and small imports of yogurt and other milk products.

Although there is a draft National Dairy Act, lacking to date are also updated regulations (or enforcement of those existing before the civil war) for milk hygiene and fresh milk adulteration practices and quality along the chain. Consumption of powder milk is also promoted vigorously by

²⁰ Tesfemariam Berhe, Eyassu Seifu, Richard Ipsen, Mohamed Y. Kurtu, and Egon Bech Hansen, "Processing Challenges and Opportunities of Camel Dairy Products," International Journal of Food Science, vol. 2017, Article ID 9061757, 8 pages, 2017

media outlets. As a result, mothers purchase imported powder milk products to feed their babies while adults are the main consumers of fresh milk (SATG and UNDP Somalia, 2016). Without any prescribed norms that stipulate testing the quality and safety of milk, no one knows the composition of, or the presence of veterinary drug residues in, fresh raw milk marketed, with the exception of some yogurt processing companies and possibly some embassies and large foreign businesses (such as private security companies) in Mogadishu that test in-house the raw milk they receive from their suppliers.

Figure 3.7 - Milk Value Chain in Somalia



The value of milk production, mostly from camels, had risen steadily since the late 1980s to reach US\$2.8 billion in 2014 (baseline before the drought), before dropping to \$2.4 billion in 2017 (because of the drought impact, see Somalia 2018). Even before the drought, productivity per animal was relatively low, with average yield for cows is 4.1L/day in the wet season and as little as 2.4L/day in the dry season (a 40% seasonal drop) and camel yield averaging 3L/day in the wet season and 2.1L/day in the dry season (with a 30% seasonal drop).²¹ These figures are low when compared with the average yield of camels in neighbouring Ethiopia that produce 8.6 liters of milk/day in the wet season and 5.8L/day in the dry season.²²

By and large the dairy trade is informal and, according to Oxfam 2018 and FAO 2019, consumers prefer to buy unpackaged fresh milk and are unwilling or unable to pay the extra costs of formal processing and packaging, making their purchase decisions largely on price. They conclude that value addition for milk may not generate returns as the market for processed milk products is a small niche segment, and most buyers are unwilling or again unable to pay for processing of any kind. Among their findings are that the key determinants of buying behaviour are first and foremost price, then home delivery and regular supply of quality (unspoilt) fresh milk.

²¹ USAID. (2017). *Somalia Growth, Enterprise, Employment & Livelihoods (GEEL): Dairy Baseline Study*.

²² Galma Wako. 2015. *Economic value of camel milk in pastoralist communities in Ethiopia: Findings from Yabello district, Borana zone*. IIED Country Report. IIED, London.



Nonetheless, in Mogadishu there are three commercial dairy factories that have grown rapidly with the support of USAID-Geel (USAID 2019), operate profitably and are on a rapid expansion path, indicating growing demand for hygienic milk and dairy products. Irmaan Dairy and SoMilk Dairy mainly focus on producing, testing, processing and packaging hygienic, including pasturized, camel and cow milk that is sold not only to hotels, restaurants, embassies and wholesale milk traders. SoMilk also produces and markets its own line of yogurt. Another dairy company, Oog Dairy established in 2016, has in fact been sourcing fresh milk from both SoMilk and its own dairy farm for making, packaging and selling to wholesale buyers mainly yogurt, but also some cheese and ghee (16 tons in 2018; see also Box 6.1 for more details on its success and business expansion plans). In Garowe, moreover, there is a micro-enterprise whose pasteurized camel milk and limited home-made yogurts sell out every day. And in Hargeisa, where all fresh camel milk is still sold only in clear plastic pouches by street-side sellers (as in the rest of the country, save for the exceptions mentioned above), there is a small company making yogurt out of imported milk powder.

3.7.2 The Meat Subsector

While production of fresh red meat was on the rise through the mid-2010s, almost exclusively to meet rising domestic demand, overall output remained well below potential both before and after the severe 2016/2017 drought, largely due to inadequate slaughtering and marketing facilities. Pastoralists sell their animals at the nearest primary town markets to either local butchers or wholesale traders, who bring them to larger urban markets for resale, including to butchers operating in the local slaughterhouses/slabs (see Figure 3.3 for a map of the meat value chain). Informal slaughtering of shoats by individuals in their backyards and larger animals by private butchers in poorly designed, open, unhygienic slaughter facilities remains the norm. These facilities, moreover, are not subject to any enforcement of sanitary regulations and/or meat inspection services, since a draft Meat Inspection Act is not yet endorsed. Electricity and cold chains are not readily available, partly due to their very high investment and operating costs and prohibitive energy costs (among the highest in Africa), resulting in high risk of disruption and monetary losses in the value chain.

Somalia's most important source of red meat is camels, supplying 52 percent of actual meat consumption. According to a study commissioned by IGAD's Centre for Pastoral Areas and Livestock Development (ICPALD 2015), furthermore, domestic consumption in Somalia averaged 4.0kg of beef, 4.3kg of small ruminant meat and 21.2 kg of camel meat per person in 2013 (ICPALD, 2015). In contrast, another study on competitiveness of the meat value chain in Somaliland (Ombui, Mogoa, & Matete, 2014) notes that the main sources of red meat for the Somaliland consumer are cattle, sheep, goats and camels, in order of importance, and that sheep and goats constitute 93 percent by number of the total animals slaughtered in all the six towns surveyed. The contradiction between these two studies is likely explained by variations in production systems (and herd composition) in the different areas that were surveyed. Analysis of meat trade in Somaliland indicated that trading in meat is relatively profitable in all urban markets covered by the study. Better margins (53-63%) were obtained when trading in camel meat and beef (51-53%), while lower margins (1.4-27.2%) were obtained when trading in sheep and goat meat (Ombui, Mogoa, & Matete, 2014).

Under the EU-funded ESOLT project (completed in 2017), the FAO conducted various feasibility studies about competitively identifying and entering into agreements with private companies to manage through public private partnership (PPP) arrangements the day-to-day operations of the municipal slaughterhouses that the project was rehabilitating. In the case of the Bosaso facility, an



investment appraisal undertaken in 2017 concluded that the business was indeed commercially viable, but that it needed to increase the fees charged and volumes handled and significantly cut down operation and administrative expenses to improve its rates of return, payback period and net present value. A limitation of this appraisal study, however, was that it did not include an assessment of the supply of animals.

Box 3.1: Limitations and Failures in Meat Processing Before the Civil War

Before the civil war, the Kismayo Meat Factory (KMF) was the only meat factory in Somalia. The plant was funded by the Soviet Union and constructed between 1964 and 1968, to produce about 9 tons a day of stewed canned meat and meat in its own juice. In 1974–75, a canned corned beef line was added and the canned meat line expanded. Daily production capacity after 1975 was 13 tons of stewed canned meat or beef in own juice (40,000 cans), 6.8 tons of corned beef (20,000 cans), and 2 tons of sausage. During 1969–76, the factory operated at close to capacity, slaughtering 200 head of cattle a day. In 1973, KMF purchased 1,291 tons of cattle for sale as frozen sides. During 1980–84, it produced only 1–700 tons of fresh and frozen cooked meat.

Canned output already began to fall in 1977. Since then, KMF operated only sporadically, working less than 50 days a year during 1983–85. Because of high cattle prices in the free market and KMF products' sale prices, which were controlled by the government, the company never covered its costs of production in any year since its establishment. Other factors hampering its operation included depletion of physical capital and the lack of spare parts from the Soviet Union. Like all other state-owned factories, its operations collapsed and its physical structures were looted right after the start of the civil war.

Source : International Science and Technology Institute (ISTI), 1986 (USAID-funded)

Other companies, with relatively modern slaughterhouses in Burao, Galkayo and Beletweyn, which had ventured in the past two decades into the chilled goat meat export business by air, went bankrupt and have ceased operations. There are multiple reasons for the failure of all chilled meat export ventures, according to Negassa, et al. (2012), the World Bank-FAO study (2018), and more recent feedback from stakeholders (see Annex 1): a ban by the UAE on old Russian aircrafts used by Somali exporters causing a major spike in air cargo costs from inland towns; management failures; uneven seasonal supply of slaughter animals with carcass weight required by importing markets (7-10 Kg); and high prices for livestock in Somali markets, due to both strong export demand for live animals exports (before the most recent import ban in December 2016) and for domestic meat consumption. Meat exporters from poor countries such as Somalia face stiff competition on weight, sanitary standards, and price from exporters in other developed or emerging economies, such as Australia, New Zealand, India and Ethiopia for fresh, chilled and frozen sheep and goats, and Brazil, Pakistan, India and Australia for fresh, chilled and frozen bovine meat.

One of the most comprehensive studies by ILRI and Terra Nuova in 2012 described the Somali chilled export meat value chain in terms of actors, institutions and practices, and profitability (Negassa, et al., 2012). The study highlighted evidence of economies of scale, and of financial advantage in species specialization (specifically, goats) by traders, and recognized a consistently applied and widely understood grading system in the country for slaughter stock. Quality requirements in importing countries were also well known throughout the chain, in that actors' rankings of attributes were consistent within and between actor stages.²³ Inconsistencies throughout the chain, however, included the nature and strength of trading relationships: long-lived

²³ FAO's SOMEAT Project (as well as its UNDP sister project) provided extensive training in meat safety requirements of exporting countries (Good Hygienic Practice and Hazard Analysis Critical Control, HACCP) to key players in the meat value chain (ministry staff, livestock traders, brokers, transporters, butchers and export slaughterhouse staff).



at exporter and agent level, but short-term and cash-based at producer level. Although exporters made payments mostly on the basis of carcass weight, agents of exporters paid based on per head of live animals.

Commercial meat processing is currently mostly limited to small quantities of whole beef chilled carcasses exported by air in cool crates and frozen carcasses sea in refrigerated containers from Mogadishu to Oman by the *Somali Meat Company* (SoMeat), which started operations in Mogadishu in 1996, plus some small quantities by two other recently established companies in Hargeisa and Galkayo targeting the local markets.

- *The Somali Meat Company* has a modern slaughterhouse, with machinery sourced from, installed and maintained by Italy, that can slaughter 100–120 animals per hour, and has recently built a second one on the outskirts of Mogadishu. It employs in total 300 workers, 260 of them seasonal, and exports 98 percent of the meat it processes, with 712.1 tons of chilled meat by air (in cool crates shipped in the cargo hold of passenger aircrafts) and 1,588.3 tons of deep-frozen meat exported to Oman in last two years (2017-2018) in refrigerated containers by sea. In regard to waste management, SoMeat at present burns most the stomach content of the animals for manure, which it sells directly to farmers who collect it from the slaughterhouse for use as fertilizer to their crops. By-products are also sold to small (mostly female) traders who in turn resell them in the city street markets. Slaughterhouse effluent is brought to a largepurpose-built reservoir far from the slaughterhouse. As of June 2019, SoMeat was still awaiting the UAE to inspect its facilities, so as to grant the company permission to export and seeking ISO 2200 international certification. It is also currently exploring exporting opportunities to Bahrain.
- *Hilibsan Meat Company*, established in the first half of 2019 to supply hygienic meat cuts to Hargeisa consumers owns land for a livestock holding ground (where animals are fed with grass and water and pre-slaughter health examinations are carried out) and uses its own refrigerated trucks for transportation of carcasses of animals slaughtered by its butchers at the Maandeeq abattoir. In its modern butcher shop with various cutting machines, large refrigerators for storage and refrigerated glass displays, it conducts modern meat cuts that are different from traditional cuts of butchers in open air markets, sells to retail customers and delivers to 15 hotels. In addition to fresh meat priced by weight, it also packages and vacuum seals meat for long-life. It currently employs 14 staff, including a veterinarian who performs pre- and post-mortem examination of animals and an Ugandan meat expert who trains the rest of the staff. Confident in the growing demand for safe and hygienic meat, the company also plans to open more hygienic butcher shops in different quarters of Hargeisa and expand its delivery business to restaurants and private residences.
- *Beira Food Processing Plc*, established in March 2019 near Galkayo, currently focuses on processing meat, beans and vegetables sourced locally or from neighboring regions to supply the local market with vacuum-packed, packaged beef stew, pasta sauce and beans and vegetable sauce, and also has plans for canning meat and exporting chilled meat. Currently, it employs 50 persons, and owns modern power, cooling, processing and packaging machinery imported from China and a food laboratory to handle necessary safety and quality controls.

3.7.3 The Hides and Skins Subsector

While hides and skins exports were before the war the third largest item in value, after livestock and bananas, recent production of hides and skins, a by-product of animals slaughtered for consumption, is comparable in to levels of the late 1980s, at about 7 million pieces a year. However, export earnings have dropped from a peak of \$8.7 million (annual average) in 2006-2014



(and \$5.2 million during 1986-1990) to US\$2.3 million in 2016 and 2017. Before the civil war various tanneries used to process the hides and skins through the wet blue stage, before selling them to local artisans or exporting them. The hides and skins trade and processing, however, collapsed in the 1990s. Currently there are a few traders who pick up the skins from villages and hides from slaughterhouses. Processing, however, has been minimal and limited today to sun-drying and dry-salting since the outbreak of the civil war and the total looting of pre-war tanneries, which used to produce wet blue leather. Domestic (rough) leather handicrafts (such as shoes and bags) by artisans is at presently negligible, largely confined to sandals and stools.

Hides and skins are a byproduct of animals slaughtered for consumption and the quality of hides and skins is directly affected by the health of animals and of their grazing environment, as well as by the flaying skills of the butchers and the storage practices of the traders (Figure 3.8 provides a representation of the hides and skins value chain, highlighting the different stages of product development, sourcing and marketing channels). Hides and skins are produced at the household level for small stock, and at a larger scale at slaughter facilities for camel and cattle hides. Small traders collect them from villages and slaughterhouses, where they are sun-dried, before being resold to wholesale traders, who cure them further, usually resulting in dry salted hides and skins before exporting them or selling them to the few artisans and tanning companies that have been operating in Somalia in the past two decades to cater solely to the domestic market for (rough) leather products. Market destinations for Somali hides and skins include Dubai, Pakistan, UAE, China, and Italy.

Hides and skins are today of poor quality, and improper processing, preservation and storage often leaves them damaged by vermin. Both the quality and supply of hides and skins is directly affected by the health of animals, the nature of the grazing environment (e.g. thorny bushes can damage the skin causing scars on it thereby lessening the quality of the skin), and by production practices. Seasonal or prolonged droughts negatively impact the industry as production of healthy and quality animals is reduced, thus yielding mostly hides and skins of inferior quality. This is underlined by the high rates of rejections by importers, because of poor quality of and damage to the hides and skins (Ombui, Mogoa, & Matete, 2014) and by the very low prices currently being fetched by producers, as low as US\$0.30 per hide or skin.²⁴

Box 3.2 - Chinese factory accused of poisoning Somaliland River water closed down.

A tanning factory, Jeronimo Group of Industries and Trading PLC, set up in the Spring of 2009 by a subsidiary of Chinese glove-making firm Phiss in the village of Da'ar Buluq, 60km outside the Somaliland capital Hargeisa, was the first foreign-owned tannery in the country post-war history. According to the company website, the factory processed 4,000 goat-and sheepskins from raw to wet blue and employed 60 local workers and 6 Chinese staff. It was shut down by the Somaliland government in 2014, following years of accusations by local people of the dumping dangerous chemicals and related health complaints.

A 2013 article in the Guardian newspaper reported that people living near the factory, had made numerous complaints about respiratory problems. A former worker at Jeronimo named Ibrahim said that one day, while mixing chromium compounds without a mask, he was overcome by the smell and fell down, hitting his head. "The company did not take me to the hospital," he says. "To this day I still have breathing problems." Other locals confirmed many health complaints had been made.

²⁴ These high rates of rejection and low prices still characterize the subsector to this day, despite that in 2017 FAO's ESOLT Project provided training on modern methods of hides and skin handling and processing to meat value chain actors and produced a training manual on Standard Operating Procedure (SOP) for improving the quality of hides, skins and leather products (both in English and Somali).



When the Guardian investigated the Jeronimo compound it found an unbearable smell, and workers with no face masks or proper shoes and sacks of corrosive material spilling onto the factory floor. Industrial waste [was being] dumped in local waterways, the company admitted, but it was adamant it had adhered to local and international rules governing the tanning industry. Livestock [in the area], which comprises up to 80% of local trade, had disappeared as animals refused to drink the water and their herders moved elsewhere, said local elders and government officials: "The water here was free; God-given. Now people have to buy it from travelling sellers. A 20-litre jerry costs US\$0.80, too much."

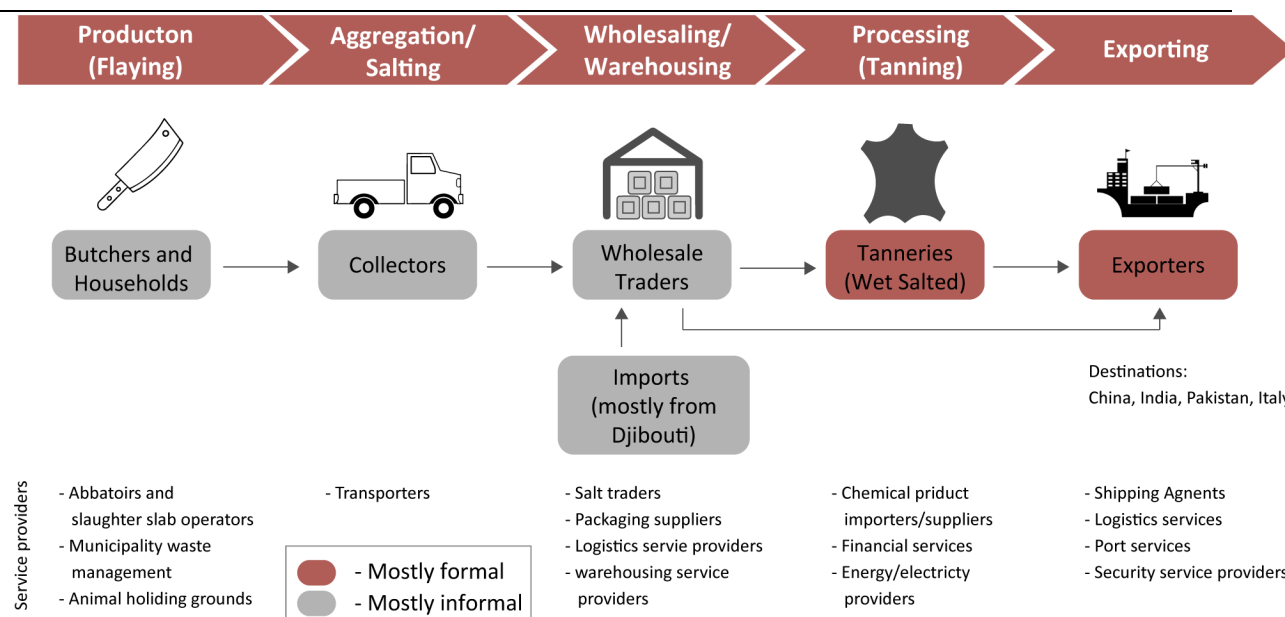
Foreign investments like Jeronimo [were and still] are seen as vital in proving Somaliland's worth as an independent nation, a point not lost on Da'ar Buluq's residents. "We've talked to ministers, deputy ministers. Each time our arguments are passed on to someone else," one village elder says. "The government considers that it is fighting a broader war internationally to attract foreign investors. So if this one is clamped down on it will have a negative impact on that." Somaliland's Chamber of Commerce secretary general, Ibrahim Ismail Elmi, said that while a seven-minister delegation was sent to inspect Jeronimo in 2008 [after its opening], the situation had been "under review" ever since. But another senior official, who asked to remain anonymous, added: "We suspect them [Jeronimo]. They use poisons and chromes, [which] are getting into the [dry] river [nearby]." Jeronimo had a \$6m agreement with the Somaliland government, the official claimed, that had to be paid in full should the firm be shut down. "We don't have the capacity to refund them, so we just give advice."

"If the government was worried about these health issues, they should have checked before we came," says Li Fai La, the factory manager. Rather than remove or recycle effluent, he said the company dumped industrial waste "in water 3km away from the factory." Li said that he had considered moving Jeronimo to neighboring Ethiopia. "You can go to the factory now," he says. "Yes, the smell is bad but trees are growing and there are fish in the water nearby." Although he did not specify what percentage of waste is dumped, the Guardian understood the factory had no waste management system in place.

Source: Sean Williams, in the "Guardian", published on Tue 2 Jul 2013

There are currently no modern facilities and tanneries in the country processing the hides and skins collected from formal slaughter facilities in towns and informal slaughters in villages to wet blue stage or leather, only traders curing them to dry salted stage and very poor-quality wet blue stage. The only post-war tannery, located between Hargeisa and Berbera and owned by Chinese investors, sustained operations for only a few years in the early part of this decade, as it was forced to cease operations by the local population enraged at the chemical pollution to soil and water sources nearby that it created (See Box 3.2).

Figure 3.8 – Hides and Skins Value Chain Mapping



Source: (Wanyoike F, 2018), simplified by World Bank for this report.

3.7.4 The Poultry and Beekeeping Subsectors

There are three distinct systems in Somalia for poultry production: (i) commercial intensive, which is rare and largely limited to the main urban centers of Mogadishu and Hargeisa; (ii) semi-intensive or backyard, which is practiced in most regional and small towns; and (iii) free range poultry production, which is common at the village level. No domestic production and consumption estimates exist, but poultry keeping is widespread among poor households, for both self-consumption and resale. Traditionally Somalis have raised village poultry for family subsistence and some poultry under the semi-intensive or backyard production system for local sales and have not bred chickens for meat or eggs on a commercially viable scale.

Commercial poultry farming, however, has been on the rise, particularly in urban areas, though almost exclusively to supply eggs to nearby towns. There are today 38 medium- and large-scale commercial poultry farms in the country, with 23 of those in the Banadir Region and surrounding areas. Intensive poultry production is highly capital intensive because it relies on (growing) imports of day-old chicks, drugs, vaccines and equipment. As a result of strong local demand by a growing diaspora community in larger towns, large remittance inflows since the civil war, and the rising price of goat and camel meat, chicken meat has become more widely available in urban markets and more affordable relative to other meats. Almost all chicken sold in Somalia on a commercial scale, however, is imported via Dubai, much of it frozen from Brazil, and most eggs are imported from Yemen, though volumes have been declining due to security challenges in the north of the country. All such imports take place without any quality control and health certification.

There is only anecdotal evidence about beekeeping, which remains a small-scale side business for most beekeepers. Until recently there were only some small beekeeping operations in rural areas throughout the country, and especially concentrated in the beekeeping belt of Bay and Bokol regions of South West state, which employ traditional hives and honey harvesting techniques. Recently, however, Hormuud (the largest telecom company in Somalia) established commercial beekeeping operations in the Tiye glo area of Bokol region and several other areas of South West state as well as in Dhusamareb in Galmudug state. Consumption is still limited to households and local markets.



3.8 Sector Policy Framework

Somalia's paramount development framework is the National Development Plan (NDP) 2017-2019, the first development plan crafted by a Somali government since 1986. The NDP stipulates the country's short and medium term strategic direction, development priorities and proposed implementation mechanisms, including the use of development aid. Its main theme is to accelerate the country's socio-economic transformation in order to achieve the stated objectives for poverty alleviation, economic revival and societal transformation in a socially just and gender equitable manner. The NDP includes a specific chapter on livestock, setting out a vision and related interventions to support the delivery of essential services necessary to achieving a vibrant and commercially competitive sector, sustainable livelihoods, and inclusive economic growth. This chapter set out four main operational goals: (i) strengthening institutional capacity and public infrastructure relevant to the livestock sector; (ii) increasing animal health services and the productivity of small and large livestock sector producers; (iii) enhancing trade in livestock and livestock products; and (iv) improving national food security, nutrition, and food safety within the livestock sector.

Box 3.3 - Status of the 2016 Veterinary Law Code

Before 2016, the legal framework for registration and licensing of veterinary services providers were based on the National Veterinary Code Law No. 20 of the 27th June 1967 and subsequent Ministerial decrees of 12th November 1969, 11th November 1968 and 15th September 1985. To foster compliance with modern food safety standards and requirements by the World Animal Health Organization (OIE) in its Animal Health Terrestrials Code, Chapter 5, as well as by the WTO and foreign governments and private sectors concerning import/export procedures, the FGS adopted the new Veterinary Law Code No: 21 of 27 June 2016.

Since then, the MoLFR has been evaluating for renewal all licenses previously issued to livestock service providers, traders and processors. Covered providers include those engaged in disease notifications and investigations; dairy processing plants and markets; slaughterhouses and meat markets; live animals, meat and hides and skins export trade and certifications; quarantine management; livestock-related agribusinesses (dairy and poultry farms), and private veterinarians. for delivering good services which comply with the new standards and measures required by the OIE, WTO. SPS and the requirements of the importing countries.

The new veterinary law code is already under enforcement and MoLFR's and states' inspection officers are instructed to apply strictly the articles and provisions of the new veterinary law code before issuing any health or inspection certificates for livestock and livestock products intended for local consumptions or export. Weak capacity, low budget allocations, and in the southern regions of the country also the current patchy security situation, however, pose the main challenges for the enforcement of the Veterinary Law Code.

Furthermore, the FGS and FMSs have endorsed a number of policies to guide the development of the livestock sector. Among these, the most important nationwide policy document is the *Livestock Policy 2015-2020* of the MoLFR (Somalia 2015). This document presented the FGS' vision for the sector, spelled out the public and private sector core functions and addressed issues and needs critical for the sector's development related to livestock health, trade, land degradation and range management, institutional framework and coordination, and infrastructure. An updated draft *Somali Livestock Development Policy* was also prepared in February 2017 (by Dr. Abdirahman Nur Qeiley), but was not formally adopted by the ministry (Somalia 2017). Among other recommendations to support production levels, it reaffirmed the longstanding prohibition under the Veterinary Law Code on export of female animals. In addition, the Ministry of Finance has been implementing a policy of



granting exemptions from customs duties on imported machinery and supplies needed for agricultural production and agro-processing, including for the livestock sector, after validation and endorsement by the relevant sector ministry regarding the type and purpose of imports for which the exemption is requested.

Other key analytical, policy, and legal documents endorsed or formally adopted by the FGS and relevant to the preparation of the Livestock Sector Development Strategy (LSDS) are listed below.²⁵

- FAO/World Bank/EU, *Somalia - Towards a Livestock Sector Strategy* (Final Report, 2004) – a seminal, detailed, and comprehensive analytical and strategic document to assist the Transitional National Government (TFG) in mobilizing and directing then-expected foreign development assistance;
- World Bank, *Somalia - From Resilience towards Recovery and Development* - a Country Economic Memorandum for Somalia, and particularly the livestock-specific sections of chapters 5 and 8 that assessed the sector's historical performance, challenges and needs;
- World Bank and UNDP, *Somalia - Joint Needs Assessment 2006 (JNA)* and especially its Productive Sectors Module, which analysed and presented the country's development challenges and needs (including those of all productive sectors, and first and foremost the livestock sector) towards the attainment of the Millennium Development Goals (MDGs) by the year 2015;
- FGS, *National Veterinary Law Code 2016* – which guides the sector to conform to current technology and international trade regulations such as those of the World Organization for Animal Health (Office International des Epizooties - OIE) and the Sanitary and Phyto-Sanitary (SPS) Agreement of the WTO;
- World Organization for Animal Health, *Terrestrial Animal Health Code*, 2011, which sets out standards for the improvement of animal health and welfare and veterinary public health worldwide, including through standards for safe international trade in mammals, reptiles, birds and bees and their products;
- FAO/OIE, *Global Strategy for the Control and Eradication of PPR*, 2015;
- WTO, *SPS Agreement*, 1995 and its requirements on trade in agricultural, including livestock, products;
- AU/IBAR, *various recommendations* for disease control policies and strategies in East Africa
- FAO/IAEA, *various recommendations* on accreditation regarding sanitary measures of inspection and certification protocols and implementation procedures for export/import of livestock/livestock products;
- FGS, *Somalia - Drought Impact Needs Assessment* (2018), including its findings and recommended short-, medium- and long-term interventions for the livestock sector's the recovery and resilience, prepared with support from the World Bank and U.N. agencies, including the FAO; and
- World Bank and FAO, *Somalia - Rebuilding Resilient and Sustainable Agriculture* (2018), including its findings and recommendations for the livestock sector's development.

²⁵ Beyond such federal and nationwide analytical and policy documents, the Somaliland and Puntland governments have also elaborated with external assistance sector guidance and planning documents as well as laws (e.g. for Somaliland: 2015 Dairy Act; 2016 Veterinary Code; 2017 Animal Health Strategy; 2017 National development Plan II 2017-2021; etc. etc. For Puntland: 2010 Meat Inspection and Control Act; 2010 Dairy Policy; 2015 Dairy Industry Act; 2017 PPR Eradication Strategy; 2017 Animal Health Strategy; etc. etc.). Other states are also in the process of preparing and adopting own regulations.



3.9 Overview of Main Development Challenges for the Livestock Sector

Despite the recent performance through the mid-2010s in livestock export volumes and earnings, Somalia continues to face several major constraints and challenges to the development of its livestock subsector. This section summarizes the main constraints to the development of the sector, as identified by the federal government in its most recent policy, strategy and planning documents (Somalia 2015; 2017; and 2016) and by public and private stakeholders during the recent consultations over the sector's strategy. These challenges (as well as opportunities and recommended interventions) are discussed in more detail for each value chain/subsector in chapter 5 and 6 and for cross-cutting issues in chapter 7 of the LSDS. Many traditional structural constraints associated with livestock production have intensified in Somalia recent decades, including:

- water shortages, due to insufficient and irregular and inadequately harvested rainfall, affecting livestock health and survival, and availability of fodder;
- lack of access to good quality pasture, due to land degradation and reduction of productive land, which is in turn the result of deforestation, soil erosion, invasion of unpalatable plant species, spread of private enclosures, and increasingly frequent recurrence of natural shocks such as drought and floods;
- lack, or poor quality, of fodder at export ports, along transit routes to them, and around urban centers;
- seasonal fluctuations in market prices for livestock and livestock products, due to the factors listed above and to the lack of storage facilities;
- transboundary animal diseases with occasional flare-up of epidemics, other livestock diseases, and pests, due to poor and inadequate animal health services and laboratory facilities;
- overreliance on only a few export markets and on live animal exports;
- persistent transport and market infrastructural and logistical weaknesses, which undermine delivery of high-quality animals to domestic and foreign markets and expansion of dairy production and processing;
- persistent insecurity in large parts of southcentral Somalia;
- weak institutional framework, including limited and largely informal organization of the livestock value chain, lack of coordination and integration among its actors, and poor linkages and ineffective negotiations and agreements with its main trading partners that also sharply limit the country's potential in the livestock commodities export trade;
- absence of necessary legislation and lack of enforcement of existing rules and regulations;
- limited availability of quality veterinary education and technical training facilities in pastoral areas; and
- financial constraints – with very low government budgetary allocations to livestock and lack of private investment in processing of livestock products (dairy, meat and hides and skins mainly) for value addition, partly due to poor access to credit and lack of incentives and encouragement by governments.



4.0 SECTOR DEVELOPMENT STRATEGY: VISION, PRIORITIES, STAKEHOLDERS

4.1 Overall Vision for Development of Livestock Sector

The vision is for Somali livestock sector by 2030 to make a major contribution to: i) better nutrition and enhanced food security of its entire population via an expanded supply of high-quality and hygienic animal proteins; ii) to improved livelihoods of the majority of the rural population via higher incomes from livestock products; and iii) to the country's economic growth, via more efficient, inclusive and environmentally sustainable production, improvements product quality, and expanded value-adding processing. This general vision for the sector, as well its development potential, objectives, principles, and priorities were already outlined in the Livestock Policy 2015-2020, were embedded in the NDP 2017-2019, and were reconfirmed by public and private stakeholders during the October-November 2018 consultations. It is thus a shared vision by the federal government, the federal member states, and the sector's other public and private stakeholders.

Despite myriad challenges, the livestock sector has the potential to become in the medium- and long-term resilient to climate shocks, to ensure the safety and quality of all its products, to reach self-sufficiency in dairy and poultry, and to export a larger surplus of higher value-added meat (frozen, and possibly also chilled, meat carcasses and cuts) and hides and skins. Underwriting this big potential across all sub-sectors are a growing local population and returning diaspora, increasing global demand for animal source foods, very low current production and productivity levels, and minimal processing.

Commercialization and climate shocks are key drivers of wealth differentiation in pastoral areas and thus require a pro-poor approach to the sector's development. Selected studies of some pastoral areas in the Horn of Africa have found that wealthier pastoral household are not only increasingly owning larger numbers of animal to the detriment of poorer households, they are also more actively engaged in their commercialization, and therefore benefitting more from export trade. Not only would a policy of promoting commercialization likely have the biggest positive impact on the already more commercial, and therefore generally better-off, households, but it can actually be a factor leading to further gradual distribution of livestock from the poorer to the wealthier households. Such a shift leads to various changes in terms of resource access and social behaviour that over time make it more difficult for poor pastoralists to build herds (Catley, 2017). Climate shocks also facilitate a transfer and consolidation of animal stocks from poorer households to wealthier households. Therefore, an important implication of this research focused on the pastoralist production systems is that a pro-poor approach to pastoral development should go beyond promoting only livestock commercialization and should focus also on preventing avoidable losses and better drought management, enabling poor herders to re-build and expand their herds and their incomes, and ensuring access to markets and rangelands (Catley, 2017).²⁶

4.2 Operational Objectives and Priorities

Toward achieving such a vision and realizing the sector's potential, this LSDS sets out a number of operational principles and objectives to address the main constraints that prevent its optimal

²⁶ In agro-pastoral and mixed farming production systems livestock also plays an important role in household income and resilience. When considering livestock development from a poverty reduction perspective, the role of animals in such systems should be better understood and documented.



performance (the constraints will be discussed in more detail in Chapter 5, 6 and 7 under each sub-sector or cross-cutting issue below). The principles to apply to institutional strengthening, capacity building, policy- and rule-making, strategic planning (as for this LSDS), sector management and project implementation by federal, state and local governments, as well as to the support of external development partners, are: consultative, participatory, and transparent processes and mechanisms; interactive, inclusive and evidence-based decision making; and consensus building with community and private sector ownership and buy-in. The LSDS operational goals are:

1. ***Preserving Animal Resources*** – Safeguarding the diversity and optimizing the use of existing animal resources is critical for both resilience and growth. Key priorities identified in recent government documents and during the October-November 2018 stakeholders consultations include:
 - rehabilitating the deteriorated environment and especially the rangelands for grazing;
 - reducing the prevalence of transboundary and infectious diseases, with improved management and control of animal diseases and pests in line with relevant international standards;
 - increasing the sector's resilience to climate change, by easing the chronic shortage of water and developing a commercial feed and fodder subsector; and
 - conserving and improving the available genetic resources available for the livestock sector, both plant and animal, including introduction of exotic stock where appropriate and promoting technically and financially recent private ventures on artificial insemination.
2. ***Improving the management of the livestock industry*** - For growth of the sector, it is also critical to ensure optimal productivity, including for the rapidly growing dairy and poultry farming subsectors, and more efficient, less costly, and expanded domestic and export trade;
3. ***Ensuring quality and hygienic livestock products in domestic and international markets*** – Food safety is needed to protect the health of domestic consumers as well as to regain lost export markets and expand into new ones;
4. ***Increasing the processing of livestock products*** – The sector's growth is dependent on higher value addition, as the current pattern of supplying domestic and export markets with live animals and minimally processed products has reached its limit;
5. ***Supporting home-grown research institutions*** - Investment in analyzing and offering solutions to current and emerging problems is necessary for both resilience and sustainable growth of the sector; and
6. ***Ensuring security throughout the country, but especially in affected areas in southern Somalia*** - Without enhanced security in the many areas which governments and external partners cannot yet access, any public and private operation for rehabilitation and development projects and for expanded and new business ventures would remain challenging and costly, thus attracting limited donor funding and private investment.

The livestock sector's overall priority development needs, further elaborated in the next sections of this chapter as well as in the following chapters, are:

- Significant capacity building of public sector institutions, including on policy formulation and review, planning, coordination and implementation of emergency and development programs and projects, drafting and enforcement of laws and regulations, ensuring quality of veterinary and animal husbandry education, and data collection and analysis (via both an animal population and production survey on the stock/output side and a feed assessment on the input side); and



- Large public investments for expansion of public services (with emphasis on strengthening veterinary services, including disease control, meeting international sanitary and phytosanitary standards and ensuring veracity and credibility of health certifications), and for the rehabilitation and expansion of critical infrastructures such as for water catchment and road transport (Malabo Declaration); and
- Adequate security and access to finance for the private sector to expand its currently minimal processing of livestock products.

Other medium-term needs for the sector's development are:

- addressing the many and vast knowledge gaps;
- adopting a modern commercialization and market orientation mindset, including for livestock owners;
- requiring external assistance to move towards a resilience (plan and prepare for shocks) and development (inclusive and sustainable growth) focus, while also ensuring emergency support in crises;
- fostering and adopting appropriate and innovative technology;
- clarifying the respective roles and responsibilities for the public and private sectors in policy formulation, planning, and implementation of projects, and between and among the federal and state livestock, forestry and rangeland ministries and agencies;
- ensuring improved coordination and collaboration on all development activities; and
- devoting adequate resources to the sector.

4.3 Stakeholders

This LSDS covers all public and private sector participants in the livestock economy, from policy and rule makers to enforcers and from producers to consumers. However, it pays particular attention to the inclusive nature of the strategy and its impact on marginalised groups. Public sector bodies that are key stakeholders in carrying out this sector strategy include federal and state livestock ministries and affiliated institutes and research bodies. Private sector stakeholders that are critical in-service provision and investment decisions include all livestock producers, input and service providers, processors, exporters and importers of live animals and animal products, and consumers that participate along the various livestock value chains. Some of these, such as private veterinarians and large livestock traders, are also actively engaged as members of professional associations that deal with livestock animal health services or with domestic marketing and exports. Other stakeholders include external development partners bringing lessons from international experiences, sharing best practices, and providing needed financing for public investment, and to private investment.

Previous livestock sector studies and policy and planning documents, however, have almost exclusively focused on the technical challenges to production, value-addition and commercialization for both public and private stakeholders within the sector. A good understanding of the social, economic, cultural and traditional characteristics of the communities that are the intended or unintended targets and beneficiaries of sectors activities, public interventions and private investments is an important pre-condition for the relevance, effectiveness and positive long-term impact of this sector development strategy.

There is a large gap in current knowledge regarding the socio-economic and cultural characteristics of the livestock sector's current stakeholders, potential participants, their dynamic and complex communities, as detailed socio-economic data and analyses on these households and their



livelihood strategies are very limited. There is an even deeper gap regarding intermediate and final consumers of livestock products. This knowledge gap poses a big challenge to policy and decision-makers in assessing the implications of their recommendations and plans of action to deliver on key development priorities including poverty reduction, food security, nutrition, resilience, gender and socio-economic distribution.²⁷

4.4 Overall Sector Governance

The livestock sector falls under the responsibility of the Federal Ministry of Livestock, Forestry and Range (MoLFR), with the engagement of the federal members state level livestock ministries. These key government institutions have linkages with various public and private research, training, monitoring and implementation bodies, professional associations, and civil society organizations that provide key services, such as education, training, monitoring of weather, water, soil, animal population, production and trade, animal health services, technical advice, and project implementation.

This LSDS supports the creation of a sector co-management framework and the collaboration of public and private institutions to deliver animal health and veterinary services, disease and pest management and control, natural resource management, market and trade infrastructure investments as well as promoting commercial investment in processing and value-addition. However, the current ad-hoc and unclear division of institutional responsibilities between the federal and state levels impedes needed coordination among key public stakeholders and needs resolution as a matter of priority. Furthermore, lack of clarity and of operational and supervisory arrangements for public-private partnerships (PPPs) impedes needed coordination among key stakeholders, resulting in low investment and poor results. For example, the export of live animals is currently hindered by the existing quarantine arrangements, which are based on privately owned and managed quarantine/holding facilities. These facilities pay veterinarians who are accredited by the FGS but formally under the control of the Somaliland and Puntland state governments to issue animal health certificates. Such arrangements provide for insufficient accountability, transparency, and credibility of the process.

In order to make these and other PPP arrangements—which include municipal-owned but privately-managed slaughterhouses—effective, efficient and accountable, the roles and responsibilities of both public and private agents must be clearly defined and agreed, to enable each partner to perform his/her respective duties. Currently, this is not often the case. In addition, the rehabilitation and improved management of holding grounds and other infrastructure have to be clearly delegated to either public or private stakeholders, providing clarity on investment and maintenance responsibilities. Comprehensive analyses of the cost of compliance and its implications for the various stakeholders involved are also needed as a basis for improvements on current arrangements.

Consultative and participatory processes and mechanisms require that livestock producers and other stakeholders have a voice through mechanisms such as representative producer/pastoralist associations/ cooperatives, civil society organizations or other welfare groups. Currently the stakeholders

²⁷ The FSNAU over the past two decades and the World Bank in the last year has gathered some data and has also conducted some general analyses related to these dimensions, but further data collection and analyses specifically related to the livestock sector by the government in close collaboration with these institutions are needed.



formally organised are the traders/exporters association and the livestock professional association, as mentioned at end of chapter 3. It is important to understand and discuss further their respective roles and to assess to what extent they already contribute or participate in decision-making and implementation. Nonetheless, in order to ensure participatory and inclusive development of the sector, awareness, capacity, power and resources of these organizations need to be strengthened with external support, so that they can participate in policy dialogues, influence policy decision-making processes and help the public sector deliver key services.

4.4.1 Capacity Needs: Federal Ministry of Livestock, Forestry and Range

The federal Ministry of Livestock, Forest and Range (MoLFR), one of the oldest and strongest governmental institutions in Somalia, has major gaps and capacity needs in order to meet its mandate. MoLFR's mandate is "to create policies, regulations and frameworks as well as provide oversight of livestock development, conservation of forests and range management to promote sustainable economic development and food security in the country".²⁸ Approved by the cabinet in April 2016, the Ministry's work plan (2016-2019) in the National Development Plan (NDP) has the following agenda, to:

- Strengthen institutional capacity and infrastructure of the ministry and the livestock sector;
- Improve livestock health services and increase productivity of small and large producers;
- Enhance livestock and livestock products trade (in local, regional and international markets);
- Improve national food security, nutrition and food safety systems in the livestock sector;
- Establish a system for range, forest and wildlife management;
- Foster South-South knowledge transfer with Somali institutions in aspects such as natural resource management and climate-smart technologies;
- Restructure and reform the current organization of the ministry, and the way it functions, in terms of both hierarchy and internal procedures (a task for the MoLFR's Director General).

The ministry comprises of six departments: Forestry and Range, Animal Health and Public Veterinary Services, Animal Production and R&D, Administration and Finance, Quarantine, Planning and Training. Capacity and level of operation vary across departments, but all are negatively impacted by insufficient policies in the livestock sector, weak internal systems, and inadequate infrastructure and resources, including staffing, technical expertise and knowledge, materials, logistical tools, supplies, etc.

There is also weak information management system and archives, nor a research unit, which limits innovative proposals for livestock development. Furthermore, other resource constraints include basic office needs; for example, all heads of departments use their private laptops and phones. Due to limited available budgetary and external funds, there is also an absence of external advisors or experts. Insecurity in various parts of the country hinders transportation of staff for monitoring, coordination with the states, and service delivery.

²⁸ This section summarizes the findings and recommendations of two studies that have assessed the organizational and capacity needs of the ministry. One of these studies was commissioned by the MoLFR to undertake a comprehensive capacity assessment on the functions of departments within the ministry, supported by AU-IBAR. The other study was conducted under the Capacity Injection Mechanism (CIM), an initiative of the World Bank to enhance efficiency of public service delivery in the country.



The MoLFR currently lacks the resources and human capacity, including highly qualified and experienced technical advisors and consultants, to establish and support badly needed research institutions that can provide livestock producers with modern breeding products to strengthen the animal population's resilience to fast changing environmental conditions. The same applies to agencies that would provide animal health and disease control services, disease monitoring mechanisms, and adequate data for planning and investment. For instance, neither the public nor the private livestock laboratories in the country have the modern equipment necessary to conduct credible analysis of various specimen for timely information to combat diseases and for assuring importing countries of the health of livestock shipments from Somalia.

Capacity needs to be built by increasing budgetary resources, improving knowledge and skills (including IT skills) of individual staff members via training, and simultaneously enhancing organization capacity. The ministry needs to formalize and establish mechanisms for internal communication, as well as basic systems and processes like effective information management, office conduct, management, monitoring, and accountability. For example, there is need to sustain and extend the collection of the production and price data that was undertaken under the ESOLT Project (May 2015-March 2018), which established a Livestock Market Information Management System in many areas of the country, including of data on disease monitoring. Finally, the ministry reorganization requires clarification of the roles of heads of departments and of the Director General, and availability of external advisors providing technical assistance.

There is also an urgent need for the federal ministry to enhance its role in regulating and managing the rangelands and other forestry resources, including the strengthening of mechanisms for dealing with land rights and disputes and enhancing the availability of water and nutrition sources for livestock. Developing federal authority over such areas will be a critical factor to ensure rehabilitation of the natural resource base for the nomadic pastoralism that Somalia's livestock sector depends on and for minimizing communal and inter-regional and inter-state conflicts.

4.4.2 Capacity Needs: FMS Ministries of Livestock

Institutional capacity needs and challenges across Somalia's federal member states vary greatly, and especially between the long-established northern states and the more recently established states in the southern part of the country, where budgets, physical facilities and staffing of their ministries of livestock (in most cases also responsible for rangelands and forestry) are still minimal. This section summarizes the key capacity challenges and priority areas highlighted in the capacity self-assessments of the five state ministries (Galmudug, Hirshabelle, Jubaland, Puntland, and Southwest) that responded to a semi-structured survey conducted as part of the October-November 2018 stakeholders' consultations on the LSDS (more details are in the annexes that summarize the findings from the consultations with each state's stakeholders). It also includes information about Somaliland gathered during the state-level consultation in October 2017.

Among the states surveyed, Somaliland's and Puntland's livestock ministries are the only ones reporting having some vehicles and own office facilities, and receiving both a budgetary allocation, albeit small, and some external financial assistance for both recurrent expenses and projects, and thus being able to maintain a fully paid ministry staff. While no budget allocation estimates were made available by the Somaliland ministry, of its 433 total staff, 76 or 18 percent are female, and 239 or 55 percent (including 52 female staff) are based in Hargeisa, with the rest in regional offices. The 2018 budget for the Puntland ministry was \$206,148, of which \$95,280 (about 46%) paid out to staff. The ministry received US\$268,080 in external assistance. All 107 staff members are paid,



including the Minister, Minister of State, Vice Minister, Director General, four department directors, and 84 professional technical staff (11 of whom are being paid from a World Bank-funded project). While Puntland is offering some basic services to livestock producers and traders, however, it offers none to other livestock products' stakeholders (such as dairy farmers or butchers).

The other four states all lack physical infrastructure and receive minimal or even no budgetary resources and external project funding to support functioning livestock ministries. The Jubaland ministry, at the better end of this group, has no vehicles nor supplies and is currently housed in shared, cramped building, but will soon move into a new, modern building. It also received no external assistance in 2018, but had a budget allocation of US\$80,288, of which US\$34,680 (about 40 percent) is paid out to the Minister, Minister of State, Vice-Minister, DG, and a veterinarian. Among the three states in the earliest institutional building phase, the Galmudug ministry with 70 staff also received no external assistance and a total budget of US\$67,430, from which only 12 staff (mostly veterinarians or veterinary technicians) are paid. Southwest listed 47 staff and no budget for the past three years, with only a staff member, the Director General, being paid from external sources. In Hirshabelle, the ministry has no infrastructure at all and received no budget or external assistance, with all its 25 staff working as unpaid volunteers.

Furthermore, all these states' ministries cite lack of information, knowledge and skills as major constraints to their ability to deliver services. Thus, their institutional capacity needs extend widely from dedicated office space to vehicles, from budgetary and external resources for paying staff salaries to covering the cost of necessary supplies (including fuel) deliver basic services and undertake simple development projects, and from access to information and knowledge to professional training.

Lack of monitoring, critical facilities, and service provision related to animal health and food safety standards is another major challenge for most states, as they affect negatively the marketability (especially for export) of the animals raised or transiting through their states, the health of consumers, and the development of higher-valued processing. Hirshabelle, for example, has no centers to monitor and no clinics or emergency teams to control outbreaks of diseases, let alone deliver any other animal health care services. Other states struggle with compliance with international regulations related to livestock trade, including limited or no availability of animal health services and both institutional and technical limitations in health inspection and certification processes the major export ports.

To meet the states' goals of effective disease surveillance and adequate health and food safety, all states need strengthening of their public veterinary services, including via training of Community Animal Health Workers (CAHWS), replacing aging veterinarians with suitably educated young ones, and manning health emergency response teams. In terms of facilities to support the implementation of both preventative and control animal health measures, states need functioning and adequately equipped laboratories, stocking and delivery of vaccines to control transboundary diseases, rural clinics, vehicles, and other physical facilities and supplies for monitoring and extension services. They also listed technical and financial needs for external support to establish forest and range reservation lands and sustainable range land management, including the development and adoption of a uniform policy and legal framework to control the migration of livestock and the expansion of private enclosures on communal pastoral lands (emphasized by Galmudug and Jubaland).



There is also a general inadequacy across all states of wholesale and retail hygienic and modern market facilities for live animals and main livestock products such as milk and meat, with Hirshabelle for instance citing no market facilities at all in the entire state. Thus, capacity needs cited across states to support domestic trade include training of stakeholders along and infrastructures and equipment for the main value chains (milk, meat, hides and skins, and poultry), more inspectors for enhancing livestock transport, and the improvement or new development of wholesale (such as milk collections centers) and retail market facilities under public-private partnerships. Puntland emphasized also the need for legislation and regulations, their enforcement, and building of cooperatives. Galmudug highlighted issues with extensive, recurrent droughts and listed the need for projects such as drilling water wells, development of water catchments, and even the development of wind and solar renewable energy.

Table 4.1: Private Sector's Primary Expected Roles and Activities in Livestock Sector

Value Chain Segment	Activity/Service
<i>Primary production</i>	➤ Animal production systems management ➤ Animal care ➤ Breeding
<i>Processing</i>	➤ Home processing (sour milk, skins, ghee, etc.) ➤ Slaughter house and storage investments, management and services ➤ Investment in food and tanning processing facilities and infrastructure ➤ Animal-related product development (e.g. dairy and hides and skins) ➤ Establishing accredited food testing laboratories ➤ Animal feed production and milling ➤ Extension
<i>Trade</i>	➤ Investment and management of livestock marketing facilities ➤ Marketing of livestock and livestock products ➤ Transport services ➤ Export of meat and live animals ➤ Sale of animals at primary, secondary and tertiary markets ➤ Import and marketing of animal feed
<i>Veterinary services</i>	➤ Veterinary clinical services (diagnosis, treatment, surgery, prophylaxis, vector control, etc.) ➤ Importation, distribution, storage and retailing of veterinary drugs, chemicals and equipment ➤ Management of herd health ➤ Participation in veterinary extension services ➤ Artificial insemination
<i>Animal Production</i>	➤ Production programs ➤ Artificial insemination ➤ Natural Resource Management ➤ Feed and fodder production ➤ Extension
<i>Financing</i>	➤ Provision of value chain and micro-financing
<i>Consumer Education</i>	➤ Demand for safe and quality products

4.4.3 Private sector: expected responsibilities

Private sector individuals and entities are actively engaged at all levels of the livestock value chain. These include production, processing, domestic and international trade, veterinary services, provision of inputs such as veterinary drugs and animal feed, and financing. The relatively stable situation in Somaliland and Puntland and the return of relative peace and stability in Mogadishu,



Kismayo and surrounding milk catchment areas is attracting new investment from local and diaspora investors. Somaliland has recently developed an investment strategy for its private sector, while Puntland is in the process of upgrading its Development and Investment Authority and establishing a one-stop-approach for private investors. In Mogadishu, the Ministry of Planning, Investment and Economic Development is also in the early stages of establishing its one-stop-shop. Various challenge and capital venture funds have also been/are supporting Somali business start-ups and expansion, including commercial dairy farms and some processing business ventures. However, issues such as high risks related to political instability and in some areas insecurity, high cost of electricity, poor access to commercial financing, and the lack of well-developed, all-year-round supply systems make the economic risk of such investments considerably high (SATG and UNDP Somalia, 2016).

The roles and functions of the private sector, including in PPP opportunities, are envisaged to remain as listed in the 2015 Livestock Policy, including in, but not limited to, the activities listed in Table 4.1. In delivering these functions, the private sector will be expected to operate within the regulatory environment provided by government. In many areas, the public and private sectors need to work closely and share important information with each other, in order for government to deliver effectively the required services and for the private sector to receive the support it needs, such as in animal health, food safety and related to bilateral commercial agreements with foreign governments or private importers.

4.4.4 Gender, youth and disadvantaged groups

Particular attention in the design and implementation of this sector strategy will be given to women and youth, whose engagement in livestock trading and marketing has been traditionally at the local and subsistence levels and whose equitable benefits and participation should be ensured. Women and youth bear substantial responsibilities in livestock production, though Somali society maintains clearly delineated gender roles and responsibilities. Men and boys are responsible for most activities related to camels and cattle, including buying, owning, grazing, milking, slaughtering, and selling them. Women and girls are responsible for most of these same activities related to small stocks (goats, sheep and household poultry) as well as for the sale and processing of most livestock products. They control at least 80 percent of milk production and virtually all of the milk wholesale collection and retail trade. Men and women are jointly responsible for sheep and goats trade and export, with men typically handling the major livestock trade for export.

Gender roles, however, have been shifting as households are confronted by crises and shocks, as men and women adopt different coping strategies to increase household resilience. Women's engagement in a range of income generating activities, including livestock trade, value addition and product business opportunities, has increased. Though the diversification of livelihoods may contribute to household resilience and women economic empowerment, trade-offs occur as women are facing increasing workloads and responsibilities. This impacts women's availability for reproductive and caring tasks and may negatively impact children's nutritional status and well-being, as they may be left in the care of others for substantial periods.

Limiting factors for inclusive and gender-sensitive sector development include inadequate awareness among politicians, officials and local elders on the need for gender mainstreaming, lack of gender-disaggregated information, and limited technical knowledge and skills and access to technical higher education by women. To enhance the income-generation activities of women and youth, awareness raising should be complemented by promotion of appropriate technologies and by



their participation and empowerment in decision-making, as well as by enhancing their access to resources, including finance and practical information.

Youth participation in the sector is a necessary condition for long-term development and stability, starting with education.²⁹ and the improvement of technical skills and knowledge needed to upgrade and promote sector growth. The live animal export value chains provide the bulk of the employment opportunities, including small-scale traders, brokers, large scale traders, as well as trekkers, feedlot operators, lorry drivers, loaders and others providing ancillary services. Large additional employment opportunities, however, lie in the value addition and processing of livestock products (outputs and inputs). The expansion of dairy, meat, and fodder production offer substantial income generating opportunities for women, as they are already involved in many tasks at different levels of the value chain (from production to sales). Besides the number of jobs, the quality of the employment (decent work) and the per capita earnings from these jobs should be considered. For many of those who have moved to (peri) urban areas and found new and different primary livelihoods, additional support is required to integrate some livestock keeping activities in their livelihood portfolio.

²⁹ A section covering the status of and recommendations for livestock-related education and human resources is in Chapter 7.



5.0 PRODUCTION AND MARKETING: CONSTRAINTS AND RECOMMENDATIONS

Several livestock subsectors are found in Somalia, varying in their relative economic importance among regions, as determined by factors such as the natural resource endowment, the availability of water, pasture and labor, and the sizes and types of livestock raised. The main value chains relate to ruminants (i.e., camels, cattle, sheep and goats) and to two critical inputs for their production and marketing (i.e., animal health services and feed/fodder). Their main products (milk, meat, and hides and skins) and two other emerging value chains/subsectors (poultry and beekeeping) are covered in the next chapter.

5.1 Strengthening animal production systems

Livestock production in Somalia is constrained by several factors linked to the nature of traditional production systems. These constraints include: (i) shortages of water, due to insufficient rainfall harvesting facilities; (ii) inadequate availability of natural feed and commercial fodder (to be expanded in a later section); and (iii) high incidence of diseases. The latter is mostly due to insufficient veterinary services and oversight, also to be expanded below. Moreover, resilience of animals to climate shocks remains low, partly because of the (iv) degradation and deforestation of the rangelands, and partly because (v) the livestock population increased greatly through the mid-2010s until the most recent severe drought, putting additional stress on the rangelands. Other systemic physical constraints to live animal trade are: (vi) inadequate and, in many cases, dilapidated, transport infrastructure which are increasing both cost and time for moving livestock from production areas to domestic urban markets or transshipment ports; and (vii) lack of adequate resting areas (including pre-quarantine holding grounds) with water, pasture and health/vaccination facilities for the large number of livestock in transit between the major livestock collection centres and the export ports. Another set of institutional constraints and gaps is related to (viii) the lack of and/or inconsistent implementation of regulations and policies, and (ix) insufficient knowledge and data about animal population, offtake rates, incidence of diseases, and domestic consumption, among others.

Somalia conducted its last national census of livestock in 1975, and subsequent animal population and offtake estimates were based on limited surveys and extrapolations. Thus, new reliable information on the livestock sector is badly needed. Box 5.1 describes current plans for a comprehensive aerial and ground animal population survey to establish a reliable new baseline and the foundation for a credible livestock statistical system, to guide the Federal Government and Federal Member States and their development partners in planning and implementing effective livestock interventions.

While local breeds are well adapted to the country's harsh geographical and arid climate, poor breeding practices are a factor contributing to their low live weight and productivity, i.e. milk and meat yields. The traditional community-based breeding system, in which the best animals were exchanged for breeding purposes, has been negatively affected by the move toward more commercial production systems, in which animals with larger frames, and therefore heavier body weights, are sold for slaughter or live export, because they attract better prices. Apart from the very limited introduction of improved exotic dairy cows in some peri-urban areas, there have been no major interventions in Somalia on breed improvement.



Experiences from similar countries such as Ethiopia, however, shows that the extensive pastoral production systems in challenging arid and semi-arid lands pose a big challenge for the feasibility and adaptability of improved exotic breeds and hybrids. In fact, centralized genetic improvement schemes, through importation of improved breeds in the form of live animals, semen, or embryos and crossbreeding have not succeeded much and, even if successful in the near term, sustainability remained a big challenge. Experiences in neighbouring countries clearly demonstrate the benefits of such community-based breeding programs. In Ethiopia the community-based breeding program (CBBP) for small ruminants has been piloted by the International Centre for Agricultural Research in Dry Areas (ICARDA) with notable achievements so far, to include: (i) higher animals' productivity, i.e more births, better growth and reduced mortality, ii) increased income (20 percent on average) among participating households, implying major livelihood improvements, and (iii) increased meat and milk production and consumption in participating communities, implying improved nutrition (ESGPIP 2008; Haile et al. 2019). As a result, the Government of Ethiopia included CBBP as one of its key livestock breed improvement strategies in its national Livestock Master Plan, which builds on existing traditional practices for the selection of best performing local breeds and broad community participation through the entire stages of selection, production, promotion and marketing.

Therefore, the traditional, community-based breeding system should remain the main core strategy in Somalia to improve productivity while maintaining adaptive traits, versus a centralized and exotic breeds-based breeding program. Cross-breeding with foreign semen for cattle, however, is recommended to improve milk and meat yields, and there is already a recently-established private company (Somali Bovine Genetics or SBG) offering such services. An animal genetic resource strategy is currently under preparation by the MoLFR, with plans for all livestock stakeholders to contribute to and validate it.

Despite the many daunting constraints listed above, strong opportunities exist for reviving and expanding the live animal trade from its peak years of 2015-2016. There is strong potential for improving the quality and resilience of Somali livestock, and its offtake, but without targeting or fostering increases in the animal population, because of the limited carrying capacity of the degraded rangelands and the broader environmental challenges discussed elsewhere. Moreover, there is an existing and growing demand for freshly slaughtered meat products from grass-fed livestock in both local and international markets. Somalia's strategic location and proximity to the Gulf countries also can be better leveraged to supply much of the large demand in such markets now satisfied by imports from elsewhere for live animals (and their products).

Livestock producers with large number of animals (mainly goat and sheep) should raise their offtake to between 10-20 % of their stock, and either save the income generated into a reserve fund for emergencies or utilize it to fund their children's education, etc. As of now, those pastoralists may run the risk of losing most (if not all) of their livestock stocks when a severe drought strikes and they will become destitute because they don't have not established a safety net to fall back. There will be also the added advantage of reducing competition to limited pasture and environmental degradation. Almost all Somalis who attended the consultations workshops in October-November 2018 [and in April 2019] were in favour of this innovative approach and recommend that awareness of pastoralist be raised through media so that the pastoralist start changing their old habit of rearing livestock.



Box 5.1 – Proposed Aerial and Ground Livestock Population Survey

The aerial component of the 1975 census comprised the use of rear seat observers counting animals from flying sample transects, but technology has changed significantly since then. The FAO, with initial input from the World Bank, worked on the terms of reference of an international consultant, who developed the methodology to conduct a livestock population survey for Somalia. The proposed design, to cost in the \$1.2 million-\$2million range, conforms to current FAOs guidelines and best practices, while considering also the lessons learned from the 2008/09 Trial Survey, which was conducted in Jowhar (only partially because of insecurity), Borama and Oodweyn districts, and which showed wide variation in accuracy. Its methodology, coverage, and timing (likely in 2020 to allow sufficient time for mobilizing the necessary funding from donors) is currently under consideration by the MoLFR.

Among the key features of the new proposed methodology are that:

- An SRF with transect spacing 20 km and sample size of 5 percent be conducted over the whole of Somalia using the new ‘oblique-camera-count’ (SRF/OCC) method;
- flights be conducted at 1,200-3,000 meters above-ground-level for security reasons and requiring two months of flying;
- interpretation of the imagery will require a further four-five months;
- livestock estimates be determined using the standard Jolly II analysis for SRF data and plotted onto a 20 km national grid;
- the aerial survey be accompanied by a ground component, with a simplified questionnaire using a stratified sampling approach to also collect information on natality, mortality, offtake and other production-related aspects; and
- a calibration testing flying at these altitudes be carried out in trials in three districts in three different regions in Somalia.

Source: FAO-Somalia

In ensuring water security for people and livestock, boreholes can play an important role, but they are also associated with environmental degradation. The yield of groundwater-fed boreholes is less vulnerable to short-term fluctuation in rainfall than other sources such as berkads, open dams, shallow hand-dug wells, and springs. This makes boreholes an important source of water in times of severe drought especially for humanitarian response and particularly in non-riverine regions of Somalia. However, heightened pressures on pasture around these boreholes during drought events can cause long-term damage to the surrounding rangeland, creating so-called ‘sacrifice zones’. Deep boreholes are also not a cost-effective solution for watering livestock and for increasing crop productivity. The costs for drilling and equipping them are high, ranging between US\$0.5 million and US\$1 million due to a combination of physical, market, and security conditions,³⁰ and furthermore their operation and maintenance costs are much higher than from shallow water sources and challenges related to water quality (for example, high salinity).

Based on recent project experience, ample low-cost opportunities exist to enhance rural communities’ access to water across Somalia’s dry lands by deploying small-scale water harvesting and storage technologies and without exacerbating conflict. Wadis in northern Somalia, and in

³⁰ UNICEF Somalia, personal communication.



areas other than the Juba and Shabelle valleys, are ephemeral with water flowing for very short periods during the seasonal rains. Following seasonal rains, water infiltrates into shallow aquifers that last for only a few months of the year. Around these shallow aquifers, there is a small but growing horticultural production base selling vegetables to urban areas and improving rural incomes.

Pilot projects in Puntland and Somaliland, including the World Bank-financed Water for Agropastoral Livelihoods Pilot Project (WALP, P152024), have demonstrated that water harvesting and storage in these dry lands can be increased through investment in small dams such as sand dams, subsurface dams, and infiltration galleries.³¹ Sand dams, in particular, are experiencing a renovated interest because of their relative simplicity and their potential in enhancing the resilience of marginal dry-land environments. Studying the impacts of sand dams in Kenya on vegetation biomass, Ryan and Elsner (2016) concluded that sand dams hold strong scope to improve the adaptive capacity of dry lands by helping sustain vegetation biomass during drought periods. Increased water availability supported by improved vegetation biomass and soil management means effective support for agricultural activities and food production, including for livestock watering and nutrition, in the arid and semi-arid zones of the country.

While animal health-, nutrition-, and export-specific recommendations are further dealt in the two following sections, priority interventions necessary (but not sufficient) to address the constraints mentioned above and to realize the main goal(s) of increasing the subsector's productivity (and exports) are to:

1. Improve communications within the sector regarding sector policies, regulations and domestic and export requirements and ensure processes are in place and resources are available to assist stakeholders in complying with these requirements;
2. Strengthen public extension services on health and animal husbandry to address rural needs (social, economic and environmental) for diversification, resilience, sustainability;
3. Incentivize the private sector to rehabilitate pre-war holding/resting grounds in strategic locations along both domestic and export trade routes, where veterinary services, fodder and water can be provided;
4. Rehabilitate and expand pre-war critical public livestock watering-related infrastructure (mainly sand dams, open dams, and reservoirs or *berkads* for rainfall harvesting, but also shallow hand-dug wells and springs) and livestock transport-related rural and main roads, the former to be managed by communities and the private sector;
5. Gather a critical set of nationwide data and information, including via a comprehensive, reliable census/survey of animal population and production, and feed and fodder inventory and balance.

In the medium-term, other recommended interventions include:

- Supporting private sector investment in modern and resilient production systems that reduce current vulnerabilities and offer better opportunities in international livestock markets;
- Developing a strategy to strengthen availability of natural feed on rangelands and to expand greatly availability of quality commercial fodder;
- Improving local breeds' production and productivity via scientific and community-based innovative programs such as: participatory genotype characterization through phenotypic features, training of communities on simple breeding concepts and improved husbandry

³¹ These technologies protect water from high evapotranspiration rates by holding the water in shallow sand aquifers that can be used to supply limited amounts of water for domestic, livestock, and agricultural uses.



- practices and developing community-based schemes for stock upgrading intensive feeding of selected breeding herds on improved fodder, and adopting effective management systems;
- Support pastoralist and agro-pastoralist IDPs return to their original areas and livelihoods ways via, among other measures, restocking programs for shoats;
 - Collect and publish an annual livestock statistics bulletin, covering animal population and offtake estimates, and market prices of live animals, as well as livestock export data and livestock product market prices, milk and meat prices, fodder/grass prices, chicken and egg prices, etc.
 - Establishing an independent Somali Livestock Research Institute (SLRI), with Somali and foreign scientists from different specializations, capable to researching the sector's challenges and needs using the latest techniques and methods and advising governments on suitable interventions.

5.2 Environmental Degradation and Natural Resource Management

The current large livestock population compared to historical pre-war levels has been putting a major strain on increasingly limited natural resources such as rangeland and water, particularly in the northern, central and southern rangelands. Its impact also needs to be assessed in light of competing resource uses, for example, agriculture production and water for human consumption and other sources of degradation such as charcoal production, wood collection and increasing frequency and severity of droughts.

Trade bans may have some negative environmental effects, by keeping the livestock that would have been taken off the rangelands on increasingly stressed land, and by vulnerable livestock keepers resorting to other activities to meet their needs, including charcoal burning for illegal exports.³² Not only are producers forced to retain animals, but the drop in their value may lead to producers having to resort to destructive coping measures in order to maintain an income and livelihood.

The potential negative environmental impacts, particularly in peri-urban zones, of intensive zero-grazing or semi-intensified milk production systems should be better understood and strategies should be developed to mitigate these impacts. According to a recent study (USAID, 2017), for instance, a high concentration of dairy animals (about 12,000 to 15,000) is kept on what have effectively become permanent pastures, causing gradual environmental degradation in the area. Furthermore, if the use of chemicals by tanning factories remains unregulated, processing of hides and skins to higher stages such as wet blue may also cause adverse environmental impacts, which should be well understood and mitigated.

Improving the natural rangelands resource base is critical to sustaining the production and productivity of livestock, in parallel with improving livestock productivity and value addition. There is a pressing need to clarify land tenure arrangements; reach consensus over policies and their implementation with local communities; improve communities' capacity to manage their natural resources in a sustainable manner; and, specifically for the livestock sector, support rangeland rehabilitation with technical assistance and public investments for intensified soil and water conservation, reforestation and afforestation, and the reseeding of pastures. Improved community-level water management is also key in improving resilience against droughts, through the protection

³² However, export volumes trends show that declines in export numbers following export bans are temporary (as alternative trade outlets are found), so negative impacts would be experienced mostly in the short-term.



of livelihoods and assets (UNICEF, Safe the Children, 2017). Somaliland and Puntland are engaged in a number of proactive initiatives, often with support from development partners, to address widespread communal rangelands degradation, including investments at grassroots levels in establishing and upgrading community-managed rangelands reserves. The impact and sustainability of these initiatives should be assessed and strategies for upscaling of successful practices should be developed.

Many stakeholders recommend re-establishing the National Rangeland Agency. However, management of rangelands through effective community and producers' associations may well be a more effective alternative, especially considering that the government has no effective oversight in all the grazing hinterlands. Participatory land use planning and enforcement, including communal control over the expansion of enclosures, are also priorities that need to be agreed and enacted upon.

5.3 Climate Change Vulnerability, Adaptation and Mitigation

There is still no consensus in the scientific literature over whether the Horn of Africa will experience more or less rainfall, but all studies and projections by climate scientists agree on a warmer future, with increased variability and frequency of extreme rainfall events, with related cycles of intensified floods and droughts. Among 167 countries ranked for overall vulnerability to climate change, the Center for Global Development places Somalia at the top of the list, at number 5 for most vulnerable to extreme weather risks in 2015 (Wheeler 2011). Therefore, regardless of whether climate change will result in absolute wetting or drying, large scale investment in watershed management and infrastructures that would mitigate the impact of extreme cycles of rainfall, floods and drought will be critical for the resilience of Somali livelihoods. A recent analysis of prehistorical trends suggests that climate conditions in the region will become even drier, more erratic, and more extreme than in recent decades and thus less favourable to crops, livestock, fisheries, and forestry-based livelihood systems, but most global and regional climate models predict higher rainfall rather than drier conditions, although the absolute value of projected changes vary greatly across models.

Climate change, moreover, is likely to affect different states differently, may also change the rainfall intensity of different seasons, and even if wetting would provide benefits to the agricultural sectors, temperature increases may well offset any such benefits. In the Lower Jubba, a recent study projected a decreasing trend in rainfall through 2030, followed by an increase in rainfall in the 2050 and 2070 scenarios (Ogallo et al. 2018). Another recent study pointed out that projected changes in precipitation are most uncertain across the Greater Horn of Africa due to local factors, such as complex terrain, large inland water bodies, land heterogeneity, and their interactions with large-scale climate forcing mechanisms consequently result in diverse spatial rainfall patterns across the region. Still, this second study also points to a wetting signal over Somalia under 1.5°C -2°C global warming scenarios (Osima et al. 2018). A third study found that most models agree on the projected increase in rainfall during October-to-December over Somalia (Endris et al. 2018). Other studies, which project a scattered pattern of intensified rainfall over East Africa in terms of locations, frequency, duration, and intensity, also point to this change occurring in Somalia during its short *Deyr* rainy season (Han et. al. 2019).

The consequences of such climatic changes on Somalia's already fragile ecosystems and for Somali livelihoods that depend on them are dire. Higher air temperatures will increase evapotranspiration and more variable and extreme rainfall on already barren soils will reduce spatial infiltration and



facilitate more runoff and erosion, resulting in less groundwater recharge and surface water for plant growth. Other likely impacts include reduction of vegetation for grazing and more variable water availability, with grave impacts on livestock herding and related livelihoods. Rising sea temperatures and acidification will also blunt fish stocks and change their distribution, potentially undermining a valuable resource such as fish waste for animal feed.

Climate change is coming on top of the threats from severe deforestation, other environmental degradation of the rangelands from poorly regulated human activity including overgrazing and enclosures, and erratic and diminishing water flows in its major rivers due to much expanded use of water in their upstream basins in the Ethiopian highlands. The combination of these factors poses a serious threat to the long-term viability of agriculture in Somalia, including its livestock sector. They also pose another huge and unprecedented challenge to the country's weak institutions, limited public capacity, and private investors. In the absence of substantial progress in adaptation to make agricultural systems, including livestock ones, more climate resilient, there is little scope for economic recovery. In such a scenario, the United Kingdom's Met Office Hadley Centre for Climate Science and Services predicts that Somalia's vulnerability to food insecurity will grow by 27 percent and 38 percent by 2050 and 2080, respectively.

In the short-term, Somalia can strengthen the predictive capability for drought and cyclone of the early warning system provided by SWALIM and FSNAU and ensure preparedness for timely and targeted emergency responses to shocks. In the medium-term, widespread water management planning and investment in rainwater and water runoff catchment infrastructure are essential, at individual household level as well as with large-scale pans, reservoirs and dams without losing too much to evaporation. Traditionally, catchments for water storage have been dug; however, with expected dramatic climatic shifts, widespread management is needed to take advantage of either additional or reduced water supply. Otherwise, soil erosion and gully formation caused by flooding and erratic rainfall, which is already taking a toll on Somali agriculture, including the livestock sector, will continue to increase. Beyond planning, expanded management requires strong institutions, infrastructure development, market access, and skills and technology development.

While many traditional practices are already in place to cope with climate variability, they need to be strengthened to cope with increasing risks due to climate change. These include moving livestock and families, keeping diverse animal species, creating reciprocal arrangements with other pastoralist groups for access to pasture and water, developing water-conservation techniques, observing and disseminating early-warning signs of impending drought, and practicing alternative livelihoods. As the country's institutions become stronger, it is also important to integrate traditional practices with government policy. Beyond resilience, there is also a need to strengthen the country's livestock emergency response capabilities, for which the Livestock Emergency Guidelines and Standards (LEGS) offers guidance through a best practice handbook and an evaluation tool.

Somalia's Nationally Determined Contribution (NDC) to the UNFCCC outlines intended project areas necessary to adapt to climate change, with some measures also for mitigation. The intended project areas are:

- Sustainable land management, to build resilient rural livelihoods and enable national food security;
- Integrated water resources management, to ensure access for vulnerable communities and sectors;
- Reducing risks among vulnerable populations from natural disasters;



- UN Joint Programme on Sustainable Charcoal Production and Alternative Livelihoods (PROSCAL), and promoting charcoal production from *Prosopis* and replacing related livelihoods with crop production;
- Rehabilitation of Fanoole Hydro-Electric Dam and associated irrigation infrastructure;
- Domestication of indigenous and introduction of new economically important plant species;
- Up scaling the Use of Solar Energy; and
- Marine and Coastal Environmental Governance and Management of Somalia

When actively promoting livestock development, including promoting a commercial feed value chain, opportunities should be considered for reducing GHG emissions, especially methane gas (CH₄), although only about 62.92 MT CO₂ or less than .12 percent of total global emissions in 2015 originated from Somalia, 95 percent of which came from agriculture (including livestock), forestry and other land use (AFOLU). Somalia submitted its first National Communication to the UNFCCC at the start of 2019. While Somalia is not obliged under the Climate Change Convention to reduce its GHG emissions, the country has opportunities to ensure sustainable development and limited emissions compared to a business as usual scenario.

Most of the emissions in Somalia in agriculture are coming from livestock in the form of enteric fermentation and animal wastes and from the fodder it consumes. Small ruminants' production for meat represents between 7 and 10 percent of the sector emissions globally and emissions from the production, processing and transport of feed actually account for about 45 percent of sector emissions. Mitigation will require a holistic approach by integrating animal manure waste management systems and harnessing biogas capture--which can be used for lighting and cooking in households. Mitigation options such as agroforestry and grassland and pasture management can increase soil carbon sequestration and offset emissions from livestock. Furthermore, use of solar and wind water pumps for irrigation and watering of livestock can replace use of diesel and petrol generators, and the use of solar energy can improve on farm productivity.

Strengthening feed and fodder systems could mitigate the risks related to dwindling natural resources and reduce livestock losses during future drought episodes. Emissions can be reduced by adopting climate-smart fodder production systems that ensures feed nutritional value is improved by integrating leguminous forages and more efficient use of crop residues through processing. The recommendation of promoting commercial fodder production (offered in Chapter 5) and other more radical ones, including to move towards zero grazing, with their associated impact on the environment and contribution to greenhouse gas (GHG) emissions, must be balanced by the fact that mobile livestock management is still regarded by experts as one of the best forms of land-use in arid sub-Saharan countries such as Somalia. This is despite that nomadic pastoralism and its associated low-input low-output production system are becoming increasingly difficult in a country with a growing population, rapid urbanization, weak governance and in a warming world. Thus, the impact on GHG of commercial feed production compared to potential benefits in other environmental aspects needs to be researched using local emission data and calculations. Caution should be exercised, moreover, when proposing the development of feed from fish products, to ensure this does not originate from illegal and unsustainable fishing practices. Rehabilitating Somalia's forests and effective rangeland management can also be a crucial GHG emission mitigation option as a carbon sink as well as blunting the impact of extreme weather events.



5.4 Land Tenure

Land tenure security is a fundamental prerequisite to long-term sustainable development, as private investments in infrastructure and the development of rangelands, food and fodder production, enclosures, etc. are all conditional on the security of land ownership and user rights. Current unclear tenure rights foster overgrazing, lack of action to combat soil erosion and gully formation, and the expansion of private enclosures on communal rangelands. Women, furthermore, are particularly affected by insecurity of tenure, reflecting complex dynamics extending from the interaction of multiple legal systems and Somali socio-cultural norms. The Provisional Constitution, however, guarantees the rights of every citizen to own land provided that it is managed in an equitable, efficient, productive and sustainable manner and the FGS is in the process of drafting a new land policy that guarantees access to all its citizens, including women and minorities.

Under Somali pastoral traditional norms, local communities collectively use and share rangelands and forests. However, grazing disputes leading to fighting between neighbouring pastoral clans have become more common since the collapse of the pre-civil war government in all parts of Somalia, including at the borders with Ethiopia and Kenya (see Box 5.2). In general, clashes involving rival clan militias that started from communal disputes remain the single most common form of armed conflicts in Somalia, comprising about 35-40 percent of total security occurrence per month. These conflicts are often resolved by local clan elders and religious leaders, however, with local district or regional authorities (including neighbouring authorities in Ethiopia and Kenya) intervening only in the case of major interclan fighting (World Bank and FAO, 2018).

Box 5.2. Addressing Conflict Risks and the Needs of Nomadic Pastoralists

In Somalia, as elsewhere in the Horn of Africa, localized conflicts between farmers and herders, and between different pastoralist groups, frequently revolve around issues of contested land use, grazing rights, and insecure access to water and pasture. Scarce and under increasing pressure, these resources must be shared with rural farming communities and the needs of growing urban centres.

Expansion in recent decades of private enclosures on traditionally open communal rangelands, especially along livestock migration routes, increasingly jeopardizes the mobility of pastoralist communities, thereby weakening their capacity to cope with adverse climate conditions. Existing tensions and conflict risks are amplified during extended dry periods—the frequency and intensity of which have increased in recent decades—when pastoralist livelihoods become particularly precarious. In the absence of provisions to assist pastoralists to safeguard their capital stock during emergencies and otherwise recover from the shock, resulting economic insecurity and deprivation increases the risk of wider violence and social breakdown. Moreover, political and socioeconomic marginalization of pastoralist communities and policies that either neglects or undermines traditional governance and arbitration mechanisms have, over time, weakened resource management and conflict resolution capacities among rural communities.

The design of future interventions to support delivery of services to rural communities in Somalia's marginalized dry-land areas should emphasize a learn-by-doing approach to generate new understanding about (a) the current dynamics of nomadic pastoralist systems in Somalia; (b) the specific needs of pastoral communities, and their relationships with other livelihood groups including landed farming and agro-pastoralists communities; (c) reforms of regulations and land tenure rights relating to access to pasture land and water for communities, especially nomadic pastoralists; and (d) mechanisms needed to ensure emergency access for pastoralists to water and pasture and fodder during droughts.

These lessons should inform downstream interventions designed to mitigate drivers of vulnerability and conflict, build trust within communities, strengthen resilience, and enhance participation of pastoralists in consultative and decision-making processes that would support customary approaches and traditional governance systems for managing competition over natural resources and conflict resolution.

Source: World Bank, Somalia - Water for Agro-pastoral Productivity and Resilience Project (P167826), under preparation.



In Al-Shabaab-controlled areas in the southern regions, their harsh rule has many detrimental effects on rural, including pastoral, communities, threatening the lives of summarily judged sinners, confiscating assets, performing abductions, forcing marriages and instilling a culture of fear and mistrust in the people, according to the findings of a study commissioned by the United Nations in Somalia (UNSOM XXXX). Al-Shabab's rule has also negatively affected the overall economy, trade, security, logistics and religious life, as well as community participation in peace initiatives. As government becomes stronger and cleaner, more transparent and equitable, effective community responses to the current security challenges can also be expected.

Efforts to improve land tenure management should therefore aim to:

1. Strengthen the Water and Land Management Section of the Forestry and Range Department in MoLFR;
2. Create formalized land commission units consisting of elders and religious leaders from various communities, already engaged in land conflict resolutions, of civil society engaged in conflict resolution, and of relevant government officials.
3. Reach consensus over policies and their implementation with local communities, tailoring these to the specific regions, being conscious of the level of insecurity and government control;
4. Promote equitable access in practice to land rights, as guaranteed in existing laws, the Provisional Constitution and the expected land policy;
5. Ensure secure pastoral migration routes, including along the riverine areas and cross-border, to enhance livestock mobility, which is key to its resilience even during the dry season and at very harsh times.
6. Support participatory land use planning;
7. Institute and support a process of land tenure formalization, particularly in relatively stable areas, by issuing titles for some cultivated farmland (and demarcating urban property boundaries).

5.5 Recovering and fostering live animals' exports

Somalia's close proximity to various export markets and favourable consumer preferences in the Gulf markets provide Somalia with a competitive advantage that can support growth of exports of live animals, but also chilled meat and other livestock products. To exploit this potential advantage, however, Somalia needs to address successfully the vexing issue of the credibility of its health certifications for exports, and to diversify and reduce both its product concentration and heavy reliance on a few export markets, which makes the live-animal export sector very vulnerable to events in Saudi Arabia for shoats, Yemen and Oman for cattle and shoats, Kenya for cattle, and Egypt for camels (Godiah et al. 2015). Moreover, camel and especially cattle overland exports to Kenya are expected to continue, as Kenya is challenged to meet its domestic beef meat demand and informal exports of animals from Somalia, as well as from Tanzania and Ethiopia, help Kenya meet this demand (Aklilu, Little, Mahmoud, & McPeak, 2013).

A deeper understanding of the dynamics and decision-making processes behind the recurring import bans by Saudi Arabia and the UAE, however, is required. These include who takes decisions, how they are communicated and enforced, how they impact on the price of various livestock types, how much of the market share is lost and to whom after taking account of the redirection of exports through third countries.³³

³³ Such studies would benefit from support and expertise of the World Organization for Animal Health (OIE), which has an international mandate as a regulatory body for animal health and disease declaration.



The ban on livestock imports has highlighted various issues that must be addressed, namely:

1. Reassessing the relationship between public and private veterinary services at quarantine stations and the roles and responsibilities of each stakeholder, in regard to (i) laboratory testing and reporting and health certifications at export points, (ii) mechanisms for government oversight and enforcement of relevant regulatory measures, and (iii) current lack of government accreditation of foreign veterinarians who deliver services in the quarantine stations;
2. Using a unified Animal Health Certificate issued only by FGS public veterinarians, as agreed among the FGS and the FMs, as mandated by the Veterinary Law Code, and also as recently demanded by Saudi Arabia, which was the main destination market for Somali livestock until its December 2016 import ban;
3. Analyzing the cost of regulatory compliance and enforcement (for example the three-week quarantine and health certificate emission) for both public and private sector players and how that cost can be distributed along the value chain with minimal upstream impact towards producers;
4. Mainstream standard methods and procedures (SMPs) for export quarantines, including for pre- and post-phases, that IGAD and AU-IBAR developed and validated with the participation of the IGAD member states.

Other recommendations for the medium- and long-terms are:

- Understanding the targeted export markets for livestock (and meat) and their characteristics, preferences, and perceptions around the quality of Somalia's exports, their competitiveness and opportunity for volumes or value optimization, as well as the cost/benefit of entry into new markets, so as formulate evidence-based effective marketing strategies;
- Analyzing the various existing regional trade channels, including cross-border trade;
- Analyzing the equity implications of promoting expansion of the live animals' trade;
- Elaborating a specific livestock export strategy focused on competitive advantage and best ways to optimize strategic proximity to Gulf States' and other international markets, especially in Southeast and East Asia, including via the establishment of FMD-free zones as suggested in the OIE Animal Health Terrestrial Code, Section 5 covering trade measures and import/export procedures;
- Understanding the differences and potential synergies between the commercial and sacrificial live animal export value chains and how to optimize benefits for both.
- Adopting regulations that can ensure traceability of live animals (and other products) exported.

5.6 Improving and expanding animal health services

While the private sector and LPAs accredited by the Ministry of Livestock Forestry and Range of the Federal Government of Somalia and funded by development partners have provided most critical veterinary services, such efforts remain insufficient to address the serious risk of animal diseases and consequent import bans. Animal diseases, especially those transboundary in nature, are common in Somalia and in neighbouring countries, and are one of the major concerns for the production and marketing of live animals.

The 2016 National Veterinary Law no. 21, which updated the preceding Law no. 20 of 1967 provides sector stakeholders with the policy and legal environment to ensure the effective prevention and control of animal diseases, including, and especially for, those which are of



transboundary nature (see Box 3.3 in Chapter 3).³⁴ In addition to animal health strategies adopted by Somaliland and Puntland, there is also an IGAD regional animal health strategy covering the period 2017-2022 (ICPALD 2017) and a May 2019 MOU between Somalia and Ethiopia on cooperation and coordination on animal health and sanitary measures. There is, however, a need for a more comprehensive legislative and regulatory national framework and effective and sustainable institutional arrangements and mechanisms for its implementation and enforcement, including:

- A Federal Meat Inspection Act, with disease notification and related information
- Emergency Preparedness and Response (EPR) laws, regulations and guidelines;
- A Pesticides Act, addressing veterinary, agricultural and environmental concerns;
- A Veterinary Code of Ethics;
- Regulations covering animal production/zoo-technical services and agribusiness; and
- Regulations to update Ministerial Decree of 1985 dealing with the import and distribution of non-prescription drugs and Ministerial Decree of 1988 dealing with private veterinary services and prescription drugs import and distribution.

To improve animal health as well as to meet international standards and requirements of importing countries, this LSDS prioritizes the following additional interventions:

1. Strengthen public veterinary services provision and oversight of private veterinarians, by (i) improving veterinary and animal husbandry education, (ii) establishing laboratory and diagnostic services, field clinics, vaccination, emergency investigation, and treatment teams at the regional and district levels, to ensure both prevention and immediate recovery of sick animals from the infectious diseases; and by (iii) re-establishing and ensuring credible government surveillance and oversight on animal health and certification processes;
2. Apply the OIE-PVS Pathway Tool to assess and monitor the performance of veterinary services, in all four components: (i) human, physical and financial resources to attract resources and retain professionals with technical and leadership skills; (ii) technical authority and capability to address current and new issues including prevention and control of biological disasters based on scientific principles; (iii) sustained interaction with interested parties in order to stay on course and carry out relevant joint programmes and services; and (iv) ability to access markets through compliance with existing standards and the implementation of new disciplines such as the harmonisation of standards, equivalence and zoning.³⁵
3. Establishment at the national level an Epidemiological Data Management Unit (EDMU) for surveillance and recording disease information and outbreaks reports (with district focal points to be established for reporting disease outbreaks and incidents to EDMU) and for conducting epidemiological studies.
4. Gather and mainstream data on livestock health, antimicrobial drug use, and prevalence of antimicrobial resistance in animals and animal products in the OIE/AU-IBAR managed information systems and within the framework of the AU-IBAR sponsored One Health Framework;

³⁴ There are three major concerns highlighted in the National Veterinary Code, these are: (i) disease notification and security; (ii) veterinary health certification of exports and imports of animals and animal products; and (iii) the approval of the independent veterinarians or not employed by the government to carry out and perform certain tasks based on the ministry of livestock, forestry and range authorization and the ministry always has responsibility for monitoring and verification.

³⁵ For more information on this tool, see <http://www.oie.int/support-to-oie-members/pvs-evaluations/oie-pvs-tool>



5. Implement the cooperation elements of the IGAD Animal Health Strategy, including sharing of data and coordinating disease prevention, emergency preparedness, and control with neighbouring countries.

For the medium- term, the LSDS offers also the following recommendations:

- Improve the administrative structure of the Public Veterinary Health Services, and its collaboration with private veterinary associations (PVAs), entrepreneurs and producer cooperatives;
- Make qualified manpower operational again in the field and as advisors to the federal and state ministries, by recalling, reorganizing, retraining and adequately funding former veterinary professionals and livestock experts and technicians;
- Monitor, prevent and control TADs not only among livestock but also among the wildlife population
- Monitor, control, and educate traders, CAHWs and livestock professional and owners on quality and proper use of veterinary drugs to ensure effective treatment and minimize the risk of AMR;
- Support the development of community-owned and managed health services.

5.7 Animal Welfare

Animal welfare concerns the humane care and treatment of animals that are raised, transported and sold commercially. It is both important per se according to Somali cultural and Islamic beliefs, for enhancing animal health and productivity, and as a means to enhance the value of marketed animals and their products. Most of animal welfare issues in Somalia occur during loading, transporting and unloading of livestock, not because of mistreatment by their handlers but because of the inadequate facilities and services that are available, such as sanitation, nutrition, water, and veterinary care, in rural areas and along transit routes. Even at the major export ports there are few loading and unloading ramps. There are also no standards for maximum transit time during transportation. The country also lacks adequate facilities for animals to rest and recuperate, before export or domestic marketing in large urban centers. It is therefore essential that strategic resting facilities be established along the major trading routes.

Box 5.3 – Five Freedoms of Animal Welfare

The OIE Global Animal Welfare Strategy was developed from lessons learned from actions taken at national and regional level and aims to be a source of ongoing guidance for the OIE's activities in this area. Adopted in 2017 by all Member Countries, it was developed with the objective of achieving: "A world where the welfare of animals is respected, promoted and advanced, in ways that complement the pursuit of animal health, human well-being, socio-economic development and environmental sustainability".

According to the OIE Terrestrial Code, animal welfare means 'the physical and mental state of an animal in relation to the conditions in which it lives and dies.' The guiding principles which inform the OIE's work on the welfare of terrestrial animals include the 'Five Freedoms', developed in 1965 to capture global societal expectations for the conditions animals should experience when under human control.

Of these five widely recognised freedoms, no. 1, 3, and 4 are those most relevant to the Somali context.

1. freedom from hunger, malnutrition and thirst;
2. freedom from fear and distress;
3. freedom from heat stress or physical discomfort;
4. freedom from pain, injury and disease; and
5. freedom to express normal patterns of behaviour.

Source: World Organization for Animal Health (OIE).



Animal welfare issues have not been well studied across Somalia, as it has not been a priority largely due to lack of exposure to global animal welfare norms and debate, except in Somaliland, where most available documentation on animal welfare comes from. Somaliland's law dedicated to animal welfare, which is very comprehensive and surpasses OIE standards, needs dissemination and implementation. There are also some civil society organizations, e.g. the veterinary faculty at Golis University and the Somaliland Veterinary Association, amongst others, that deal with animal welfare and whose objectives and activities include promotion of community awareness and education on animal welfare, advocacy for animal welfare legislation, and mobile veterinary clinical intervention. However, no one in the state ministry or among livestock stakeholders has been trained on international standards such as the OIE Improved Animal Welfare Program.

Even though livestock transport trucks are modified specifically for this purpose, a welfare assessment at a key market in Somaliland rated it poorly. Inadequate water and feed for animals during most of the year was recognised as a weakness, and drought a major threat. Loading facilities at the port of Berbera have been upgraded, but handling of animals leaves a lot of room for improvement. The overall conclusion from IGAD is that there is an interest to improve animal welfare in Somaliland, and where there is a will there is a way. Other parts of Somalia face much worse conditions and still needs to start by establishing the will and then by searching for the way to ensure better animal welfare.

5.8 Improving and expanding animal nutrition (water, feed and fodder)

Ensuring year-round affordable good quality feed by enhancing fodder production and feed systems development is important to improve animal quality, to mitigate lack of pasture and browse at times of seasonal or prolonged severe droughts, and to reducing the pressure of a large animal population on increasingly scarce natural resources. The feed subsector, however, has always been and is to this day characterized by a widely practiced traditional low-input grazing and browsing system throughout Somalia, subject to seasonal shortages of pasture, and very limited commercial feed/fodder production. In recent decades, the underdevelopment of this subsector has been compounded after the outbreak of the civil war by the breakdown of government institutions (such as the defunct pre-war National Range Agency) and traditional management institutions, as well as by degradation, deforestation and continuous overgrazing resulting in major soil erosion, desertification, and reduction in edible biomass. Other constraints and gaps are insecurity and difficult access to some parts of the southcentral rangelands, the uncontrolled expansion of enclosures on previously open rangeland, low levels of technical knowledge on fodder production and storage, poor access to fodder seeds and other production inputs, absence or inadequate water, storage, and transport infrastructure, and unsettled land tenure issues in southern Somalia.

The 2016-2017 major prolonged droughts that led to high mortality of livestock across the country has stoked interests of investors and traders in producing, commercializing and storing improved feed and fodder, from grasses, crops residues, and also from cereal grains and oil seeds. The nascent commercial forage production and marketing systems, however, are for the moment very limited and inadequate. It is also not very well studied, as no state-level quantitative information exists on the amount of commercial fodder grown, on the availability and uses of various inputs, and on the number and type of actors involved.

Development opportunities in this subsector can rely on much increased demand for fodder by expanding commercial livestock rearing operations (throughout the year), field testing by some research institutes (e.g. SATG) of promising better grasses, and existing large-scale commercial

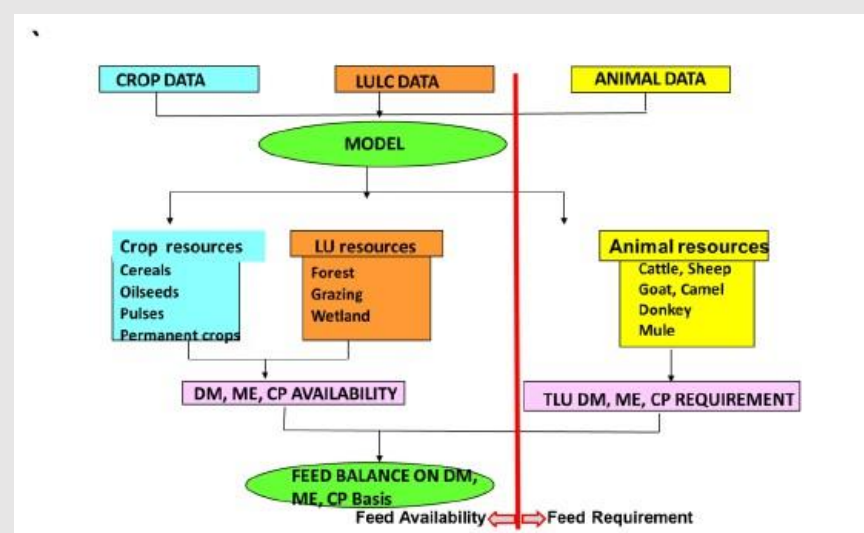
crop production in some states of cereals and oil seeds. Furthermore, they can leverage the availability of suitable land and water in some states (e.g. South West and Jubaland, but also in Somaliland and Puntland where seasonal rivers flow in abundance but drain into the sea without their waters being utilized and of some water catchment facilities on the rangelands that only need rehabilitation.

Pastoral destitution is principally driven by feed and water scarcity, as the natural resource base in the rangelands is shrinking fast. To address this, and also to enable realistic improvements in livestock production, feed resources ought to be considered in the broader development perspective and not just as an emergency response to drought, as is the case now. Institutionalization of a feed security system is therefore requisite such that the country is aware of its needs, resource availability, gaps, and how the gap can be filled within the country, the region or beyond. A feed strategy can only be developed following an assessment of the national feed inventory, which includes measuring actual and potential feed sources, and of the national feed balance, which compares feed sources to feed needs, as illustrated in Box 5.2.³⁶

Box 5.2: National Feed Inventory and Balance

Over the past decades, droughts and environmental degradation, combined with a growing human population are leading to bigger challenges to Somalia's livestock system. Available evidence indicates that pastoral destitution is principally driven by feed and water scarcity, as the natural resource base in the rangelands is shrinking fast.

To address this, and also to enable realistic improvements in livestock production, feed resources ought to be considered in the broader development perspective and not just as an emergency response, as is the case now. Institutionalization of a feed security system is therefore requisite such that the country is aware of its needs, resource availability, gaps, and how the gap can be filled within the country, the region or beyond. A national feed inventory and balance assessment, which includes measuring actual and potential feed sources compared to the feed needs, is illustrated below.



Source: FAO-Kenya

³⁶ A recent report assessing the value chains for various crops in southcentral Somalia shows that it is feasible to estimate potential yields of crop by-products for livestock feeding, as part of a broader feed inventory assessment (FAO 2016).



In view of these constraints and opportunities, the subsector would benefit from the following priority interventions related to water and fodder, as suggested in recent research (UNICEF, Safe the Children, 2017) and in stakeholders' consultations (see Main Annexes 1-2 and Online Annexes):

1. Rehabilitate the existing water reservoirs and increase the number of water catchment facilities on the rangelands (see Box 5.X);
2. Conduct a feed and fodder analysis that looks at availability (inventory) and current and prospective feed needs (based on reliable estimates expected from the proposed livestock population survey);
3. Study potential production capacity and develop a livestock feed and rangeland strategy;
4. Analyze feed and fodder value chain linkages with all relevant subsectors, access to production areas and markets and transport linkages between them, and consider market support schemes to stimulate and facilitate the inclusion of more distant communities in the fodder value chain;
5. Support technical training on feed and fodder production;
6. Support private sector and cooperative investment in feed and fodder production;
7. Re-establish a rangeland management agency and enforce pre-war and/or more recent policies and laws (and traditional systems) to rehabilitate and manage the rangelands;

In the medium-term, implementation of the following recommendations would further benefit the development of this subsector:

- Support integrated production systems, leveraging the complex interlinkages and synergies between crop agriculture and (peri-urban) livestock rearing systems;
- Develop feed production contract farming, to supply feed to the quarantine and animal holding centers and to ensure availability for pastoral communities of fodder during dry seasons/drought periods;
- Promote drought- resistant fodder varieties that can withstand harsh weather conditions.
- Develop a land use strategy, with strong participation and involvement of local communities and authorities, in order to minimize conflicts over land tenure and user rights;
- Align and harmonize policies and institutional arrangements among various line ministries, as animal feed production is at the nexus of crop agriculture, livestock, and environment;
- Re-establish a forestry and range training institute, with branches in all states.



6.0 VALUE ADDITION: CONSTRAINTS AND RECOMMENDATIONS

6.1 Livestock-related Nutrition and Food Security

Despite widespread livestock ownership, the great majority of Somalis, whether rural or urban, have a diet composed more of cereals, sugar and oil complemented by consumption of animal products, which provide the most important source of protein. In pastoral zones, households are highly dependent on markets, where they purchase cereals (sorghum, rice, and wheat flour), oil and sugar – the basics of the pastoral diets. Somalia usually needs to import more than half of its basic food, especially rice, wheat flour and sugar from the world market and sorghum and maize from Kenya and Ethiopia (though the import requirement changes from year to year, see FSNAU, 2016).

Being highly dependent on basic food imports, Somalia is very vulnerable to market disruption caused by conflict, and also to severe global price hikes in commodities coming in from the world market. A flourishing livestock export sector is crucial in financing the substantial annual food imports required to ensure food security for many Somalis. Disruptions in exports, most recently as a result of lingering effects of the drought and the livestock import ban imposed by Saudi Arabia, therefore have negative effects on pastoral households' food security as they rely on animal sales to purchase their food. Poorer households are believed to be particularly affected, especially during drought episodes, as they are often less able to continue their commercial activities thereby reducing their income.

Although many pastoral households are highly dependent on markets for their food stuffs, a large share of dairy products (fresh milk and ghee) are also for self-consumption. Camel milk and to a lesser extent goat milk complement their diet, especially during the wet seasons when milk availability peaks. Poor households have less access to milk than middle income and better-off households, as their herd sizes are smaller and include fewer lactating camels. Some do, however, receive gifts of milk from wealthier neighbors.

According to FSNAU data, consumption of own meat or milk constitutes on average 10% of the diets of poor pastoral households, compared to an average of around 20% for wealthy pastoral households. Promoting more varied dairy and by-products of meat production and consumption, including from poultry, would have positive nutritional outcomes for the country's population, besides being an additional source of income for producers. Comparing pastoral households in Somaliland and Puntland, the former appears to consume more of their own meat and milk, with own livestock product consumption of poor households making up around 10% of their diet, against around 5% for Puntland's poor pastoralists. Wealthy households in Somaliland also consume more of their own livestock products as compared to Puntland. In the Juba pastoral and Southern-inland pastoral livelihood systems, consumption of own meat and milk in family diets is significantly higher: up to almost 30% for poor households and just under 40% for wealthy households.

Furthermore, a recent econometric study of food demand showed that, while the basic staples of cereals and oils are income inelastic, the more expensive food categories such as meat/fish and dairy are highly income elastic (1.43 and 1.45, respectively; see Hussein et. al 2019). Thus, the demand for meat and dairy products is likely to increase well above the expected rise in average per-capita income, by 43-45 percent more in fact. Furthermore, increased processing, storage and consumption of readily available by-products of meat production, such as offal, would provide additional cheap sources of protein for domestic consumers.



6.2 The Dairy Subsector

The main constraints that have kept the dairy sector from reaching its potential are both institutional and structural, and include most of those affecting the health and welfare of live animals. Under the first category, the dairy subsector suffers specifically from lack of regulations and from poor or no enforcement of low and outdated pre-war hygiene standards. Under the second category, the list of major constraints specific to the dairy subsector include lack of processes to preserve milk quality, inefficient marketing channels fraught with excessive intermediaries, the poor state of transport infrastructure, rapid spoilage from high temperatures and humidity, lack of sufficient quality commercial fodder and the absence of economies of scale. Other constraints reported in the literature (World Bank and FAO, 2018; Younan, Gure, & Schulze, 2013) and by stakeholders are the low and highly seasonal productivity, due to lack on modern management skills and knowledge on animal nutrition and feed supplementation. Poor hygiene and a complete absence of cooling system in the camel milk value chain also cause rapid souring of the milk, leading to losses in market value of 20-40 percent (Younan, Gure, & Schulze, 2013).³⁷ Stakeholders also highlighted limited disease control and surveillance, low levels of technical knowledge on commercial milk production, high transport costs from remote production areas, and, as noted above, lack of access to finance.

In all states, according to studies (e.g. Oxfam 2018, USAID 2017) and to stakeholders, demand for fresh milk and other processed dairy products will continue to grow beyond the natural population growth, which alone will provide over a million new potential consumers in the next decade. According to the most recent Oxfam 2018 and FAO 2019 studies, moreover, most consumers prefer fresh raw milk to packaged milk and most buyers are unwilling or again unable to pay for processing of any kind, thus processing may not be profitable. The FAO-Investment Centre recent study on peri-urban dairy value chains, funded by the World Bank and based on case studies conducted by Somali researchers in Hargeisa and Burao in Somaliland, Bosaso and Galkayo in Puntland, Mogadishu in Banadir and Kismayo in Jubaland focused on the fresh milk subsector and its recommendations for its development have been integrated in the broader list below.

Dairy-subsector stakeholders, however, see much stronger prospects than the Oxfam and the FAO-Investment Centre studies for value-addition via processing of milk supported by an appropriate public policy and robust private investment. Their view expressed during the October- November state-level consultation workshops and during the June 2019 validation workshop in Mogadishu was rooted in the current poor and inefficient state of the subsector, the large number of available milk-producing animals, the commercialization trend that has already started, and the high, growing and more sophisticated domestic demand from increased and still increasing urbanization, higher incomes, and diaspora returnees. This view is also buttressed by the success and current expansion plans by the three dairy processors in Mogadishu that have been mentioned at the end of Section 3.6 in chapter 3 and highlighted in Box 6.1, as well as by international comparisons and trade statistics, in particular the large bill in recent years for imported dairy products, ranging between US\$60 million and \$82 million, mostly for milk powder and long-life packaged milk (see Fig. 3.6 and the last paragraph of section 3.6 in chapter 3 in this report). The recent study by Hussein (2019) that found a high-income elasticity of dairy (1.45) also provides confidence in a steady expansion of dairy demand, including of quality, safe processed products.

³⁷ Shelf life of refrigerated camel milk is about 7 days, after which the milk become sour, though still consumed as such.



Box 6.1 – OOG Dairy's Processing Success and Expansion Plans

The OOG Dairy company, which produces mostly yogurt, started in July 2016 among family members with \$5,000 as seed money, with the business later scaled up gradually to a cumulative investment of about \$160,000 (including other non-family shareholders) as of November 2018.

OOG Dairy has been buying raw cow milk from the Somali Milk Co. (SoMilk), but only produced by cows not treated with antibiotics (as that would not allow yogurt-making), which is a common practice on dairy farms near the Shabelle river because tse-tse is endemic in the area. The company also sources cow milk from its own dairy farm, where it built a shed housing 10 local-breed cows on 4 ha of land at the outskirts of Mogadishu. The reason for the choice of local breed cows (9 of which produce 12 liters/day, with the remaining one 9 liters/day) instead of importing higher yielding Holstein cows was better adaptability to local hot/humid climate, resistance to diseases, and minimal treatment needs with antibiotics and drugs. Since families buy cow milk and its products mainly for children, minimizing pharmaceutical treatment is a plus for marketing.

The company does its own testing for antibiotics in the milk, as well as for density, fat content, and adulteration (added water). It then pasteurizes the milk (at 95 C) which is processed for yogurt through its plant, including heat exchanger, fermentation tank, and recently acquired chilling equipment comprised of freezers, chillers, homogenizers and incubators. The latter was acquired thanks to a grant from USAID/GEEL (before, it used a cooling tank).

Current capacity is 800 liters per day of drinkable yogurt in bottles, 200 l/d for classic yogurt in cups, 100 l/d for fermented milk (lala/laban in Somali) and 120 kg/d for ice cream. In 2018 it produced 16 tons of yogurt (mostly), including very small quantities of fermented milk and ice cream. In June 2019 it also started producing yogurt from camel milk, for which marketing is about to start. The company offers two types of yogurt in various flavors (natural, vanilla, mango and strawberry), a classic variety in modern plastic cups and the drinkable variety in sturdy plastic bottles of varying sizes, the latter having become by mid-2019 its best-selling item. It does not sell directly to the public, but to a distributor that places its 25-yogurt packs in supermarket, hotels, small shops, and preschools in and around Mogadishu (as far as Afgoi). The company has two small delivery trucks, and its distributor also has its own delivery truck, all with OOG Dairy logos on them.

Revenues exceed variable costs, but no provision for depreciation of assets has been yet made (or accounted for), so whether the business is profitable or not is uncertain. Its biggest cost is raw cow milk (around \$1 per liter, representing 40% of total variable cost), followed by electricity (\$0.37/Kwh, representing 25%). Laborers and drivers are paid \$150/month, the finance clerk and the production manager \$350/month and the CEO \$450/month. Licensing fees are \$500 annually with MLFR and the company also pays \$200 annually as a local tax to sell their products in Afgoi. The USAID grant allowed both improved quality and the addition of 15 new jobs, for a total by mid-2019 of 31 direct-hire staff and 200 indirect beneficiaries.

Since the company believes it has sufficient spare capacity to meet any further growth of demand for yogurts, its next expansion plans are focused on adding in 2020 to its current small processing footprint an automated line for the production of 1,500 liters/hour extended shelf life (ESL) fresh milk. Such an expansion is expected to require the hiring of 69 additional workers on a full-time basis. Proper training has been and will continue to be provided to workers who operate the facility, to ensure plant and food safety, improve product quality, reduce raw materials' losses, and increase utilization of by-products, and thereby increase the customer base and profitability.

Among its strengths, Oog Dairy lists the college-level technical and business education of its founders and current top managers, the active involvement in its business by its main shareholders, its brand recognition in the local market, and its vertical integration with a cow dairy farm. It believes its success has crucially depended on its ability to ensure a regular supply of milk and other essential inputs, on its thorough quality checks at different stage of the production process, on available cold chain for the distribution of its products, and on its aggressive marketing efforts. Among its opportunities are strong, unmet and growing demand for quality milk and for dairy products and availability of suitable land for the planned factory expansion, including a milk collection facility. Constraints to further expansion of capacity are limited access to capital and availability of trained electricians and mechanics to operate its plant. The major risks it sees are the general insecurity in and around Mogadishu, which can easily disrupt the timely distribution of its products, the competition from imported dairy products, and shortages of milk supplies.

Source: Interviews with Oog Dairy's CEO and Production Manager in November 2018 and June 2019.

A number of knowledge gaps have been identified in the existing body of literature on Somalia's dairy value chain that need to be urgently addressed in order to enable evidence-based decision-making regarding the best way forward. This subsector's development planning thus needs to start by improving the understanding of the following knowledge gaps:



- the environmental impact of current practices, with particular focus on peri-urban systems;
- data on the number of dairy farmers, lactating animals and productivity, and understanding the dynamic between peri-urban and extensive pastoral systems;
- a comprehensive nationwide survey of existing micro- and larger enterprises processing milk;
- the nature and volume of market demand for fresh milk and other dairy products;
- consumers' preferences and perception of risks and costs (economics) of zoonotic diseases that pose a risk to public health, since the emergence of MERS-CoV (a viral respiratory infection passed from animals to humans), as an example, is believed to have been driven by the development of peri-urban dairy farms;
- analysis of existing dairy state-level policies and regulatory frameworks and levels of enforcement.

The key challenges identified in the most recent study cited above by the FAO-Investment Centre are:

- **Competitive marketing needs improved business management skills.** As competition intensifies, dairy farmers' competitive advantage rests largely on their reputation for freshness, safety and quality. With a highly perishable product such as fresh milk, peri-urban farmers have to be increasingly skilled at managing price fluctuations and mitigating losses from low prices during the wet season.
- **High cost of electricity prevents the establishment of a cold chain and hinders growth.** Dairy farmers lack means to utilize renewable energy such as wind or solar systems. There are few peri-urban farmers who have their own cold storage as the majority have not been able to invest in such facilities and need to be able to sell their highly perishable product fast.
- **Poor road networks and lack of cold chain transport facilities impact milk quality and lead to losses.** The peri-urban dairy businesses proximity to the city markets is an advantage. However, it comes at the cost of an increasing need for intensification during the dry season that leads to higher production costs. Moreover, relocating activities further from town markets in order to access free grazing is costly in terms of transport difficulties, delays, road taxes and blocks that often result in losses.
- **Lack of financing limits needed investment for drilling to improve access to water.** Like with other larger investments, many of the farmers lack the capital to be able to drill boreholes or provide other water management infrastructure.
- **The lack of livestock technical extension and knowledge on modern and improved livestock management practices reduces productivity.** The smaller peri-urban farmers, except the ones owned by big companies like Al-Maxruuf, lack basic marketing and food handling skills. In addition, livestock-rearing and farm management continues to follow traditional practices, with stagnant productivity.
- **Under-developed fodder markets slow down intensification of dairy farming.** Most available commercial fodder is from humanitarian emergency stocks or imported from Oman, with the subsector vulnerable to supply or price shocks outside of its control and at risk of being oligopolistic. The latter risk is increasingly visible, as competition among fodder importing companies remains very limited and there is little private to produce fodder locally.
- **Deficient ability to control animal disease leaves the livestock sector highly vulnerable.** There are very limited government veterinary services available to dairy farms for preventing and controlling disease outbreaks, while the quality of private veterinary clinics remains unclear.
- **Institutional fragmentation of the sub-sector undermines the effectiveness of governments' and donors' interventions.** There is no sector body such as an association or a dairy board that could advocate on behalf of and coordinate the subsector.



- **Lack of a clear legal and regulatory framework for dairy food safety poses a public health risk and impacts negatively growth potential.** No quality assurance mechanisms are put in place and incidences of contamination or diseases transmitted through the milk by the animals remain high.
- **Increasing price of peri-urban land prevents outsiders from being able to obtain title and invest in the sector.** While the use of land is free under agreement between the local clan and members of the same, land title is a necessary requirement for the security of the investment in infrastructure and other facilities and assets, when land is to be exploited for an extended period.
- **The peri-urban dairy sub-sector has high entry costs that prevent pastoral herders from entering this new type of business, which is more often than not in the hands of city businessmen owning other ventures, such as hotels and restaurants.** Investors in dairy farms are mostly better off individuals, with more influence and with considerable financial means or access to financing. Therefore, the subsector does not seem benefit lower socio-economic and poverty profile groups, including pastoralists.

Despite many challenges, the market-oriented peri-urban camel and cattle production system is emerging around many towns in across Somalia, where the sale of raw milk is already a very important economic activity. The growth opportunities for milk producers in the peri-urban context are based and depend upon:

- The remaining availability and accessibility to fertile rangeland in some peri-urban areas allows for further expansion of the sub-sector without the need for intensification and the transition to zero-grazing livestock systems, however this is context specific;
- Use of renewable energy like solar or wind power could help reduce energy costs and allow for expansion of cold chain and storage facilities that would improve outputs and economic performance;
- The availability of camels for sale allows newcomers to enter the business as long as they can secure access to peri-urban land and water;
- Strong traditional mechanism for management of use and access to land and rangeland prevent conflicts and secure investments;
- Increasing demand for meat offer new opportunities for diversification of farm income
- Improved livestock management and feeding practices that can impact productivity by reducing seasonal fluctuations. This requires locally available cheaper and more diversified sources of fodder is needed to avert risks of dependence on imports and minimize risks of monopoly;
- Professionalization of the camel dairy sector through improved herd management practices and improved animal husbandry techniques can positively impact productivity without need for expanding herd size.

In addition to improving subsector specific knowledge, this LSDS recognizes the need to address the challenges listed above as well as to foster milk processing to develop this important sub-sector, by strengthening its organisation and management and drawing private investment for its expansion and modernization. This will enhance the subsector's contribution to the socio-economic wellbeing, as well as the nutritional and health status, of the Somali population and enhance inter-sectoral links between food security and nutrition interventions. The immediate priority interventions proposed for the dairy sub-sector from the literature and the stakeholders are the following:



1. Improve veterinary and extension services to enable producers to control the spread of animal diseases, improve the health of the peri-urban herds, and meet also hygienic standards for milk production, handling and transportation and monitor the prevalence of zoonotic diseases;
2. Strengthen government oversight and auditing of milk production, handling and transportation facilities to ensure food safety and quality standards are applied effectively, including by establishing quality control centers at milk collection points and markets;
3. Support the establishment of a Dairy Board to represent peri-urban investors and serve as an entry point for discussion and private sector participation in policy dialogue;
4. Adopt a two pronged approach to the sector's development: promote higher raw fresh milk production and marketing for the rural population but with added public health safeguards (e.g. hygienic handling and retail outlets) and processed milk and higher value-added products, including yogurt made from camel milk, for urban and diaspora consumers (whose total demand currently may be small but set to grow);
5. Foster widespread adoption, including by facilitating access to credit, of dairy preservation and processing techniques, such as cold chain infrastructure, technology, and business management—to support expansion of products with a long shelf-life (e.g. extended shelf life milk, ultra-pasteurized milk, milk powder, yogurt, ghee, cheese), which can reduce spoilage during the wet season when there is oversupply of fresh milk, enhance food safety and higher value addition, and reduce dairy imports.

In the medium- and long-term, government, dairy farmers, processors, and the donor community should also:

- Design interventions in the dairy subsector to link with development plans for fodder, water, and transport;
- Adopt a gender strategy that protects and promotes the interest of female milk traders, as the peri-urban dairy subsector becomes more organized and shifting toward male-dominated employment;
- Design all development interventions by the government and by external partners to reflect the current role of youth in the sub-sector, and support its expansion;
- Support the establishment of commercially oriented dairy farmer's association that can organize the value chain actors and improve information sharing and capacity-building;
- Maintain current policy of exempting needed agro-processing machinery from customs duties and consider other policy incentives such as preferential energy rates or reduced road taxes that can stimulate private investment in modernization and cold storage facilities;
- Pilot public private partnerships for the establishment of collection points with cold storage/cooling facilities using alternative energies to reduce running costs and improve economic viability
- Collect and publish statistics on dairy sector production and consumption, in addition to existing monitoring (by FSNAU) of milk retail prices; and
- Explore opportunities for the niche export of processed and powdered camel milk.

6.3 The Meat Subsector

The meat value chain is still currently almost exclusively focused on supplying the local market (BORESHA, EU, 2018), but with increasing commercialization. Beyond traditional and widespread home-based slaughtering and meat production for shoats, commercial meat production for camels and cattle and meat trade in Somalia has been increasing, in response to growing domestic demand for fresh meat as well as for traditional meat products, such as *mugmud* and *otka* (popular dried meats in ghee). Open-slab slaughterhouses in some large towns, often owned by local governments and managed under often unclear PPP arrangements by private butchers' associations, supply



carcasses to local butchers who retail their meat cuts directly in their shops or indirectly through a network of market sellers, predominantly women. Many small and medium-size towns, furthermore, have no slaughterhouses at all.

Most such slaughterhouses (and meat markets), however, are mostly unhygienic, in various states of disrepair, and lack drainage systems. In most cases, meat is not inspected because state and local governments' veterinary departments either lack a legal framework to enforce (though meat hygiene legislation has been enacted in Somaliland and Puntland and the federal government is in the process of adopting the same, with assistance from the FAO and the EU), or they are unable to enforce regulations because of lack of meat inspectors. The poor hygiene conditions in most meat markets leave meat vulnerable to contamination. In addition to the lack or poor hygienic conditions of slaughterhouses and meat markets, other major challenges related to past failed export ventures of chilled meat were the high transport costs for export by air and the difficulty in meeting international standards. Only SoMeat, a private company equipped with two modern and hygienic slaughterhouses meeting international standards at the outskirts of Mogadishu, is exporting by air chilled and by sea frozen beef carcasses to Oman, while exploring also other markets, including Bahrain.

Promoting meat production and expanding and upgrading local processing and value addition is not only needed to meet the growing domestic demand, but it would also contribute to improved food safety, to local revenue generation, and to employment creation, especially in urban and peri-urban areas, and including for women who dominate its domestic retail trade. Prospects can rely on growing domestic and regional markets for quality "grass-fed" meat from free-range livestock, and on the country's proximity with Gulf states and other North-African (such as Egypt) and West Asian (such as Iran) markets. It should focus on building on existing positive experiences, such as the Burao slaughterhouse renovated under a public-private partnership (PPP) tripartite arrangement involving the municipality, which owns the facility, a private firm, and the community.³⁸ Similar developments under PPP arrangements are under way in Kismayo and Bosaso slaughterhouses. A subsector development strategy should also explore opportunities for expanding the marketing of traditional home-cured (or sun-dried) meat products, such as *mugmud* and *otka*, both within the country and outside.

Only scattered and anecdotal statistics on meat production and consumption and information on stakeholders exist in Somalia, from local meat production facilities for the domestic market to commercial abattoirs for export. However, three meat processing companies already operate in the country. The oldest, the Somali Meat Co. (SoMeat) in Mogadishu has two modern slaughterhouses and exports frozen cattle carcasses to Oman (and in small quantities also to Qatar) in refrigerated containers, with plans to invest in the machinery required for producing and exporting cuts. The other two operations on the outskirts of Hargeisa (Hilibsan) and Galkayo (Beira Food Factory), processing and selling meat in the local markets in a hygienic way, have been recently established but have plans to expand, and in the case of Beira Food Factory also to can meat for both domestic consumption and export. There is also opportunity to sell both fresh and chilled meat to AMISOM troops in Mogadishu and other southcentral locations, which are currently unexploited mainly due to food safety concerns for local meat preparation, storage and transport.

³⁸ After its rehabilitation the Burao slaughterhouse has been handed over in 2018 to the private sector under a PPP agreement and soon will start exporting chilled meat to Malaysia. A similar arrangement has been agreed for rehabilitation and operation of the Kismayo slaughterhouse for camels.



Table 6.1: Opportunities in Selected Animal and Meat Export Markets

Country	Market size	Live animals	Meat
Egypt	- Population to reach 98 million by 2020, coupled with strong GDP growth - Increasing per capita meat consumption. Beef consumption to reach 645,000MT by 2018 - Cattle imports to remain stable at 250,000 heads. Beef imports to grow to 280,000MT	- Significant import market for live animal, with preference for fattening stock and ready to slaughter animals - Renewed opportunities for camel exports - Egypt actively sourcing for animals (cattle) from African countries.	- Consumer preference for fresh meat, mostly beef - Chilled meat imports mostly from India (buffalo) and Brazil (beef) - Imports from Somalia are negligible - Focus on affordability
United Arab Emirates	- Population around 8.9 million and expected to increase to 9.5 million people by 2022 - Average poultry p/c consumption: 33.9 kg/yr; lamb/mouton/goat: 9.2 kg/yr; beef: 5.4 kg/yr - Projected total demand for meat by 2020: 513 million kilos	- Upward trend in live animal imports - Imports dominated by goats, then by sheep, camels, cattle - Somalia had very high share in live imports in UAE, with comparative advantage in bovine animals and shoats (respective market shares for 2015 were 96% and 24%), before import ban - Most animals from Somalia to the UAE are shipped via Bosaso, Puntland - Large poultry imports, but none from Somalia - UAE is actively promoting offshore livestock production, backed by investments in African and Asian countries	- Highest per capita spending on poultry as compared to other types of meat - High reliance on meat imports - Imports from Somalia are negligible (1% for sheep and goats meat –chilled or frozen) - Imports meat from a range of countries - Increase in camel meat and milk consumption - High per capita spending, large expat community - Rising demand for halal meat
Saudi Arabia	- Population to increase to 35.6 million by 2022 - Demand in 2017 was 1.742 million kgs; to increase to 2.069 million kgs by 2022	- High reliance on live animal imports for supply of meat and religious activities (with imports from Somalia of sacrificial animals still at 1.2 million annually) - Before latest import ban, Somalia had significant share of Saudi livestock trade: 36% for shoats; 37% for cattle; 31% of all live animal trade - Most animals exported from Somalia to Saudi Arabia are shipped via Berbera	- Meat imports were estimated to contribute to just over 1/3 of consumption - Slaughter of live animals contributes to about 60-70% of consumption - Somalia is not capturing opportunities for meat exports to Saudi Arabia
Yemen	Population at over 25 million in 2016 and growing	- As a non-GCC member, Yemen has less stringent SPS requirements and does not participate in GCC bans. - Yemen imports large numbers of animals, many unreported, for re-export to GCC countries with strict SPS regulations - Dominance of informal unrecorded trading from Somaliland ports, often through clan relationships - Relaxation of grading requirements during Haj (for re-exporting to Saudi Arabia)	- Meat deficit country - Low consumption levels of imported meat

Source: Adapted from FAO (2017) Assessment of selected export markets for Somali livestock; FAO-Somalia on Oman.

The opportunities for exporting meat to Gulf States and in the broader East African region, where demand is projected to increase, have been highlighted in a recent study by the FAO (2017) and in other reports by external development agencies. More information on demand of livestock and livestock products in four of Somalia's most important importing Arab countries is presented in table 6.1. Other countries in the Gulf already importing frozen meat from SoMeat include Oman and Qatar, while Yemen remains an unexploited market. Other unexploited regional markets include the Democratic Republic of the Congo (DRC), Uganda, and most importantly Kenya, which is only about



70–75 percent self-sufficient in meat and whose import and re-export markets are also positioned to grow.

Nonetheless, meat exports prospects need to be further analyzed, including quality perceptions, competitiveness and food safety issues of Somali meat in the Gulf, Middle East, North African, West and South-East Asian and regional markets, vis-à-vis other countries like Pakistan whose exports have already displaced Somali meat in the UAE market. It also imperative to understand better the issue of year-long availability of sufficient supplies to conquer these markets, addressing e.g. seasonality, domestic livestock prices, competition with live animal exports.

Furthermore, an expansion of meat exports beyond Mogadishu (where frozen meat exports in refrigerated containers by sea are feasible) may not be cost-effective, due to high air cargo rates for chilled meat, except for frozen meat exports from Berbera, the only other container port at this time in the country. Any such export-oriented expansion, in any case, would require considerable technical and financial support to overcome the reasons for previous high investment failure rates, due to poor operational management, livestock supply constraints, and lack of compliance with Hazard Analysis Critical Control Point (HACCP) and other international quality and food safety standards.

The LSDS recommends the following priority interventions for the meat subsector's development:

1. Approve and implement a Federal Meat Inspection Act, which has also an integral component of disease notification and related information;
2. Support public and private investments and PPP arrangements for the establishment and management of modern hygienic slaughterhouses and cold chains in all urban centers, including to prevent the congregation of stray dogs and cats around slaughter sites and to remove and dispose properly of waste (bones, skulls, hides, skins, etc.) which litter slaughter sites and nearby beaches;
3. Clarify roles and responsibilities of slaughterhouse investors, supervisors, customers under PPPs;
4. Improve integration of traditional producers with local slaughterhouses and cold chain systems;
5. Study the constraints and opportunities for developing a thriving offal industry;
6. Offer training on proper food (meat) handling practices for stakeholders across the value chain.

Other recommendations for the medium-term to support the development of the meat subsector are:

- Improve public services provided by government to meat handlers and processors, or reduce state and local taxes on meat processing;
- Maintain current policy of exempting needed agro-processing machinery from customs duties and consider other policy and tax incentives that can stimulate private investment in modernization, higher value addition (e.g. for meat cuts), cold storage facilities, and waste disposal;
- Strengthen government oversight of food (meat) safety practices to comply with international standards;
- Build capacity of the Somali veterinary system to ensure competency and credibility in (meat) certification and in meeting international trade standards and requirements;
- Offer technical support and facilitate access to credit for private companies already operating and new entrants in this subsector on: (i) environmental impact and handling of by-products,



- (ii) alternative power sources to reduce the current high cost of electricity, (iii) producing cuts and boneless meat instead of only carcasses as at present, and (iv) waste disposal;
- Collect and publish comprehensive and consistent statistics on the meat subsector, beyond prices; and
- Study, and disseminate findings on, market demand and structure, competitiveness, and entry requirements in major Gulf and regional markets related to frozen and chilled meat exports

6.4 The hides and skins subsector

Hides and skins, cured mostly to sun-dried or dry-salted stages with only some processed to wet-blue stage, though of very poor quality, barely register today in the trade statistics because of the extremely low price they fetch. They are nonetheless still exported to Dubai, Pakistan, UAE, China, and Italy, suggesting potential international demand for a revival of exports from Somalia. Animal slaughter facilities, skills and flaying practices (of skins from the animals), however, are so poor that they result in poor quality of hides and skins, thus in a high rate of rejection by importers and/or in very low prices (e.g. US\$0.3 per skin). Diseases such as sheep and goat pox, lumpy skin, mange and parasites including ticks also contribute to rejection rates as high as 50% due the pre-slaughter defects. The situation is worsened by pervasive cultural practices, including branding and coloring of animals after sale in markets which is destructive to their coats (Wanyoike F, 2018). These quality issues are largely attributed to the fact that hides and skins are handled by non-specialized labour, whether in small quantities at the household level or at larger scale in slaughterhouses and processing facilities that use mostly second-hand and poorly maintained equipment.

To the challenges mentioned above, the sub-sector faces many others that negatively impact current prospects for international competitiveness and growth, related to slaughtering, processing to dry salted, wet blue stages and leather stages, handling associated chemical waste, and marketing. They include lack of a policy and regulatory framework to guide business investment; skilled manpower, technical knowhow, technologies, modern machineries, and financing to support commercial operations; and grading, quality assurance, certification and knowledge export markets requirements to help with marketing (FAO 2009). Furthermore, only scattered information exists on the hides and skins subsector in Somalia, including on the number of actors and production levels at household level, traders and processing facilities. The potential international market demand for leather and leather products from Somalia beyond the pre-war wet blue processing stage for its hides and skins in its traditional and in new (e.g. China) export markets is also neither documented nor understood.

Nonetheless, improving the quality of current production and marketing of hides and skins would provide significant opportunities to support and sustain livelihoods, especially in rural areas, as was indeed the case before the war. In 2017, the FAO commissioned under the ESOLT Project three techno-economic assessments, one of which mapped value addition opportunities in the Somali livestock industry with a case study on hides and skins opportunities for Puntland. This assessment highlighted the importance of the subsector in all regions, and its high potential in generating economic benefits and employment, if skins were treated again as valuable commercial assets and if private stakeholders would establish suitable business structures.³⁹

³⁹ The other two assessments covered: a) the livestock holding ground at the port of Mogadishu and b) animal shed for Hargeisa.



The LSDS priority recommendations for reviving this subsector combine both those from the recent FAO assessments and the October-November 2018 Somali public and private stakeholders, as follows:

1. Undertake value chain studies that look at the economics of production and marketing of hides and skins, as well as assessing requirements and preferences in existing and potential export markets;
2. Collect more statistical information about the subsector (starting under the planned livestock population census/survey), as well as about the constraints of attracting foreign investment by importing countries
3. Formulate a policy and regulatory framework to guide both private investors and public regulators of a possible future tanning and leather industry, while also taking into account its environmental impacts;
4. Formulate subsector standards that are based on international requirements, and ensure optimization of local product quality and improved pricing of products;
5. Support and strengthen organizational capacity of hides and skins trading associations
6. Encourage private investment in technology upgrading of existing and new slaughterhouses and tanneries, with government playing a facilitating role, including with tax incentives or rebates for modern equipment;
7. Expand access to finance for traders and processors in hides and skins trade, using also targeted incentives.

Other recommendations for the medium-term development of the subsector are to:

- Establish inspection and grading systems and enforce standards (by the federal and state ministries, in close collaboration with states' chambers of commerce) to ensure that local suppliers and traders of raw hides and skins value and respect standards to meet foreign requirements;
- Provide training and extension services to value chain actors in appropriate techniques and practices, including modern flaying techniques and tannery technology to enhance product quality, value and international competitiveness;
- Encourage value addition through tanning of more hides and skins into finished leather and the manufacturing of leather products such as footwear for the domestic market;
- Assess international market demand for Somali hides and skins processed beyond wet blue before proposing public or private investments for advanced-stage leather exports.

6.5 The poultry subsector

Poultry production and marketing in Somalia has been mostly neglected. The sub-sector today, despite potential, remains underdeveloped, with poor linkages between producers, market and consumers. At present the domestic poultry meat sector cannot compete on price with imports of frozen chicken meat in large urban markets, but there is a niche market due to strong preference by local consumers for indigenous breeds and more naturally-fed chicken meat. As demand is projected to continue to increase, several new businesses have invested in poultry farms (some broilers but mostly layers in the Benadir region, with the largest egg production in the country, and also in Lower and Middle Shabelle regions of southcentral Somalia to supply Mogadishu consumers; layers in Somaliland). Because of these recent developments, the poultry sector has gained a significant role in Somalia's food security and protein availability.

Both semi-intensive poultry farmers and these nascent large-scale poultry farms face many challenges with access to affordable feed, electricity, equipment, drugs/vaccines, cold storage,



packaging, transport, water, and modern production practices. Poor knowledge of good poultry farming techniques, limited availability of improved breeds adapted to the local environment, and poor marketing information were other constraints mentioned by stakeholders. In addition, there is no systematic or up-to-date analysis of the Somali poultry sector in the literature and thus the knowledge of this subsector is very poor.

Poultry production and marketing in Somalia has been mostly neglected. The sub-sector today, despite potential, remains underdeveloped, with poor linkages between producers, market and consumers. However, because of an expanded commercial production and related consumption, poultry has become increasingly important to Somalia's food security and protein availability. At present the domestic poultry meat sector cannot compete on price with imports of frozen chicken meat in large urban markets, but there is a niche market due to strong preference by local consumers for indigenous breeds and more naturally-fed chicken meat. As demand is projected to continue to increase, several new businesses have invested in poultry farms. These produce some broilers but mostly layers in the Benadir region, which has the largest egg production in the country, in Lower and Middle Shabelle regions to supply Mogadishu consumers, and in Somaliland to supply Hargeisa consumers.

Both semi-intensive poultry farmers and these nascent large-scale poultry farms face many challenges. These relate to poor access to affordable feed, electricity, modern equipment, drugs/vaccines, cold storage, packaging, transport, water, and modern production practices. Poor knowledge of good poultry farming techniques, limited availability of improved breeds adapted to the local environment, and poor marketing information were other constraints mentioned by stakeholders during the October-November 2018 consultations. In addition, there is no systematic or up-to-date analysis of the Somali poultry sector in the literature and thus understanding of this subsector is very poor.

While the NDP references poultry as a possible growth sector, there are no targets or estimated investments allocated to it; and the fourth Peace-building and State-building Goal (PSG) among the economic foundations of the New Deal for Somalia agreed by the federal government with the donor community in the mid-2010s did not prioritize poultry as a growth sector. As a result, there has been limited government support to develop the business enabling environment for the poultry sector. Few poultry-specific services have been developed by the private sector, and there are few, if any, lead firms operating on a commercial scale, providing limited opportunities to play a facilitative role in implementation. Thus far, there has also been minimal donor support or research in the poultry sector to build upon and there is no reliable data to provide any evidence that the sector will be able to compete with imports in the near to medium term future, which risks that interventions promoted by the government may not be scalable.

Despite the poor knowledge of the sector and notable challenges mentioned above, government and private stakeholders believe that the available high market demand underlies a strong potential for expanding and upgrading the poultry subsector. Such development will have widespread benefits, to include higher income for producers and traders; higher revenues for all levels of government and poor households; creation of employment opportunities for women; and improved food security, quality and safety for consumers.

Amidst such major economic and knowledge challenges, to realize a vision for this subsector to play a big role in contributing to food security and to supplement traditional Somali livestock as a second form of livelihood, various interventions can nonetheless be recommended on the basis of



stakeholders' suggestions and available studies (see Annexes 2-8; DFID/DAI 2016; and World Bank and FAO, 2018). These are mainly focused on modernizing and professionalizing the entire value chain, as follows:

1. Create an enabling legal and regulatory environment for private investment in the sub-sector;
2. Explore PPP arrangements for the development of a local chicken feed mill industry to reduce costs of feed;
3. Promote investments in hatcheries, if it is determined that there is sufficient demand;
4. Expand poultry health services, by working with regional specialists to provide advisory support;
5. Develop appropriate policies, regulations and guidelines for more formalized poultry sector hygiene, production and packaging standards;

In the medium-term, this LSDS recommends the following priority actions:

- Conduct a full-scale poultry sector mapping and feasibility study, in order to assess the potential economic, environmental and social sustainability of poultry sector development in Somalia (layer, broiler, and traditional system where appropriate), considering the recognized constraints to sector development caused by shortage of production inputs (including production of day-old-chicks, and compound feed), technical support and skilled labor, as well as the stiff competition faced by imports.
- Revive and expand the production of maize, sorghum and promote that of soya bean for poultry feed, as poultry development goes along with poultry feed production (grains and soybean meal) and improved breeds (semi-scavenging and/or commercial layers and broilers).

6.6 Honey production and beekeeping

As already mentioned in chapter 3, there is only anecdotal evidence about beekeeping, which remains a small-scale side business for most beekeepers, and about honey, whose consumption is limited to local markets. There are no large-scale/modern honey production farms, but only a few small beekeeping operations in rural areas throughout the country that use traditional hives and honey harvesting kits. Small-scale beekeepers are not organized in associations, with limited knowledge of good beekeeping practices, and no access to affordable modern production and personal protection equipment. Other constraints are lack of feed (deforestation) and water (in dry season).

Aggregation of honey through formation of honey producer groups/cooperatives coupled with standardized honey processing through refining and commercial marketing has potential to greatly improve the local and national economy. A rapid assessment of the impact of improved beekeeping interventions introduced by the FAO in Somalia over the last six years has demonstrated several socio-economic benefits of beekeeping, especially to rural women (FAO 2015). Unlike other agricultural enterprises, beekeeping is less affected by climate-change and greatly reinforces women's income sources from honey sales during drought periods when there is no other source of income. Beneficiaries use income from honey sales to purchase food, invest in education and other household items, thus diversifying their livelihoods options.

Furthermore, honey is a popular product, widely believed to possess medicinal properties and thus locally in high demand. Prices for raw honey in recent years have been very high, thus profitable for beekeepers. Stakeholders' recommendations for the development of this subsector are for the public sector to:



1. provide extension services to current and prospective beekeepers on good production, handling, and preservation practices as well as on market opportunities beyond nearby towns;
2. facilitate their access to modern production and protective equipment;
3. support the establishment by the private sector of honey refineries, to add value; and
4. launch an awareness campaign to educate the wider public on beekeeping as a supplementary source of alternative income and livelihood, and of its benefits in pollinating trees and crops.

6.7 Domestic marketing of livestock products

There is currently no comprehensive analysis done on the consumer markets across the country, including for livestock products, defining the qualitative and quantitative characteristics of the growing urban market demand or the formal and informal market supply network that provides for consumers across the rural areas. Livestock product markets, nonetheless, play a critical role in the Somali economy, affecting both the food security and livelihood of the population. In rural areas, pastoralists, agro-pastoralists and agriculturalists all depend on markets to varying degrees in gaining access to income and food. People rely on the sales of livestock and livestock products for a significant percentage of their cash income and to cover substantial parts of their food needs through locally and imported cereal purchases.

Markets and trade (both domestic and foreign) have, in general, proven to be dynamic and resilient since the collapse of the Somali State in the early 1990s. While the trade in live animals is closely integrated with neighbouring countries and within the greater region, most trade in livestock products takes place locally and domestically. Transport and domestic marketing of livestock and its products are entirely private sector affairs, but state and local authorities as well as non-state actors intervene in collection of taxes and fees for services (such as health certification following inspections). Livestock and livestock products marketing, aside from supply challenges, is constrained mainly by the lack of necessary access roads, storage and market infrastructure, as well as by lack of financing.

As the draft livestock sector strategy of 2004 already noted, taxes on livestock exports and milk and meat transport and sales are the major source of public revenue for the areas where there are functioning administrations at both regional and municipal levels (FAO et. al, 2004). Stakeholders, however, recently reported an appalling lack of budget for livestock ministries and related salaries and services. These administrations, thus, should be encouraged, perhaps through some of the financial instruments and agreements that the international community is expected to participate in to re-invest a percentage of these revenues into the livestock sector by funding important services that will promote efficient and smooth exporting and upgrading of public food safety services and facilities. Potential direct beneficiaries or future “managers” of these taxes would be a possible Somali Livestock Board, Chambers of Commerce, municipal slaughterhouses, professional associations. “Unofficial” taxes (roadblocks and checkpoints) imposed along major routes in areas where public administration is weak should be minimized and eliminated.

Domestic marketing and trade of dairy, ruminants’ meat and poultry products are the priorities that need targeted support to enable producers to access the market and satisfy a growing national food demand, with the domestic market believed to be able absorb much higher dairy, ruminants’ meat and poultry production than at present. Major goals for such support are to foster competition at all levels of the market chain, while reducing risk and inefficiencies at all critical points. Other key objectives are to encourage more diversification of markets and especially an



increase of value-addition via processing. Among the priority interventions recommended are the following:

1. Ensure coordination with relevant trade authorities, such as the ministries of trade and affiliated bodies in the collation of livestock and livestock products relevant trade data and research;
2. Survey domestic demand trends and the current state of market infrastructure, including mapping of key markets, related facilities, main players (companies, agents, etc.), and investment needs;
3. Review the extent of state and local taxes, as well as “unofficial” charges, on traders of livestock products, with a view at re-directing at least part of the official tax revenues from livestock to enhancing essential services and infrastructures for the sector, and eliminating all “unofficial” charges at roadblocks;
4. Provide clear guidelines related to the establishment of public-private partnership arrangements for investment in and for management of critical market infrastructure and services;
5. Support and facilitate the coordination of stakeholders, through the establishment of representative consultative bodies that can engage all types of traders.

In the medium term, other recommendations to strengthen domestic marketing of livestock products are to:

- Conduct a comprehensive review of key imports that compete in domestic markets with local produce and where deemed viable consider introducing policy incentives towards import substitution; and
- Ensure strong linkage of strategies and interventions in production with strategies and investment in domestic marketing and market access improvements.

6.8 Quality standards and food safety of livestock products

The biggest hurdle Somali livestock and livestock products face is compliance with food safety requirements. Regulations such as the Meat Hygiene and Inspection Act and Dairy Hygiene Act have been enacted, but these regulations are not enforced by the government through inspections or audits of food business’ food safety practices. A Food Safety Act, however, has yet to be drafted to provide guidance to businesses on applying effective control throughout the food chain and on ensuring withdrawal from markets of unsafe food.

The responsibility of maintaining food safety and hygiene falls on the food industry which is however constrained by the lack of trained veterinarians and food scientists with the required expertise to manage their quality assurance departments. The exception is the Somali Meat Company (SoMeat), which is currently seeking ISO 2200 food safety certification for its meat processing facilities. None of the other food industry players in Somalia have gained any international food safety certification. With support from various donors, the emerging milk processors in and around Mogadishu are requesting their suppliers to provide milk free from contamination, by conducting themselves quality control tests to incoming milk. These processors, however, are still struggling to secure their supply chain and ensure the safety and quality of their milk sourcing practices.

Questions asked by potential importers, for example during the 2018 Gulfood Expo, include availability of documentation as proof of the quality assurance of the Somali product, as well as HACCP product certification. Support services such as rebuttable certification body and ISO 17025



certified food testing laboratory, however, are non-existent in the country, thus depriving the Somali food industry of a system to verify the quality and safety of its products.

To enhance the competitiveness of Somali livestock and livestock product in both local and international markets, the following interventions should be undertaken:

1. Develop and enact a Food Law and establish National Food Safety Commission;
2. Establish a Food Inspectorate to strengthen enforcement of food regulations;
3. Establish a National Food Safety Laboratory to conduct both microbiological and chemical analysis;
4. Increase supervision by the federal MoLFR over the activities of all private quarantine stations;
5. Provide training to laboratory staff who should undergo proficiency testing; and
6. Train food industry staff in the preparation and implementation of pre-requisite and HACCP.

7.0 CROSS-CUTTING ISSUES

7.1 Access to Finance

Somalia's financial sector is poorly developed except for its payment system, as many years of civil unrest and lack of functional government destroyed the foundation for financial intermediation. There are many limitations in financial, technical and managerial capacity to provide basic financial services, and most of the rural population is still unbanked, with overall penetration rates of banking services at 16 percent. So far, eight banks have been registered to offer financial services in Somalia, but three of them are yet to start operations as at January 2019. There are several microfinance institutions (MFIs) operating in Somalia either as affiliated subsidiaries of commercial banks or registered as nongovernmental organizations (NGOs). In addition to the six main MFIs, there are many other local and international NGOs providing microfinance products. Microfinance regional coverage seems to mirror affiliated commercials, while the NGO-based MFIs are usually concentrated in major urban centers such as Mogadishu, Hargeisa, Garowe and Kismayo. Both domestic and international payment services, however, are well developed, with over 67 percent of the population relying on money transfers via mobile network operations (MNOs) and 11 money transfer businesses (MTBs), the latter relying on agents spread all over the country including in rural areas and transacting over US\$2.7 billions per month.

Formal financial institutions in many developing, including neighboring, countries mostly exclude pastoralists from their services due to pastoralists' inability to pledge collateral, and livestock insurance is not available at all. Furthermore, traditionally women in pastoral and agro-pastoral communities have faced even greater barriers than men in accessing finance. Pastoralists' needs are unlikely to be accommodated by offering the same financial products as other groups of the community (Getu, 2017). Financial inclusion in pastoral areas through formal institutions is constrained by limited financial illiteracy, pastoralists' mobility, high transaction costs, tribal conflicts, and limited financial capacity. On the other hand, interest of financial institutions to operate in rural areas (at least via their agent and outlets), expanding mobile and interest free banking services, and start of sedentary way of life promise better financial inclusion for livestock keepers (Getu, 2017).

In fact, few financial services, with the exception of deposit taking and money transfers, are available to Somali livestock sector owners and traders, whether men or women. Those involved in livestock trade and other related business activities continue to rely on informal sources to fund their business activities, limiting the speed at which those businesses can grow. Informal groups and



institutions, often organized along clan lines, are less affected by cost considerations and other challenges that deter formal institutions from providing finance to livestock keepers and entrepreneurs. Groups can also provide a savings option – a service many micro-finance institutions may not offer – and groups can tailor loan sizes, procedures and terms to the preferences of their membership. However, institutions such as Amal Bank’s micro-finance section claim that they don’t provide loans for livestock related investments because of high risk of animal loss due to droughts and disease outbreaks. There is also limited understanding of loans and insurance services, which contributes to low levels of using such services.



Box 7.2 - Kenya Livestock Insurance Program (KLIP)

Pastoralism plays an essential role in Kenya's ASAL region, but the region is vulnerable to recurrent drought. The livestock sector accounts for over 80% of household's incomes in Northern Kenya. The region is exposed to recurrent droughts which decimate livestock herds, increase poverty and increase the incidence of stunting. During the catastrophic droughts in 2008 to 2011 the Kenyan economy lost an estimated US\$12.1 billion, over 70% of which in the livestock sector.

Index Insurance solution. For the past four years, the Government of Kenya (GoK) partnered with the private sector insurance companies to rollout a livestock insurance program targeting vulnerable pastoralists using index insurance (KLIP). The Government pays insurance premiums for pastoralists, and private insurance companies develop the insurance policy and pay claims when they arise. KLIP was launched in 2015, has provided protection to over 31,930 households reaching over 200,000 people and has paid out three times a cumulative amount of US\$7.2 million. KLIP now covers 8 counties and target all 14 pastoral counties by 2020.

Simplicity, innovation and speed. KLIP insures pastoralists against the loss of their livestock in case of weather events such as drought. Insurance payout is triggered using satellite imaging to monitor the level of pasture on the ground. When the level falls below an historical average (as determined in the insurance contract), the payout is triggered automatically. KLIP provides protection starting at the beginning of a season, and if a drought strikes, payouts happen quickly to beneficiaries to allow them to buy water or animal feed to keep the livestock alive. It is all about prevention, rather than cure, and speed of intervention. Index insurance allows extending access to insurance services to people who are de facto excluded, as it eliminates the need for insurance agents to be out in the field monitoring forage and animals, which, given the remote regions involved, would make livestock insurance logistically and financially impossible to provide. KLIP also leverages from Kenya's innovations in the financial sector with M-PESA (mobile money) and agent banking. Insurance companies make payment directly to the beneficiaries primarily by M-PESA, and by using the network of agents developed by some banks in the northern counties.

Crowding-in private sector. By using the private sector insurance companies to manage the risk of payouts, KLIP is crowding-in private sector capital as an effective risk management tool. Instead of facing large and uncertain expenditure from their budget during drought years, GOK now pays a fixed annual premium and pass the unknown cost to the private sector, who are better equipped to manage.

Significant Impact. KLIP has successfully responded to the recent El Niño drought in northern Kenya counties stretching from October 2015 to 2019. An impact study of KLIP released in March 2018 showed that KLIP had a significant impact on communities. Many beneficiaries reported using the payouts on household needs such as buying food for household or paying for school fees (94%) and on their livestock needs such as buying supplementary feed, medicine, water and restocking (82%). Sharing of payouts with community members or spending them for digging a well for the entire community was reported. Another beneficiary used part of the payout to buy a male goat to serve the entire village by mating their female goats.

Lessons learned. Success was due to: i) government supporting premium payments; ii) private insurers processing and paying approved claims quickly to beneficiaries; iii) existing payment infrastructure. Going forward, awareness creation of pastoralists needs to be stepped up, to stimulate the voluntary uptake of the product and ensure commercial sustainability. Other countries (Rwanda, Uganda, Ethiopia) are now interested in rolling out a similar program.

Providing inclusive financial services to all livestock and livestock products stakeholders in Somalia, including the poor, is an important element for tackling poverty and bringing about inclusive development. Access to finance would open opportunities for pastoralists to diversify their livelihood options by facilitating restocking of lost livestock, purchasing inputs, establishing small livestock-related microenterprises, and considering other sedentary economic activities, including value addition. There is a need to develop an appropriate strategy to increase financial access throughout the sector's value chains.



Opportunities for improving access to finance in the livestock sector include:

1. Increased interest by financial institutions and mobile money providers to expand their business. Banking services and mobile money usage in Somalia is increasing rapidly, providing opportunity to bring more people and businesses to formal financial services. The systems for providing appropriate financial services are developing smoothly. Such development is important for general access to financial services including the livestock sector. There exists deliberate move to sort the challenges that limits access to finance including issuance of digital Identification Cards which will facilitate opening and managing bank accounts. However, a deliberate effort to support financial institutions and mobile money operators is necessary including grants for supporting financial innovations.
2. Improving financial literacy and awareness creation. The collapse of financial services in the late 1980s (with the collapse of the bankrupt and corrupt Commercial Bank of Somalia, the only retail banking institution in the country) left the majority of the population with limited option on financial intermediation leading to reliance on informal financial mechanism. Investment by *hawale* companies and mobile money providers after the civil war has made deposit and payment services available again, but people are still reluctant to start trusting and using formal financial services.
3. Digital financial services provision. While the usage of mobile money reaching 76 percent of the population provides a huge opportunity for digital financial services (DFS), they are currently dominated by deposit taking and money transfers with very limited or no other financial services like savings, credit and insurance. With financial literacy and awareness creation, the saving and borrow culture could be stimulated creating market for such products.
4. Adoption of risk transfer instruments like livestock insurance. Risk inherent in livestock production limit access to financial services, especially credit. Livestock sector is highly exposed to drought shocks which destroy the foundation for income generation. Risk transfer mechanism is proving to be useful tool for managing drought shocks in pastoral areas. For example, the Kenya Livestock Insurance Program (KLIP), a livestock insurance scheme implemented under private public partnership pays the insured before drought become disasters (see Box 7.2). The insured households have financial resources in opportune moment to take active measures to protect their livestock. Under the request of the FGS, the World Bank is exploring the technical, operational and financial viability of such scheme for Somalia, as well as the need for extensive education of potential beneficiaries on insurance, from basics to specific available options.

7.2 Human Resources Development

Building and strengthening capacity of both public and private actors, including educational policy, regulations and institutions, is a key element of the livestock sector strategy. However, as elaborated in a background paper by Mohamud and Keilie (2018) that reviewed the state of livestock-related education and training across the country, technical secondary and higher veterinary and animal production related education lack institutional regulations and standards usually set by governmental authorities responsible for education at the local, state or federal level. Neither the federal government nor its federal member states have a higher education commission currently in operation. Higher education in Somalia currently lacks the institutional infrastructure, resources and systemic governmental and institutional checks and balances that ensure effective institutional operations. As a result, the quality of teaching and learning is relatively weak and inadequate to meet the educational demands of the Somali society. The education system lacks the necessary resources to train or hire qualified instructors, secure adequate infrastructure.



The Somali National Education Act is currently under review and consideration in the Federal Parliament and may take some time before it is enacted. There is thus no effective education policy that guides livestock educational institutions' management and administration, curriculum development, student enrolment and qualification of faculty.

Most schools of veterinary sciences and animal husbandry lack the human, budgetary, and physical capacities to prepare qualified graduates with skills, field training and laboratory experience necessary to function independently. As a result, the vast majority of the professionals overseeing today veterinary clinics, quarantine stations, vaccination and treatment programs, disease surveillances, poultry and dairy farms are former veterinary technicians trained under the pre-civil war government of Somalia.

Unlike other fields of education, veterinary schools have the added disadvantage of not attracting large enrolments, and especially female students. Veterinary school concentration in urban centers adds to the existing disconnection between the pastoral and agro-pastoral communities and veterinary schools training technicians or treating their animals. Fostering female enrolment and bringing veterinary schools and clinics closer to the pastoral and agro-pastoral communities will close existing gaps in veterinary service delivery.

Regarding further qualifications, the ESOLT project, with technical support by the FAO and funding support by the European Union, supported the training on quarantine management practices and operations as well as laboratory diagnosis and interpretation of results of common transboundary animal diseases. This targeted Somali veterinary personnel from federal and state line ministries, export quarantines, diagnostic laboratories and university veterinary graduates and has contributed to improved knowledge of professionals on sanitary, food safety and quality standards. In addition, under the ESOLT project, 120 butchers and meat traders all over Somalia were trained on quality improvement of hides, skins, leather and leather products

The following challenges need to be addressed:

- inadequate facilities such as classrooms, laboratories, libraries and research centers;
- limited qualifications and professional and academic experience of the teaching staff;
- absence of technical veterinary institutes or colleges in rural community areas;
- rampant unemployment for veterinarians graduating from most universities;
- large number of academically unprepared university students graduating from secondary schools;
- inadequate veterinary clinics for field practice;
- limited budgetary resources for veterinary educational programs;
- insufficient number of internal and external student scholarships, and especially so for veterinary sciences and animal husbandry studies; and
- poor information sharing, coordination and networking among livestock-related educational institutions

To improve the skills of the officials, professionals and technicians involved in the livestock sector, this LSDS offers the following priority recommendations in order to address the above-summarised current constraints:

1. Enact national education legislation and establish higher-education commissions at the federal and state levels to strengthen the role of government in education and enhancing the quality of programs offered, including in veterinary science and animal husbandry.
2. Standardize the academic curriculum;
3. Enhance monitoring, evaluation and enforcement of regulations over educational institutions;
4. Build capacity of faculty to improve quality of teaching and learning;



5. Provide professional development opportunities and enhance veterinary faculty networking among Somali institutions to improve teaching and research staff and graduating student knowledge;

Other recommendations for the medium-term are to:

- Strengthen employment capacity in both the public and private parts of the livestock sector, including by establishing modern veterinary research institutes, enhancing the capacity of veterinary laboratories for disease investigation, diagnostics and research, and promoting modern veterinary clinics, and field practice sites with animals;
- Establish new veterinary technical secondary and promote veterinary colleges near pastoral and agro-pastoral communities, to reduce high concentration of veterinary programs in urban centers; and
- Establish veterinary education associations for information sharing, networking, research and professional development and a central library with technical and scientific books on veterinary science and technology.

7.3 External Assistance for Sector Development

This section summarizes either the evaluations or the summary self-assessments of project and program experience by major international development partners, in an attempt to provide an overview of the focus, objectives, relevance, implementation challenges, and lessons learned from external development assistance during the last decade. The organizations listed below contributed either project or program evaluations, or summary self-assessments based on a common template.⁴⁰

Project Evaluations		
<i>Project</i>	<i>Organization</i>	<i>Document</i>
ESOLT	EU/AU-IBAR	report
MVC	EU	report
RAHS	EU/AU-IBAR	report
LIP	GIZ	template
GEEL	USAID	template
Regional works	IGAD-ICPALD	template
10-year review	FAO	report

7.3.1 Project Objectives, Focus, and Outcomes

External assistance during the past decade was concentrated on Somaliland, Puntland, and the federal government, with only one project covering southcentral Somalia.⁴¹ The projects aimed mostly on strengthening animal health systems, the policy and regulatory environment, and private sector development, contributing to building resilience, introducing technical innovations, facilitating successful emergency responses, enhancing supply chains, and aiding in access of markets. The main topical areas were animal health and disease surveillance and control

⁴⁰ A template with general questions about the project, including project overview, outcomes, and lessons learned, among a few others, was sent out to the major project implementers. Only three responded using the template while the others sent their own evaluation reports that were completed previously. Thus, seven program self-assessments and evaluations were collected and reviewed, with IGAD-ICPALD sending an evaluation that covered various country and regional projects and the FAO sending a 10-year review of 60 livestock or livestock related projects that it implemented in Somalia.

⁴¹ However, the FAO program review did not provide information on the geo-political areas covered.



(IGAD/ICPALD, RHAS), trade and market linkages (IGAD/ICPALD, GEEL, ESOLT) the feed and fodder subsector (GEEL, LIP), livestock productivity (LIP), emergency response to disasters (LIP), resilience (LIP, ESOLT, MVC, WALP), improving compliance of standards, increased certification, and expanding and upgrading small-scale infrastructure (e.g. RHAS, ESOLT, WALP).

Activities to accomplish these goals included training sessions, workshops, knowledge and technology sharing, bringing together stakeholders, capacity building, creation of cooperatives, and trade facilitation. Projects often provided technical and financial support to ministries to strengthen linkages and collaboration with livestock professional associations, training establishments and institutions, and the private sector to enhance service delivery and training. Other financial support often included infrastructure or equipment for trade facilitation, central and satellite veterinary laboratories or inspection posts, cold chain infrastructure like solar based village milk cooling facilities, new livestock holding grounds and quarantine facilities, and infrastructure for better water management. For example, the MVC project strengthened the dairy sector through improved governance and management of 25 dairy cooperatives (of which 13 were newly created) through trainings, and improved market linkages through trainings on milk hygiene and cold chain facility operation and management, along with the establishment of 6 satellite cooler centres/milk hygiene units.

While evaluations and self-assessments often cited outputs such as development of small-scale infrastructure, they also listed intermediate outcomes such as improved standard methods and procedures and compliance, strengthened disease early detection and surveillance, and improved trust and transparency across borders. They did not claim however, achieving broader final outcomes such as improved market access and linkages, strengthened government capacity, increased trade and incomes. The intermediate outcomes were due to strengthened capacities of public institutions to deliver and regulate animal health services through animal disease contingency plans, veterinary boards, animal health strategies, public, private and community partnerships in animal health services delivery and improved surveillance and control systems, and improved access to water. The WALP project, for example, facilitated the Government building of eight water sources that reached an estimated over 42,000 people with “improved water sources” and capable of providing water to 200,000 livestock, with the additional achievement of building key institutional capacity through technical, administrative, and community mobilization skills.

Other general outcomes include more private sector engagement through capital investment, strengthened dairy and feed and fodder sectors, and community-led resilience building through innovative programs such as a camel leasing model (GEEL), improved market access, and increased incomes from improved production and sales of livestock. Other outcomes include improved livelihoods of beneficiaries through education, knowledge sharing, and workshops as well as increased collaborations among pastoralists and commercially-oriented private sector enterprises. For example, the ESOLT project cited enhanced networking, negotiation skills, and trade linkages for 115 Somali livestock value chains across all project areas.

While evaluations boasted success in several objectives, however, most of the projects focused only on certain areas (i.e., Somaliland and Puntland) and often on only a few districts within those states. Thus, while general federal and state level capacities have indeed improved, benefits have reached only a very limited number of pastoralist communities.



7.3.2 Relevance and Sustainability of Program Assistance

The objectives of external assistance outlined above were and are indeed still relevant to the development needs of the country, the livestock sector, and the targeted beneficiaries. When surveyed, stakeholder views on relevance were also in agreement. However, some major binding constraints have remained unaddressed. For instance, while many projects focused on animal health or market linkages, they did not address environmental issues such as degradation and deterioration of pastures and rangelands, even though these issues were widely recognized across projects. Similarly, few (none in southcentral Somalia) projects focused on rehabilitating watering facilities or fostering commercial feed supplies and markets.

External development partners also express high confidence in the sustainability of outcomes of their interventions. However, given the many daunting implementation issues faced and the lack of rigorous impact evaluations and slow progress on domestic resource mobilization, the sustainability of project benefits remains unclear. Donors' optimism is mostly related to private sector partner engagement, overall goodwill from state and federal authorities, gradual though slow improvements in the quality and numbers of ministerial staff, and good security in the northern states and in the capital city. Often projects had specific components that focused on creating ownership, e.g. training of cooperatives and individual members on long-term business strategies or handing over plans of specific components, thus strengthening their sustainability. The WALP project, for instance, had capacity building and creating ownership by getting government ministries to lead key aspects of the project as two of its objectives. Similarly, many projects included components for developing public-private partnerships (PPPs) to ensure effective and sustainable management of the assets handed over.

Most evaluations, reviews and self-assessments, however, qualify this optimistic view about the sustainability of external interventions as subject to continuing adequate financial support being available, which is in fact hardly ever assured. External financial assistance for productive sectors, including the livestock sector, has been and continues to be very limited and subject to wide fluctuations over time. Moreover, domestic resource mobilization at all government levels is still minimal and mostly earmarked for security and administrative needs, rather than for providing other essential services and public investments in the productive sectors.

7.3.3 Implementation Challenges

A variety of challenges presented themselves during project implementation, some more specific to certain projects than others. The impact of interventions was often weakened due to weak capacity, poor human resources, narrow project design, inadequate governance arrangements, including stakeholders' engagement and institutional and donor coordination, and insecurity and lack of infrastructures. Below is a summary of the main issues cited in project evaluations and program reviews undermining their effectiveness and sustainability.

Weak Institutional Capacity. This was the most highlighted and binding constraint to project success. There are overall poor communication and coordination between federal and state ministries, largely due to a lack of resources, incentives and institutional knowledge (with many staff being relatively new). Inadequate cost recovery for services provided and no agreement on revenue sharing among federal, regional, and port authorities result in high dependence by the sector on external donor support, resulting in vulnerability to shocks when support is inconsistent or is abruptly cut off.



Poor Education/Skills Development. There has been a high staff turnover after training or workshops, thus resulting in recently acquired knowledge and skills moving outside the livestock sector. There is also often lack of qualified technicians for maintenance and implementation. Moreover, very low levels of information among stakeholders on animal husbandry best practices (e.g. ruminant feeding), and on food storage and hygiene (e.g., for milk and meat) have undermined interventions to improve animal health and food safety. Finally, there is only limited and fluctuating awareness among regulators of animal and products' standards required by the OIE and importing countries, thus limited ability to ensure traders' compliance to avoid import restrictions.

Narrow Project Design and Limited Focus on Hand-Over and Management. Historically, a piecemeal approach to the sector and lack of advance attention and clear plans for post-investment hand-over and management of assets provided by donors and multilateral development agencies, such as fodder sheds, dairy markets, slaughterhouses, fish-processing plants and cold chains in the northern regions, have resulted in long delays in their operations and in some cases have undermined sustainability of initially positive outcomes. Focusing interventions solely on some segments of the supply chain, moreover, results in poor or limited upstream and downstream linkages.

Poor Donor Coordination. There is in general poor communication and collaboration across experts, project designers and managers, and donors in the sector, resulting often in different projects addressing similar segments and gaps. There needs to be a more holistic multi-donor and multi-disciplinary approach to development of the livestock sector in Somalia.

Limited Beneficiary Involvement. Another common challenge was the limited ownership of projects' activities and assets by beneficiaries, which undermined the achievement of the interventions' objectives and hampered long term sustainability. Lack of input by beneficiaries and inadequate involvement of communities at the design stage resulted often in poor identification of needs and solutions. Furthermore, multiple, parallel projects (as a result of poor coordination) resulted in limited beneficiary or stakeholder engagement in any single one.

Poor Sector Governance. Inadequate policies, strategies, and legal/regulatory frameworks compound the weak human resource base. Also, the FMSs should recognize and respect the coordinating role of federal authorities, including on international negotiations, regulatory enforcement, supervision, and health certifications. Low or no budgetary allocations for staff compensation, moreover, results in high staff turnover, minimal or no service delivery and extension staff, and poor or ineffective outreach to remote rural areas and communities.

Insecurity and Dilapidated Infrastructure. Security and access remain problematic, especially in southcentral Somalia. Often activities cannot be implemented because of lack of access, due to insecurity, exacerbated by insufficient public sector infrastructure and lack of vehicles and communication equipment for field activities. All such factors also often hampered information sharing among project implementing partners.

7.3.4 Lessons Learned

The following are other lessons learned based on a review of donor assessments reviewed:

- There needs be always a major skills and capacity building component in all projects.
- Long procurement delays and slow delivery should be expected, but local procurement and project implementation, though even slower, should be encouraged wherever possible, for capacity building.



- Using local consultants, sourcing in-country expertise and engaging private stakeholders and local authorities in project design and implementation also enhances ownership and chances for project success.
- Since insecurity often restricts what public officials and foreign personnel of donor organizations can do and where they can go, partnering with private sector and local community organizations is often necessary for delivering key services and monitoring project implementation and results.
- A collaborative approach among different institutions that can leverage different expertise and strengths and enhance technical coordination is also critical for building synergies with other country and regional projects, while avoiding duplication of interventions, and thus for project success.
- Value chains are complex and often require good understanding of social dynamics, as relationships among producers, traders and retailers can be more valued by industry stakeholders than pure economic gains.
- Selection of project sites must be transparent, evidence-based and technical criteria-driven. Criteria ought to include access, existing facilities and resources, and realistic chances to utilize new ones.
- There needs to be a continuing strong awareness of regulators and other key value chain actors of SPS and quality standards prescribed by the OIE and/or required by importing countries, to avoid trade restrictions.
- A pooled and programmatic approach that layers different activities, with long timeframes, broad scope and high flexibility, is essential in fragile and complex environments such as Somalia.
- Hand-over and management issues after construction of facilities, especially those of productive and commercial nature as mentioned above, should always be considered and planned for during project design;
- Shocks and disasters should be expected and addressed in project design and resource allocation.
- Multi-sectoral approaches and interventions to strengthen resilience and disaster management are more effective than just a one-sector (livestock-only) approach.
- Suitable technology that is geographically, environmentally, climatically, economically and culturally appropriate should be used to ensure resilience, reduce risk and increase sustainability.
- Local species, while less productive, are better adapted to survive the conditions of the local environment than foreign breeds, for which required inputs are often not available or sustainable.



8.0 CONCLUSIONS AND NEXT STEPS

As elaborated in chapters 2 and 3, the livestock sector remains key to economic growth and poverty reduction, but its performance and prospects are undermined by considerable structural weaknesses, severe exposure to downside risks, and vulnerability to shocks from climate change that can impose major damages and losses and from animal health emergencies that can lead to import bans. The main constraints for the sector's development were advanced in a summary form in section 3.8, chapter 3 and elaborated in more details in the subsequent chapters 4-7. These can be grouped around seven major themes:

1. Weak institutional and policy framework, including limited and largely informal organization of the livestock value chain; lack of coordination and integration among its actors; weak human capacity; minimal or zero budgetary resources; lack of enforcement of existing rules and regulations; and poor linkages and ineffective negotiations and agreements with its main trading partners that also sharply limit the country's potential in the livestock commodities export trade;
2. Dilapidated water and transport infrastructure, causing water shortages due to insufficient and irregular and inadequately harvested rainfall and higher transport and marketing costs than necessary, all of which affect livestock health and survival, availability of natural fodder, and undermine delivery of high-quality animals to domestic and foreign markets and expansion of dairy production and processing;
3. Lack of access to good quality pasture, due to land degradation and reduction of productive land, which is in turn the result of deforestation, soil erosion, invasion of unpalatable plant species, spread of private enclosures, and climate change, which has been bringing increasingly frequent recurrence of natural shocks such as drought and floods;
4. Poor veterinary health services and transboundary animal disease control, with occasional flare-up of epidemics, other livestock diseases, and pests, compounded by limited availability of quality veterinary education and technical training facilities in pastoral areas;
5. Overreliance on only a few export markets and on live animal exports;
6. Financial constraints – with very low government budgetary allocations to livestock and lack of private investment in processing of livestock products (dairy, meat and hides and skins mainly) for value addition, partly due to poor access to credit and no access to insurance, and lack of incentives and encouragement by governments; and
7. Persistent insecurity in large parts of southcentral Somalia.

Despite current daunting challenges, there is in fact strong scope for the sector's recovery and growth in the medium- and long-term, if managed properly, in both live animals and livestock products. Success in tackling the sector's constraints and harnessing its potential, however, will require a clear vision of the priority policy reforms and interventions needed to alleviate the most binding constraints and concerted action across federal and state levels to mobilize and guide public and private investment, with strong, coordinated support by its development partners.

This livestock sector development strategy (LSDS) reflects the key needs and strategic priorities of the Federal Government of Somalia, its Federal Member States, and the sector's key stakeholders for medium- and long-term inclusive sector growth, trade, food safety and food security beyond those related to the sector's humanitarian emergency needs and short-term recovery from the recent 2016/2017 drought. To implement this LSDS, further work is necessary to establish quantitative and qualitative targets related to the objectives and recommendations discussed in the



preceding sections of chapters 4 through 7, and to explore the institutional mechanisms, procedures, interventions and monitoring and evaluation framework for their implementation. The key goals this LSDS has endorsed are to:

1. Strengthen the production and marketing systems for live animals, via improving environmental stewardship of the country's fragile natural resources, and in particular of its rangelands, including to adapt to an increasingly threatening climate change; health services, including the credibility of health certifications; and feed and fodder availability;
2. Develop the many value-added opportunities available in livestock products, via promoting investments in processing and quality improvements, while ensuring food safety.

The MoLFR has already taken a step in the direction of implementing this LSDS by identifying a number of preliminary operational objectives (milestones, reproduced below) and activities with timeframes and cost estimates, as its inputs to the National Development Plan Roadmap:

- Construction and rehabilitation of livestock export centers, two fodder storage facilities, genetic banking and artificial insemination and animal feeding centers;
- Enhancing livestock trade by targeting 4 million animals to be exported and completing vaccination of 28 million animals annually;
- Develop animal production through capacity building (training and equipment);
- Construction of national quarantine station for livestock, animal health centers, and completion and furnishing of the central veterinary laboratory;
- Development of forestry resources, wildlife protection and reforestation of degraded land by planting 1 million trees at national level (2019 – 2020), with 200,000 trees planted by the end of 2019;
- Ending charcoal exports and reducing charcoal use in the country; and
- Development of rangelands and water for livestock.

The FAO has made already advanced plans for piloting a proposed methodology for estimating a new animal population baseline through a combination of aerial and ground surveys in three districts in Somalia. Based on an improved methodology yet to be validated by the FGS and the FMSs, a full nationwide animal population survey would follow in 2020, depending on the availability of needed financial support from donors.

A comprehensive and ambitious Master Plan for the sector's development, however, needs to be developed in order to operationalize this LSDS' priority and medium- and long-term term recommendations. Quantifiable targets for the achievement of the LSDS objectives, moreover, are also needed, as well as specific interventions (policy and institutional changes, technical assistance and investment projects), with estimated costs, proposed timelines, and roles of government and external development partners, that would be a more concrete translation of the often-general recommended interventions in chapters 4-7 of this LSDS. Under the recently approved "Water for Agro-pastoral Productivity and Resilience or *Biyoole* Project, the World Bank is providing financial support as of July 2019 to the MoLFR's for the preparation of such a Master Plan and related project proposals.



These interventions should clarify and address the following areas:

- Overall sector governance and institutional reform, including assignment of responsibilities among all levels of governments; capacity building for the Federal Government and Federal Member States; participation in policy formulation, institutional reforms, and public investment project design and implementation by all stakeholders', including women, youth and disadvantaged groups; and effective aid coordination and partnership by all levels of government with international donors and development agencies;
- Natural resource management, including of the rangelands and including land tenure, and climate change adaptation and mitigation;
- Animal production and marketing systems, including effective rangeland management; animal health services (and credible health certifications for exports), nutrition, and welfare; and necessary data and analyses for evidence-based decision making, among others;
- Production and marketing of dairy, meat, hides and skins (for exports), poultry and beekeeping products, to promote value added and ensure food security and safety;
- Access to modern financial services, including insurance;
- Human resource development, including capacity building for the public and private stakeholders and for technical and higher educational institutions in veterinary, food and animal husbandry sciences.

Finally, to attract external funding to carry out such intervention, the FGS and FMSs also need to clarify effective implementation mechanisms and a monitoring and evaluation framework, that can ensure and assess the effectiveness of interventions (outputs and outcomes) and the performance of implementing agencies, domestic and external. Considering the many fragmented efforts in the livestock sector by various development partners and the limited capacity of federal and state ministries, the establishment of an advisory technical committee would help in the preparation and implementation of such a Master Plan, including via remote links. To carry out all such necessary next steps, and also to implement much-needed animal population survey and feed inventory and balance, the MoLFR is seeking both financial assistance from its development partners for a new technical assistance activity during the second half of 2019 and 2020. This will also ensure adequate input for the most important sector of the Somali economy into the National Development Plan 2020-2024 currently under preparation.



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Table SA1.1: Land Cover / Land Use by district

DISTRICT	IRRIGATED CROP	RAINFED CROP	NAT.VEG. CLOSED to OPEN	NAT.VEG. SPARSE and HERBS	BARE AREA	BUILT UP	WATER BODIES	Total Area (Km2)
Adan Yabaal	0.00	734.88	2,365.80	119.37	743.34	1.49	0.00	3,964.88
Afgooye	902.93	1,306.02	1,239.25	382.58	36.51	76.02	0.25	3,943.55
Afmadow	32.23	6.45	23,073.65	3,575.75	154.72	2.98	34.96	26,880.75
Baardheere	103.66	436.55	13,323.72	1,455.66	5.23	1.74	28.90	15,355.47
Badhaadhe	0.74	60.76	8,833.80	742.13	22.05	3.43	21.32	9,684.23
Baki	36.33	10.90	549.37	2,671.78	147.52	0.73	0.00	3,416.63
Balcad	489.36	1,474.75	1,702.03	135.48	416.56	2.96	2.47	4,223.62
Banadir	0.00	72.39	9.20	29.11	36.82	64.93	0.00	212.45
Bandarbeyla	0.98	0.00	340.62	5,409.13	5,641.04	0.00	0.74	11,392.50
Baraawe	123.30	47.52	2,541.72	298.36	218.08	1.00	1.25	3,231.24
Baydhaba	0.00	2,210.24	10,600.65	77.21	0.50	18.49	0.25	12,907.35
Belet Weyne	396.34	201.44	12,244.65	1,951.29	51.55	19.87	1.26	14,866.40
Belet Xaawo	0.00	6.83	3,683.15	197.46	1.95	0.00	0.49	3,889.87
Berbera	12.09	0.00	534.24	6,775.39	2,884.04	11.37	7.98	10,225.11
Borama	28.82	182.85	812.07	2,020.12	14.77	15.74	1.70	3,076.08
Bossaso	13.80	0.00	141.42	4,916.94	2,842.16	16.47	0.73	7,931.51
Bu'aale	176.89	28.77	5,125.21	131.60	5.75	0.25	26.02	5,494.50
Bulo Burto	189.28	1,034.81	13,659.59	1,127.21	16.98	11.49	3.50	16,042.86
Burco	92.37	0.24	3,624.99	11,700.20	599.05	35.09	0.97	16,052.92
Burtinle	1.97	0.00	8,022.49	1,601.15	67.95	2.95	0.25	9,696.75
Buuhoodle	0.24	0.00	3,655.82	3,018.95	102.52	4.40	0.24	6,782.18
Buur Hakaba	0.00	1,153.41	16,636.18	222.14	4.75	0.75	0.25	18,017.47
Cabudwaaq	0.00	6.93	4,407.87	1,735.18	49.50	0.74	0.00	6,200.22
Cadaado	0.00	13.60	5,124.22	1,241.11	32.15	6.43	0.00	6,417.52
Cadale	0.00	1,949.79	2,230.15	97.63	1,522.98	0.25	0.00	5,800.80
Caluula	5.08	0.00	787.05	3,475.27	329.73	1.45	0.73	4,599.32
Caynabo	0.00	3.41	1,612.28	4,864.34	1,583.56	5.60	0.00	8,069.20
Ceel Afweyn	8.24	0.00	2,289.90	10,562.61	2,819.36	0.97	3.15	15,684.22
Ceel Barde	0.00	52.30	7,083.29	540.40	5.55	1.39	0.00	7,682.94
Ceel Buur	0.00	906.63	11,265.73	1,120.19	60.09	5.46	0.00	13,358.12
Ceel Dheer	0.00	974.83	5,499.25	932.27	2,885.42	4.23	0.50	10,296.50
Ceel Waaq	0.00	2.55	7,673.86	165.51	1.16	6.96	0.23	7,850.27
Ceerigaabo	73.94	2.18	2,469.21	10,576.99	12,903.89	11.39	0.48	26,038.10
Dhuusamarreeb	0.25	178.43	10,716.41	1,741.96	43.37	10.90	1.98	12,693.30
Diinsoor	0.00	981.72	8,540.67	211.22	1.49	2.99	0.00	9,738.09
Doolow	30.78	15.51	1,426.54	141.32	0.25	5.42	12.06	1,631.88
Eyl	0.99	0.00	3,523.01	5,057.69	730.38	0.74	0.74	9,313.54
Gaalkacyo	1.97	0.00	11,473.02	3,493.85	211.22	20.73	0.00	15,200.79
Galdogob	3.69	0.00	1,423.62	529.24	1.23	1.97	0.00	1,959.75
Garbahaarey	32.04	109.15	7,526.03	611.11	17.27	7.76	26.04	8,329.41
Garoowe	9.32	0.00	2,486.07	4,912.78	1,204.28	10.55	0.00	8,623.00



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Gebiley	43.18	959.83	909.37	2,297.32	31.53	28.14	0.49	4,269.85
Hargeysa	69.80	1,239.43	3,460.25	8,312.66	533.13	66.40	0.49	13,682.16
Hobyo	0.00	613.81	16,211.18	7,194.67	1,373.64	5.95	6.69	25,405.94
Iskushuban	6.34	0.00	461.03	10,301.63	7,742.55	2.44	19.98	18,533.97
Jalalaqsi	97.92	280.53	2,533.03	199.35	7.74	1.75	1.25	3,121.58
Jamaame	391.43	11.28	1,629.40	2.51	93.28	0.75	6.52	2,135.17
Jariiban	0.00	0.00	7,631.66	2,960.33	309.92	1.24	0.74	10,903.89
Jilib	278.58	45.76	4,686.41	183.06	22.01	1.25	27.51	5,244.58
Jowhar	727.53	1,393.85	1,775.10	701.55	17.74	4.00	13.99	4,633.75
Kismaayo	124.13	42.55	8,688.33	435.07	30.72	8.06	23.92	9,352.77
Kurtunwaarey	390.85	19.49	1,969.75	162.44	0.00	2.00	0.00	2,544.54
Laas Caanood	6.85	0.24	3,929.35	5,455.89	2,171.21	8.32	0.00	11,571.86
Laasqoray	3.39	0.00	1,700.49	5,228.52	5,302.22	4.85	0.97	12,240.45
Lughaye	0.00	0.00	118.62	2,888.41	317.93	0.00	0.24	3,325.20
Luuq	37.22	65.57	7,563.33	573.67	2.53	9.87	21.77	8,273.97
Marka	518.58	7.02	331.60	99.00	70.93	5.76	0.25	1,033.15
Owdweyne	68.72	5.34	1,434.89	5,596.76	1,230.91	4.61	0.00	8,341.24
Qandala	0.48	0.00	628.68	6,162.64	679.11	1.21	0.48	7,472.60
Qansax Dheere	0.75	942.39	2,318.21	5.75	0.00	0.25	1.25	3,268.60
Qardho	4.65	0.00	1,743.22	10,297.86	6,677.12	8.07	0.00	18,730.92
Qoryooley	563.51	262.02	2,350.45	28.97	0.00	6.24	0.00	3,211.20
Rab Dhuure	0.00	343.90	2,350.95	606.40	3.26	5.26	0.00	3,309.77
Saakow	62.12	418.80	7,284.66	282.54	1.00	0.25	12.52	8,061.90
Sablaale	85.24	1.50	5,832.75	67.24	2.75	0.00	3.00	5,992.48
Sheikh	57.22	0.00	981.17	1,829.23	89.70	2.91	0.00	2,960.23
Taleex	4.16	0.00	329.21	4,166.54	4,114.36	0.73	0.00	8,615.01
Tayeeglow	0.00	127.41	6,030.76	428.93	0.00	4.50	0.00	6,591.59
Waajid	0.00	21.50	2,734.70	44.50	0.00	2.75	0.00	2,803.45
Wanla Weyn	0.00	1,796.74	3,355.97	538.22	6.49	8.99	1.25	5,707.67
Xarardheere	0.00	567.30	2,103.05	1,720.79	678.47	2.24	0.00	5,071.84
Xudun	1.46	0.00	513.24	4,872.88	3,722.23	2.20	0.00	9,112.02
Xudur	0.00	346.73	4,467.86	608.13	0.99	5.19	0.00	5,428.90
Zeylac	0.48	0.00	66.06	5,459.24	536.71	0.96	1.21	6,064.67
TOTAL	6,312.2	23,678.9	338,402.6	190,282.4	74,186.9	605.2	324.0	633,792.2
%	1.0%	3.7%	53.4%	30.0%	11.7%	0.1%	0.1%	100.0%

Source: SWALIM



Statistical Annex 2: Livestock Population Before and After Recent Drought

Table SA2.1: Livestock Population in 2014

Region	Total Camel	Total Cattle	Total Goats	Total Sheep	Total Livestock Population
Total All Regions	6,611,835	3,930,383	28,703,379	13,647,910	52,893,507
Lower Juba	322,042	620,654	732,224	492,673	2,167,593
Middle Juba	165,335	520,175	478,247	393,329	1,557,086
Gedo	770,894	336,629	1,825,849	750,202	3,683,575
Bay	361,562	800,964	1,171,477	117,007	2,451,010
Bakol	617,905	369,601	1,459,008	408,830	2,855,344
Lower Shabelle	286,770	535,447	981,022	464,193	2,267,433
Middle Shabelle	156,138	185,540	1,099,778	521,759	1,963,214
Hiran	638,935	347,044	1,995,619	680,917	3,662,516
Galgadud	461,495	33,978	2,031,000	850,953	3,377,426
Mudug	437,672	13,070	2,057,841	881,057	3,389,640
Nugal	377,872	0	1,959,593	1,217,801	3,555,266
Bari	86,649	0	1,496,383	745,638	2,328,669
Sool	236,260	0	1,541,657	1,267,790	3,045,707
Sanaag	233,942	0	2,842,832	2,044,901	5,121,676
Togdheer	496,815	5,018	1,952,918	582,689	3,037,441
North-West (Woqooyi Galbeed)	564,659	96,567	2,745,465	1,139,224	4,545,914
Awdal	396,890	65,696	2,332,466	1,088,945	3,883,997

Source: FSNAU-MoLFR estimates as of October 2017 for the DINA report (Somalia 2018)

Note: These estimates exclude the animal population in Benadir region (in and around Mogadishu), which is believed to be sizeable.

**Table SA2.2: Livestock Population in 2015**

Region	Total Camel	Total Cattle	Total Goats	Total Sheep	Total Livestock Population
Total All Regions	7,020,482	4,703,340	31,799,888	14,830,082	58,353,792
Lower Juba	318,520	682,293	835,296	556,134	2,392,242
Middle Juba	163,689	578,714	547,646	440,007	1,730,057
Gedo	751,398	360,927	1,719,902	695,237	3,527,465
Bay	402,039	1,022,628	1,564,509	163,657	3,152,833
Bakol	685,183	487,191	2,034,564	576,674	3,783,611
Lower Shabelle	349,147	716,914	1,264,876	579,288	2,910,224
Middle Shabelle	170,476	244,999	1,263,059	580,789	2,259,324
Hiran	731,457	393,570	2,386,125	816,685	4,327,836
Galgadud	477,889	35,519	2,279,783	967,451	3,760,641
Mudug	437,406	13,305	2,304,958	1,002,650	3,758,319
Nugal	375,805	0	2,203,698	1,359,942	3,939,445
Bari	86,620	0	1,612,942	815,605	2,515,167
Sool	252,662	0	1,574,829	1,282,787	3,110,279
Sanaag	246,315	0	2,848,783	2,055,383	5,150,480
Togdheer	536,343	5,018	2,068,707	617,904	3,227,972
North-West (Woqooyi Galbeed)	607,783	96,567	2,882,861	1,195,573	4,782,784
Awdal	427,750	65,696	2,407,351	1,124,315	4,025,112

Source: FSNAU-MoLFR estimates as of October 2017 for the DINA report (Somalia 2018)

Note: These estimates exclude the animal population in Benadir region (in and around Mogadishu), which is believed to be sizeable.

**Table SA2.3: Livestock Population in 2016 (pre-drought)**

Region	Total Camel	Total Cattle	Total Goats	Total Sheep	Total Livestock Population
Total All Regions	7,173,988	5,319,533	30,998,566	13,685,565	57,177,652
Lower Juba					
Middle Juba					
Gedo					
Bay					
Bakol					
Lower Shabelle Benadir					
Middle Shabelle					
Hiran					
Galgadud					
Mudug					
Nugal					
Bari					
Sool					
Sanaag					
Togdheer					
North-West (Woqooyi Galbeed)					
Awdal					

Source: FSNAU-MoLFR estimates as of October 2017 for the DINA report (Somalia 2018)

Note: These estimates exclude the animal population in Benadir region (in and around Mogadishu), which is believed to be sizeable.

**Table SA2.4: Livestock Population in 2017 (post- drought)**

Region	Total Camel	Total Cattle	Total Goats	Total Sheep	Total Livestock Population
Total All Regions	6,302,967	4,918,342	26,400,917	11,319,747	48,941,974
Lower Juba					
Middle Juba					
Gedo					
Bay					
Bakol					
Lower Shabelle Benadir					
Middle Shabelle					
Hiran					
Galgadud					
Mudug					
Nugal					
Bari					
Sool					
Sanaag					
Togdheer					
North-West (Woqooyi Galbeed)					
Awdal					

Source: FSNAU-MoLFR estimates as of October 2017 for the DINA report (Somalia 2018)

Note: These estimates exclude the animal population in Benadir region (in and around Mogadishu), which is believed to be sizeable.



MAIN ANNEXES

Main Annex 1: Overview of State-Level of Stakeholders' Consultations, Oct-Nov. 2018

Main Annex 2: Overview of Stakeholders' Consultations at Validation Workshop, June 2019

Main Annex 3: Somalia' Livestock Genetic Resources

Main Annex 4: Somalia's Livestock Statistics

ONLINE ANNEXES

Online Annex 1: Benadir Consultations and Capacity Needs

Online Annex 2: Southwest Consultations and Capacity Needs

Online Annex 3: Somaliland Consultations and Capacity Needs

Online Annex 4: Puntland Consultations and Capacity Needs

Online Annex 5: Galmudug Consultations and Capacity Needs

Online Annex 6: Hirshabelle Consultations and Capacity Needs

Online Annex 7: Jubaland Consultations and Capacity Needs

ONLINE BACKGROUND STUDIES

Online Background Study 1: Johnston, Rouja (2019). *Peri-urban Dairy Value Chain Development – Somalia*. April 2019. Based on submissions by three sets of local researchers from Hargeisa, Garowe and Mogadishu. Funded by the World Bank.

Online Background Study 2: Mohamud, Abdinur and Abdirahman Nur Keilie. (2018). *Review of Livestock-related Education and Training in Somalia*. October 2018. Funded by the World Bank.

Online Background Study 2: FAO-Somalia (2019), Design of a Methodology to Estimate the Livestock Population of Somalia. Funded by UK-DFID.



Main Annex 1: Summary Overview of October-November State-Level Consultations

Introduction

From October 14th to November 6th, 2018, the federal Ministry of Livestock, Forestry and Range (MoLFR), the FAO and the World Bank carried out consultation workshops with public and private stakeholders from the Benadir region and all six federal member states, as part of the joint background preparation for a Livestock Sector Strategy (LSS) for the next 10 years.⁴² The workshops took place as listed below:

Regions/States	Location	Dates	Somali Participants (of which Women)
Benadir/Federal	Mogadishu	14-15th Oct	30 (6)
SWS	Mogadishu	16-16th Oct	20 (3)
Somaliland	Hargeisa	20-21st Oct	33 (5)
Puntland	Garowe	23-24th Oct	30 (4)
Galmudug	Mogadishu	28-29th Oct	20 (3)
Hirshabele	Mogadishu	30-31st Oct	20 (1)
Jubaland	Kismayo	3-4th Nov	30 (3)
Total Participants			180 (25)

The objectives of these consultations were to:

- build ownership by the federal and state governments of the development of the LSS, as well as strengthen the leadership of the federal government and the federal-state dialogue and coordination in the process
- capture federal and state-level long-term (2030) 'visions' regarding the development trajectory for the livestock sector
- identify critical performance issues, challenges, opportunities, and development needs of state-level livestock production, marketing, and processing systems (be it extensive or intensive), including the priority value chains (VCs) in each state
- identify sources and gaps of knowledge and public and private sector stakeholders in each state, and collect available qualitative and quantitative information.

For each state's 2-day long workshop, World Bank and FAO staff made five presentations to share with participants analytical findings from recent or ongoing studies and to guide the subsequent group discussions and presentations by the participants. Furthermore, senior FMS livestock ministry officials (federal in the case of the Benadir workshop) gave either a Powerpoint or an oral overview presentation of the sector's performance, challenges and needs. This was followed by identification/prioritization of value chains/subsectors as summarized below (with the exception of Benadir and SW State), and by breakout groups' discussion and presentations (mostly in Somali) for the top value chains (VCs)/subsectors, that in most cases were live animal, feed/fodder, dairy and meat. Poultry, hides and skins, and beekeeping followed in order of economic importance and potential at considerable distance.

⁴² Some state-level consultations could not be held in their state capitals due to security considerations, and in such cases state officials and stakeholders were flown to Mogadishu.



Rank	State level VC Prioritization						
	States						
	Benadir	SWS	Somaliland	Puntland	Galmudug	Hirshabele	Jubaland
1	Dairy	Meat	Live Animal	Meat	Live Animal	Live Animal	Live Animal
2	Fodder	Poultry	Dairy	Live Animal	Dairy	Meat	Meat
3	Live Animal	Dairy	Meat	Dairy	Fodder	Dairy	Dairy
4	Poultry	Fodder	Fodder	Fodder	Meat	Fodder	Fodder
5	Honey	Honey	Poultry	Hides & Skins	Hides & Skins	Poultry	Poultry
6	Hides & Skins	Hides & Skins	Honey	Poultry	Honey	Honey	Hides & Skins
7	Animal Health	Animal Health	Hides & Skins	Honey	Poultry	Hides & Skins	Honey

Cross-sectoral General Findings

While the output of the consultation workshops can be found in the Online Annexes for each state and for each of the main VC/subsectors, the general cross-cutting findings of these consultations were that:

1. Despite the many challenges, the livestock sector has big potential across all sub-sectors, because of a growing local population and returning diaspora, increasing global demand for animal source foods, very low current production and productivity levels, and minimal processing;
2. To realize this potential, what is required is:
 - A. significant capacity building of public sector institutions, including for:
 - i. policy,
 - ii. development and enforcement of regulations,
 - iii. ensuring quality of veterinary and animal husbandry education, and
 - iv. data collection and analysis (via an animal population and production survey and a feed inventory and balance assessment);
 - B. large public investments for:
 - i. expansion of public services (with an emphasis on strengthening the veterinary services, including disease control, meeting international sanitary and phytosanitary standards and ensuring veracity and credibility of health certifications), and
 - ii. rehabilitation and expansion of critical infrastructures such as for water catchment and road transport (Malabo Declaration),
 - C. security and access to finance for the private sector to expand its current minimal processing of livestock products;
3. The sector's development and growth also needs:



- a. addressing the many and vast knowledge gaps;
- b. adopting a commercialization and market orientation mindset, right down to the livestock owner level;
- c. moving towards a resilience (plan and prepare for shocks) and development (growth) focus of external assistance, in addition to ensuring emergency interventions at times of crises;
- d. fostering appropriate and innovative technology;
- e. ensuring coordination and collaboration among all stakeholders and external partners, including clarification on roles and responsibilities under PPPs; and
- f. devoting adequate resources to the sector.

For the four main VCs/subsectors that workshop participants identified in almost all states, the key cross-state findings are summarized below.

Live Animals: General Findings

Current situation

- Traditional production systems (and associated limitations incl. fodder, water, diseases)
- Increased population of animals (but high vulnerability to shocks)
- Reduced exports due to droughts and bans from KSA and UAE (exception of Hajj)
- Current export market is Oman, Yemen and small shipments to Egypt.

Constraints and gaps

- Insufficient veterinary services and oversight including on certification
- Failure to meet international standards and requirements of importing countries
- Insufficient knowledge and data
- Inconsistent implementation of regulations and policies
- Overdependence on traditional production systems and limited export markets

Opportunities and potential

- Existing and growing demand for grass-fed livestock in local and international markets
- Somalia's' strategic location and proximity to the Gulf markets.

Recommendations from state participants

- Provide extension services to livestock producers (on health and animal husbandry)
- Rehabilitate holding grounds and other critical infrastructure such as roads (role of public vs private?).
- Improve laboratory and diagnostic services
- Establish credible government oversight of health certification processes
- Conduct comprehensive, reliable census/survey of animal population and production.



Meat: General Findings

Current Situation

- Only one private sector international standard abattoir is operational (SoMEAT only to Oman).
- Increased local market for consumption of fresh meat
- Many slaughterhouses in poor state or repair

Constraints and Gaps

- Recurrent droughts
- Most abattoirs and slaughterhouses not functioning properly.
- Difficulty in meeting importer country standards
- High transport costs for exports

Opportunities and potential

- Growing global, regional and domestic markets for quality meat
- Large "grass-fed" livestock population
- Well positioned for Gulf states and other markets.

Recommendations from state participants

- Build modern slaughterhouses in cities and establish PPPs for their management
- Strengthen government oversight and auditing of meat safety to comply with international standards
- Reduce taxes on meat or improve services provided by government
- Collect and publish statistics on the meat subsector, beyond FSNAU retail prices.

Dairy: General Findings

Current situation

- Milk production mainly from traditional pastoral system (subsistence, poor commercial)
- Camel milk is the most in demand (+ cow and also some goat milk)
- Large-scale importation of milk and milk products
- Lack of finance means very little processing or storage of milk.

Constraints and gaps

- Low and very seasonal production levels under pastoral system
- Limited disease control and surveillance
- Lack of feeds and feed formulation
- Low levels of knowledge on commercial milk production
- High transport costs from remote production areas
- Low Access to credit and finance.

Opportunities and potential

- Large number of milk producing animals
- Commercialization has started
- High and growing domestic demand, from increasing urbanization and higher incomes.

Recommendations from state participants

- Improve veterinary and extension services to producers (incl. milk hygiene and management)
- Support private sector investment in milk processing (incl. cold chains)
- Strengthen government oversight and auditing of milk (and in general food) safety, including by establishing quality control services at collection centers and markets
- Add value to milk especially during the wet season when there is plenty of supply



- Aim to reduce milk imports rather than become an exporter (with possible exception of processed or powdered camel milk)
- Collect and publish statistics on the dairy sector, beyond FSNAU retail prices.

Animal Feeding and Fodder: General Findings

Current situation

- Traditional grazing system, (minimal input and seasonal shortages of pasture compounded by breakdown of traditional management institutions)
- Limited commercial feed/fodder production
- Continuous overgrazing and reduction in edible biomass
- Deforestation resulting in erosion and desertification.

Constraints and gaps

- Little or no effective grazing management (defunct National Range Agency)
- Insecurity and low access to rangelands in parts
- Expansion of enclosures (and failure to control)
- Low levels of knowledge on fodder production and management.
- Low access to fodder seeds and other production inputs
- Poor infrastructure (storage, road, water)
- Land tenure issues.

Opportunities and potential

- Increased demand of fodder and feed supplies (Commercialization, improved breeds)
- Existing large-scale crop production in some states
- Plenty of suitable land and water available in some states (e.g. SWS, Jubaland)
- Previous rangeland investments (e.g water catchments) only need rehabilitation

Recommendations from state participants

- Carry out a feed inventory and feed balance and develop a feed and rangeland management strategy
- Support training on feed and fodder production
- Rehabilitate and manage rangelands by reestablishing a rangeland management agency and enforcing policy and laws (including traditional systems where relevant)
- Support private sector and cooperative investment in feed and fodder production.



Main Annex 2: Summary Overview of Validation Workshop, June 17-19, 2019

Livestock plays a major role in Somalia; almost half of the population living in rural areas rely on it and it remains the largest contributor to Somali GDP and livelihoods, economic growth and poverty reduction. The sector's performance and prospects, however, are undermined by various production constraints such as poor animal nutrition, diseases, eroded genetic resources, and poor natural resource management besides other structural and institutional weaknesses.

This event marks the latest in a broad-based and broadly-inclusive consultative process that was built on consensus across all levels of the GoS around a single vision and a set of growth and development priorities for the livestock sector. The outcomes of these combined activities will inform and guide near-term interventions toward recovery and medium- to long-term programming, policy reforms, and investments for growth of competitive and sustainable Somali livestock sector.

The World Bank in collaboration with FAO is engaged in supporting the FGS in the development of a multi-year livestock sector strategy that will contribute to the improvement of local livelihoods based on a climate resilient and sustainable growth of the sector.

The workshop which took place during the period of 17-19 June 2019 at Mogadishu's Aden Adde International Airport (MIA) with the participation of a broad spectrum of sector stakeholders including federal and state government officials, private sector leaders, professional/industry associations, non-governmental organizations, academia/research institutes, livestock development partners and international donor community. A total number of 160 participated the validation workshop.

The overall objective of the workshop was to present, discuss, and validate a draft document of Livestock Sector Development Strategy (LSDS) and to initiate a forward discussion on how to mobilize its implementation. The workshop drew participants from all the states and Federal government of Somalia and was officially opened by The Prime Minister of the Federal Republic of Somalia HE. Hassan Ali Kheire. He commended the organizers for their effort in organizing such a well-attended and all- inclusive workshop and requested for workable solutions emanating from the workshop to drive the livestock sector to a notch higher. Because of the importance to livestock to Somalis, the Prime Minister requested from Minister of Livestock to table the outcome of the workshop and LSDS document at upcoming Ministerial cabinet meeting.

The FGS Minister of Livestock, Forestry and Range (MoLFR), Hon. Hussein Mohamud Sheikh, the World Bank country Representative to Somalia, Mr. Hugh Riddell, and the FAO Representative to Somalia Mr. Serge Tissot in their opening remarks all highlighted the importance of livestock to economic development of Somalia. They lamented on the drawbacks from drought to livestock development and impacts on livelihoods, especially loss of GDP and livelihoods. They commended the development of the Livestock Strategy document as a significant milestone given that its actualization will have far reaching impacts in overall development of the country in terms of improving livestock sector more sustainable and competitive in the global markets, competitive against a backdrop of other emerging challenges such as climate change and increased cross-boundary diseases and new requirements in the international livestock markets. Furthermore, they pleaded with the donor community to assist mobilize resources to actualize the strategy across the different value chains. Finally, they recommended that the strategy would require development of workable implementation frameworks along these chains and be integrated into the National Development Program and called for all to share their experiences.



Highlights of strategy based on sectoral presentations

The MoLFR/FGS technical team presented on the ministry's milestones in the National Economic Development Roadmap of the FGS. Key gaps for which donor community/donors could assist were highlighted. Areas covered included Animal Production including Feed Resource development; Animal Health including certification for Somalia Livestock Export and National Livestock Quarantine improvement; Livestock Genetic Resources; Value-addition of Livestock Products; Livestock Marketing; Rangelands development; Livestock census, Livestock Infrastructure development and Capacity development

Despite current daunting challenges, there is in fact strong scope for the sector's recovery and growth in the medium- and long-term, if managed properly, in both live animals and livestock products. Success in tackling the sector's constraints and harnessing its potential, however, will require a clear vision of the priority policy reforms and interventions needed to alleviate the most binding constraints and concerted action across federal and state levels to mobilize and guide public and private investment, with strong, coordinated support by its development partners.

This livestock sector development strategy (LSDS) reflects the key needs and strategic priorities of the Federal Government of Somalia, its Federal Member States, and the sector's key stakeholders for medium- and long-term inclusive sector growth, trade, food safety and food security beyond those related to the sector's humanitarian emergency needs and short-term recovery from the recent 2016/2017 drought.

To implement this LSDS, further work is necessary to establish quantitative and qualitative targets related to the objectives and recommendations discussed in the preceding sections of chapters 4 through 7, and to explore the institutional mechanisms, procedures, interventions and monitoring and evaluation framework for their implementation. The key goals this LSDS has endorsed are to:

- 1) Strengthen the production and marketing systems for live animals, via improving environmental stewardship of the country's fragile natural resources, and in particular of its rangelands, including to adapt to an increasingly threatening climate change; health services, including the credibility of health certifications; and feed and fodder availability;
- 2) Develop the many value-added opportunities available in livestock products, via promoting investments in processing and quality improvements, while ensuring food safety.



Main Annex 3: Somalia' Livestock Genetic Resources

Somali livestock are of a broad range of breeds that are well adapted to a nomadic way of life, limited feed resources and intermittent water supply. Because Somali people live in several countries other than geographical Somalia, some of their traditional livestock breeds are also found in Djibouti, Ethiopia and Kenya.

Camel breeds. Somalia, one of the world's largest dromedary camel populations are kept by nomadic pastoralists under traditional management. Surveys conducted during the years 1982/86 listed two main types of camels in Somalia: (i) northern camels and (ii) southern camels. The northern type is divided into three main sub-types: a) Coastal camels -of northern Somalia; b) Guban camels -of the mountain areas of Somalia.; and c) Hawd camels -of the Nugaal area. The southern type is also divided into three main sub-types: d) Hoor of central Somalia, e) Siif-da'ar of south central Somalia and f) Eydimo of western Somalia.

Cattle breeds. The cattle of Somalia are mainly the East African Zebu type of which the following types are recognized, the Somali Boran, Dauara and Surqo (Rege and Tawah 1999; Nur 2005). The Somali Boran are believed to be a descendant of the first introduction of zebu into Africa from West Asia and are thought to have evolved following the migration of Ethiopian cattle into Somalia and Jubaland along the Somalia-Ethiopia border (Rege and Tawah 1999). The Surqo breed is a zenga breed (Rege and Tawah 1999; Nur 2005). The zenga breeds are breeds that resulted from zebu-sanga crosses that came about following the introduction of zebu cattle into Africa from Asia. The Surqo breed is a crossbred of the Boran of Somalia or Ethiopia with an unknown sanga population (Rege and Tawah 1999). The Zebu cattle of Somalia mainly belong to a group designated as "Small Zebus of the Somalia". The main "breeds" are North Somali, Gasara and Garre. Boran cattle similar to those of southern Ethiopia and northern Kenya are found along Somalia's border with these countries. The Jiddu or Tuni is an Intermediate Sanga/Zebu type with long horns and possibly has a certain degree of tolerance of trypanosomosis.

Goat breeds. The main types of goats found in Somalia, are the Long-eared Somali goat, the Short-eared Somali goat, and, to a lesser extent, the Somali Arab goat. The Somali Long-eared goat is thought to be a descendent of the Somali, Arab goats, which were introduced from Arabia (FARM-Africa 1996). The Somali Arab goat, as the name implies, was introduced into Africa by Arabian traders. Almost all Somali goats are short eared. Breeds of this type are classed as "Somali" In the traditional Somali descriptive systems the 'short eared/ degiar' is smaller than the 'degwein' and has short prick ears in contrast to the forward inclined Pendent ears of the latter. The goat is of small to medium size with males weighing 30–50 kg and females 25–40 kg. A further short eared goat is the Arab (which in this context is a Somali clan name). It is the smallest of the Somali goats and is kept by coastal dwellers in the vicinity of towns. The goat is prolific with many twin and not a few triplet births being the rule. Some long eared ('deguein') milk goats are kept in villages and towns. These goats are usually known as Benadir and mainly occur in the southern half of Somalia.

Sheep breeds. The Somali or Black Head Somali is the main sheep breed. This fat-rumped meat breed is found throughout Somalia. It is highly appreciated as a meat animal by people in the countries to which Somalia exports livestock. Somali sheep are the immediate ancestor of the Black Head Persian which was developed in South Africa in the late nineteenth and early twentieth century's and that has been widely used for crossbreeding in many parts of Africa and elsewhere in the tropics. The classic color, as suggested by the name, is white with a black head. The fat rump is the most characteristic feature.



Main Annex 4: Livestock Sector Statistics

The last livestock census in Somalia was done in 1975. Until the late 1980s, the government conducted reliable sample surveys of the livestock population, which was estimated at about 40 million for 1988 (Somalia 1989; FAOSTAT). Since then, animal population estimates have been based largely on projections and extrapolations by external agencies with very limited current information or access to many parts of the country. One estimate suggested a livestock population of 39 million in 1998 (FAO, World Bank and EU 2004).

New baseline estimates for 2014 were prepared by the FSNAU in October 2017 and agreed to as pre-drought baseline figures by the representatives of MoLFR, the federal Ministry of Planning and International Cooperation, the livestock ministries of FMSs, and their international partners for the Drought Impact Needs Assessment (Somalia 2018a). These desk estimates are reproduced in Statistical Annex SA2.1 of this report (LSDS). They placed the country's livestock in 2014 at 52.9 million head, including 28.7 million goats, 13.6 million sheep, 6.6 million camels, and 3.9 million head of cattle. The FSNAU figures for goats were much larger and the figures for camels and cattle somewhat smaller than previous estimates by other sources and their regional distribution is also different (see World Bank-FAO 2018b, section 5.7, pp 56-58).

In addition to livestock exports which are easier to monitor, a significant number of animals are marketed domestically, generating employment for the local population, especially women, who often sell domestic meat and produce animal byproducts, including milk, soaps, and jewelry from bones. In the early 2000s, the FAO estimated offtake percentages at 27.3 percent for sheep, 23.3 percent for goats, 11.3 percent for cattle, and 1.6 percent for camels (FAO 2004). Livestock product guess-estimates are also available in Somalia 2018 and chapter 5 of World Bank-FAO 2018.

All such point estimates, however, are highly unreliable, given the high uncertainties about demographic parameters, recurrent droughts (and associated mortalities), and frequent but unpredictable cycles of destocking and restocking. They are also mostly inconsistent with previous ones, as their methodology cannot be used to generate plausible time series. For instance, the October 2017 point estimate by FSNAU of livestock population were derived by taking very disaggregated estimates of human population and household sizes (from UNFPA 2014 and from NDP 2017-2019) and multiplying household size by the average number of animals per household, as found in various livelihood surveys of pastoral households in limited geographical areas conducted at different times over the past decade (FSNAU and FEWS-NET 2016). Applying the same parameters and assumptions to smoothly growing estimates of human population would result in equally smoothly growing estimates for livestock, without regards to higher mortality due to recurrent shocks like droughts or higher asset accumulation due to unpredictable export bans.

The FAOSTAT database shows figures, including those for livestock products, annually over a long time, but the economic team working on World Bank-FAO 2018b found them unreliable and implausible. They often appear arbitrary and/ or smoothly extrapolated or flat. Furthermore, no FAOSO staff has any knowledge of who in government has been reporting these figures to FAOSTAT in Rome or how these figures have been validated. They seem to have been reported by (former) government officials directly to FAOSTAT in Rome even during the civil war years when no (recognized) government was in power. Other livestock (e.g. TLU) estimates in any other national or international databases are equally unreliable.

Between February and March 2016, the World Bank in collaboration with Somali statistical authorities conducted a first wave of the Somali High Frequency Survey (HFS) in all accessible areas of nine regions within Somaliland, Puntland and southern Somalia (World Bank 2016). The survey interviewed 2,882 urban households, 822 rural and 413 households in Internally Displaced People (IDP) settlements. The sample was drawn randomly based on a multi-level clustered design. The second wave was conducted in



2018 and its overall analysis has become available in early 2019 (World Bank 2019; Questionnaire for Wave 2 is available online). This dataset includes comprehensive information on assets and consumption, to allow estimation of poverty rates. Its questionnaire also included 39 questions related to livestock (Section G), but whose responses save two have not yet been analyzed.

A Pilot Livestock Survey limited to two districts was undertaken by the FAO during 2008-2009. It had been planned to cover two contrasting districts, Oodweyn in Somaliland and Jowhar in southern Somalia, but escalating insecurity and cost-saving considerations forced a suspension of activities in Jowhar and forced a shift of the second district to Borama, also in Somaliland. The pilot survey combined use of aerial and ground survey teams, with the former designed to estimate the number of households/ *bomas* in the districts plus the numbers of animals.

Through the implementation of this pilot project that undertook wet and dry season surveys in two contrasting districts, the methodology for data collection, entry, analysis and evaluation was tested; qualified Somali experts were integrated into the implementation team; government authorities (i.e. ministries of livestock and planning and local district authorities from Borama and Oodweyn) took part in the implementation of the project; a total of 47 personnel as ground teams were identified and trained to undertake the ground surveys and thereby building capacity for future such surveys; questionnaires were developed and tested during the pilot study; and data analysis and evaluation of the methodology was conducted. Finally, stakeholder meetings were also held in Hargeisa, Mogadishu and Nairobi to present the findings and obtain feedback. The pilot survey's final report and its annexes do highlight the strengths and weaknesses of the aerial and ground surveys and how both can be utilized for a more comprehensive, national baseline animal population survey.

The general lessons from the Pilot Survey's final report that will contribute to an improved and more robust methodology in the design of a future national survey were as follows (for original lessons, see FAO 2010, p. 17):

- Unforeseen possible deterioration in security conditions requires that future project's scope, methodology and timelines be flexible, robust, and realistic;
- Advance simulation is needed to determine the sample size for the aerial survey for better accuracy of estimating animal numbers and smaller gaps in comparison with ground survey estimates;
- Given the shortage of skilled workers in the country, including trained personnel on data collection, entry, analysis and evaluation at all levels, detailed training of field personnel (ground teams) is needed for a better understanding of data collection methodology;
- Adequate field-testing of the questionnaires is important.
- Data analysis is to be standardized for better understanding of results presentation.
- Funding should be adequate and cost-sharing arrangements should be clarified to carry out all the necessary steps to finally have reliable animal numbers.

More specifically, other lessons recommended by the University of Bristol to improve quality of responses and data from ground surveys and which would make use of more advanced statistical methods a possibility were as follows:

- Some questions from the ground surveys will need re-writing and it is particularly important that the livestock number questions are thoroughly tested before they are used in the future.
- It is essential that respondents and fieldworkers understand questions and range of expected responses.
- Fieldworkers and data processors should be fully trained to check for data inconsistencies and the entry of data should be thoroughly checked before being sent for analysis.
- Much time and effort were invested in removing the many errors from the data. Some errors could be avoided by investing more time in the design of the database, for example, with more drop-down boxes and validation rules incorporated. To reduce the time required to clean the data and to allow for random checks where validation is not possible, the person analyzing the data should receive



copies of the completed paper survey forms. One particular error was the usual survey problem of confusing missing values with zero. While some missing values were recorded as zero, the main problem for this pilot study was that many zeros had been recorded as missing values. Training and a code book is needed for the data processors containing the possible responses to questions under different scenarios.